

Analysing and Modelling Spatial Woodland Structure

Habilitationsschrift (D.Sc. dissertation)

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by

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Pommerening, Arne:

Dadansoddi a Modelu Adeiledd Coed Gofodol

Analysing and Modelling Spatial Woodland Structure

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I Gymru, oherwydd yr hyn y mae hi'n ei olygu i mi.
Diolch o galon.

I Jonathan Ole ac i Lowri Angharad, fy mhlant annwyl,
am eu cariad a'u hir amynedd.

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Abstract. This Habilitationsschrift (D.Sc. dissertation) discusses approaches to spatial woodland analysis and modelling in the general context of spatial statistics with particular emphasis on point process statistics. It explores how these approaches relate to each other and considers important differences in concepts and assumptions.

Six publications (I-VI) by the author are included in this Habilitationsschrift and present recent advances in: the development of summary characteristics of spatial woodland structure (I), edge-correction methods (III), non-parametric modelling of spatial woodland structure (II, V), the application of point process methods to problems of woodland management (IV) and allometric modelling of tree crowns (VI).

Crynodeb [Dadansoddi a Modelu Adeiledd Coed Gofodol]. Mae'r Habilitationsschrift (traethawd D.Sc.) yma yn ymdrin â dulliau dadansoddi a modelu adeiledd coed gofodol yng nghyd-destun cyffredinol ystadegaeth ofodol gyda phwyslais arbennig ar ystadegaeth proses pwynt. Archwilia'r traethawd sut y cyfeiria'r gwahanol ddulliau at ei gilydd a beth yw'r gwahaniaeth mewn cysyniadau a thybiaethau.

Cynhwysa'r Habilitationsschrift chwe chyhoeddiad (I-VI) gan yr awdur a chyflwyna ddatblygiadau newydd ym maes ystadegaethau cryno adeiledd coed gofodol (I), dulliau cywiro effaith min (III), modelu adeiledd coed gofodol gyda dulliau amharedrig (II, V) a hefyd dulliau ystadegaeth proses pwynt wedi eu cymhwyso i broblemau rheoli coetiroedd (IV) a modelu alometrig brigau coed.

Zusammenfassung [Analyse und Modellierung Räumlicher Waldstruktur]. Die vorliegende Habilitationsschrift diskutiert Ansätze zur Analyse und Modellierung räumlicher Waldstruktur im Gesamtzusammenhang räumlicher Statistik. Der Schwerpunkt der Abhandlung liegt in der Punktprozessstatistik, und es werden Gemeinsamkeiten und wichtige konzeptionelle Unterschiede der betreffenden Ansätze untersucht.

Die Habilitationsschrift umfasst sechs Publikationen (I-VI) des Autors, die Neuentwicklungen im Bereich statistischer Größen zur Waldstrukturbeschreibung (I), der Randkorrektur (III) und der nicht-parametrischen

Waldstrukturmodellierung (II, V) darstellen. Ferner werden Punktprozessmethoden auf Probleme des Waldbaus (IV) und der allometrischen Modellierung (VI) von Baumkronen angewendet.

Резюме [Анализ и моделирование пространственных структур леса]. Настоящая диссертация на соискание учёной степени доктора наук посвящена подходам к анализу и моделированию пространственных структур леса в общем контексте пространственной статистики. Главное направление работы лежит в области статистики точечных процессов и посвящено исследованию общих черт и существенных концептуальных отличий различных подходов и предположений.

Диссертация содержит шесть научных публикаций (I-VI) автора, которые дают обзор новых идей в области статистических величин для описания структур леса (I), методов поправки краевого эффекта (III) и не-параметрического моделирования структур леса (II, V). Автором описывается применение методов точечных процессов в лесоводстве (IV) и аллометрическом моделировании древесной кроны (VI).

Preface

*Hi hen, eleni ganed.
It is old. This year it was born.*

The consideration of the spatial nature of data by forest scientists is both old and new.

Since the end of the 19th century silviculturists and forest yield scientists have related growth and timber quality to initial spacing and tree density. At about the same time forest statisticians realised that the performance of sampling methods depends on the spatial distribution of trees in a forest. With the advent of soil science at the beginning of the 20th century, the existence of localised differences in microsites was acknowledged as relevant. Later, geo-statistical methods allowed the estimation of soil properties and other site characteristics for any point in a forest from sample data. This is an important pre-requisite for biogeochemical modelling. More recently tree growth models and other plant scientists have developed competition indices to account for the change in growing space and resources available to trees over time.

During the last 50 years spatial statistics has emerged as a special branch of statistics with a wide range of methods that have considerably improved our ability to identify and quantify spatial patterns and opened the possibility of linking them to environmental processes. Most of these methods were first devised in mathematical statistics and later found applications in various fields of natural sciences. Forest scientists, such as Matérn, have influenced modern point process statistics and forestry data have become a natural field for application of spatial statistics. This development has been paralleled with significant progress in mapping and surveying as well as in general computer technology. During this same period foresters and ecologists have independently developed various unrelated quantitative concepts to describe and to model spatial plant patterns such as measures of structure, diversity and competition.

This Habilitationsschrift is a dissertation in fulfilment of a D.Sc. degree at the University of Natural Resources and Applied Life Sciences, Vienna, Aus-

tria. It consists of a Rahmenschrift (p. 1 - 36) and six research papers, which are reprinted in the appendix. The Rahmenschrift discusses, in a systematic way, the most important approaches to spatial woodland analysis using terms and definitions of general spatial statistics to explore the relationship between traditional forestry approaches and concepts of spatial statistics. This provides a foundation for understanding the similarities and dissimilarities between the various concepts and gives suggestions for future methodological research.

There are many people to thank for their support and advice and sadly I have to restrict myself to only a few names. Among those who encouraged me to advance my academic career with this habilitation I wish to name especially my parents, Ursula and Baldur Pommerening, and my mother in law, Александра Алексеевна Шмуратова. To their continuous support this work owes more than I can express.

Three academic mentors, Professor Klaus von Gadow, Professor Hans Pretzsch and Professor Dietrich Stoyan, excited me with spatial statistics at the beginning of my research career and provided me with a thorough scientific education. For their many lively and inspiring discussions I cannot thank them enough. Among these colleagues and friends I particularly wish to mention Professor Dietrich Stoyan whose enthusiasm and continuous ready support throughout the years have been a great asset and a privilege. This work owes a lot to the many fruitful discussions we had. The mutually beneficial knowledge transfer from statistics to forestry and vice versa has had a significant positive impact on both our lives.

My colleague Professor Hubert Sterba kindly acted as my BOKU mentor and has taken much interest in my work and in our exchange between Vienna and Bangor, for which I am very grateful. Together with his wife Veronika he also helped the family and me to settle during the weeks we spent at Vienna while I was working on the habilitation. Professor Hubert Hasenauer provided advice on habilitation matters and he and his family contributed to pleasant stays in Vienna.

I am also grateful to the Executive of the School of Environment and Na-

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Particular thanks to Juliane Dahl for the cover picture and to Heledd Wyn Owen for designing the cover of this book and for taking care of its production. Diolch o galon i ti.

And finally I cannot close without a special word of thanks to my partner Natasha for her sacrifices that enabled me to pursue my dreams: Спасибо за твою поддержку, твоё терпение и все твои лишения ради моей научной деятельности, которая для меня очень важна. Я очень тебе благодарен.

Amen, dyn pren.

Bethel/Bodorgan, 30.04.2008

Arne Pommerening

1 Introduction

Many if not all data collected in forest science have both a spatial as well as a temporal dimension. These characteristics suggest the use of spatial statistics for analysis of forestry data and the quantitative description of spatial forest structure has recently become an important element of modern sustainable forest management (see e.g. Gadow and Hui, 2002; Pommerening, 2002).

The importance of spatial patterns can be seen in production, environmental and social aspects of forestry. For example the availability of growing space determines to a large extent survival, growth and the morphology of trees (see Assmann, 1970; Oliver and Larson, 1996; Pretzsch, 2002, for an overview). Also, woodland habitats are characterised and influenced by the spatial arrangement of trees (Pitkanen, 1997) and spatial woodland structure has an impact on the recreation and amenity value of forests (Pukkala *et al.*, 1995).

In all three examples the properties of the whole system, the forest, and the interaction between its components, the trees, are dependent on the microstructure of the system. There are similarities with other branches of science, for example in materials science, where corresponding relationships are described as *structure/property relationships* and their formulation and prediction is of great importance (Torquato, 2002).

The recognition of the spatial nature of forestry data has given rise to many studies using spatial methods and Jansen *et al.* (2002) stress the enormous diversity of concepts and techniques published in the forestry literature.

For example Karafyllidis and Thanailakis (1997) and Li and Magill (2001) model the spread of forest fires using a technique based on cellular automata. In another study Chen and Gadow (2002) use a spatial variant of simulated annealing for optimising forest harvesting planning. Ecological investigations increasingly focus on the interactions between pattern and process with a wide variety of different methods (see e.g. Reich *et al.*, 2004; Lancaster, 2006; Perry *et al.*, 2006). In many papers point process statistics is being used for the description and modelling of spatial tree patterns (e.g. Penttinen *et al.*, 1992; Moeur,

1993; Pommerening, 2002). The development of tree growth models which quantify intra- and interspecific competition based on the spatial positions of single trees highlight the practical significance of spatial methods (see Hasegauer, 2006, for an overview). Geostatistical regionalisation methods, which make use of the spatial nature of environmental data such as precipitation or soil properties have also been applied in forest science (see for example Jansen *et al.*, 2002).

The common denominator of all these applications is the reference to space and adjacency. The methods and models used, however, can vary widely and many spatial approaches are only an extension of conventional modelling by incorporating spatial constraints.

Since most spatial-statistical methods used so far to describe and model spatial woodland structure are based on points, this Habilitationsschrift mainly concentrates on applications of *point process statistics*. Two other important and related areas of spatial statistics, *geostatistics* and *random set statistics*, and their contributions to forestry applications are briefly characterised.

1.1 Objectives

Applications in forest science relate to all three fields of spatial statistics and the objectives of this Habilitationsschrift are **1.** to give a brief overview of their theories, **2.** to describe point process statistics in some detail and **3.** to show how the approaches to quantifying and modelling spatial woodland structure relate to them; **4.** to identify common themes which provide new opportunities for future research and modelling and can help to assess the usefulness and appropriateness of various summary characteristics; finally, **5.** to provide a synthesis of the author's own published work which is listed in the appendix on page 37f.

2 Main areas of spatial statistics

Spatial statistics is a methodology for describing and modelling spatial uncertainty. More explicitly spatial statistics aim to provide quantitative descriptions of natural variables distributed in space or in time and space. While Cressie (1993) and Stoyan *et al.* (1995) presented the whole spectrum of methods in spatial statistics, this Habilitationsschrift concentrates on the three main fields, *geostatistics*, *point process statistics* and *random set statistics*.

2.1 Geostatistics

Geostatistics is that branch of statistics which studies the variability and prediction of *regionalised* variables (Chilès and Delfiner, 1999; Wackernagel, 2003). Such variables can take values anywhere in the two- or three-dimensional space under study, for example in a forest stand or forest block. Examples are precipitation and altitude. Stem diameter is not a regionalised variable, since it can be defined only at tree locations. Geostatistics has found a wide range of applications in forest science (see for example Jansen *et al.*, 2002), typically in the analysis of covariates of forest point patterns (see section 2.2). Sample data for a regionalised variable $Z(x)$ are gathered at discrete points, y_i , which are chosen by the investigator, often following a systematic grid or lattice.

One of the key applications in geostatistics is *spatial interpolation*. The value $Z(x)$ at a location of interest, x , different from the sampling locations, y_i , can be estimated through a number of interpolation methods which include kriging, splines and weighted moving averages as well as regression techniques (Cressie, 1993; Chilès and Delfiner, 1999; Jansen *et al.*, 2002). Spatial interpolation is an important pre-requisite for producing maps of the whole study area and some of these methods are implemented in Geographic Information Systems (GIS). Kriging is a least squares prediction method using a special second-order summary characteristic (see section 3.3), the *(semi)variogram*, $\gamma(\mathbf{r})$. It is the mean squared difference of the values of the regionalised variable at x and $x + \mathbf{r}$

divided by 2, where $\mathbf{r}$ is a difference vector of length r :

$$\gamma(\mathbf{r}) = \frac{1}{2}E(Z(x) - Z(x + \mathbf{r}))^2 \quad (1)$$

The variogram characterises the spatial variability of a given regionalised variable $Z(x)$ at points x and $x + \mathbf{r}$ (Wackernagel, 2003; Illian *et al.*, 2008). Its values for different $\mathbf{r}$ give information on the behaviour of $Z(x)$ for arbitrary spatial scales.

Two important assumptions in the realm of geostatistics are that **1.** the sample points, y_i , are independent of the regionalised variable and **2.** values measured at points close together in space tend to be similar. Violations against the first assumption can lead to estimation bias.

Three examples, of the many applications in forestry, are given here. In a case study Jansen *et al.* (2002) use geostatistical methods to improve the ecological site description for forest planning in the Harz mountains (Lower Saxony, Germany). Humus content, rooting depth and mean annual air temperature were some of the regionalised variables estimated for the whole mountain range from sample points. In New Zealand Payn *et al.* (1999) apply variogram and kriging methods to monitor changes in forest productivity and nutrition using site index and basal area index as regionalised variables. Finally, Zawadzki *et al.* (2005) review the application of geostatistics for investigations into forest ecosystems using remote sensing imagery. These authors concentrate on three main aspects, **1.** the spatial variability of features identified from digital images, **2.** the estimation of biophysical forest parameters and of variables of forest ecosystem structure and **3.** forest classification methods.

2.2 Point process statistics

Point process methods are applied to the statistical analysis of point patterns aiming to analyse their structural variability. Of particular interest in the analysis are short-range correlations (Diggle, 2003; Illian *et al.*, 2008). The most important example in forestry are the patterns created by the positions of tree

stem centres, but tree tops or the positions of other flora and fauna can also be studied. Often it makes sense to analyse additional characteristics of the objects which are associated with the points. For instance these additional characteristics may describe the size or species of a tree and are usually referred to as marks, m_i (Diggle, 2003; Illian *et al.*, 2008). In many cases the marks are dependent on the points or vice versa. Spatial point processes are mathematical models for point patterns.

In some studies point process and geostatistical methods are combined, for example when regionalised variables control point patterns as so-called covariates, e.g. elevation, precipitation or soil properties (Møller and Waagepetersen, 2007). Covariates are used in such studies to explore their influence on the variability of tree positions.

Publications involving point process statistics in forestry include Matérn (1960), Penttinen *et al.* (1992), Moeur (1993), Gavrikov and Stoyan (1995) and Pommerening (2002), which give various examples using tree positions and tree attributes as points and marks, respectively. Comas and Mateu (2007) give an overview of recent applications of point process methods in forestry including modelling.

2.3 Random set statistics

Random sets serve as mathematical models for irregular geometrical patterns. Simple examples of random sets in forestry are the part X of a landscape covered with forest or the part X of the forest floor occupied by vascular plants and shrubs. In some sense, a random set is simply a special case of a regionalised variable, one that takes only the values 0 or 1. Fundamentals of the theory of random sets are given in (Stoyan *et al.*, 1995).

Random sets can be described by various summary characteristics. Practical examples are characteristics which statistically describe the components or objects forming X , such as mean area or perimeter, which were used for example by Kleinn (2000), who, by the way, never used the term “set”. If such com-

ponents are difficult to define, set-theoretic characteristics show their power. Simple numerical characteristics of a set-theoretic nature are area fraction (i.e. the percentage of area covered by X) and specific boundary length (i.e. the total boundary length of X per unit area).

A more sophisticated characteristic is the *spherical contact distribution function*¹, H_s . It gives the distribution of the distance from an arbitrary test location outside X to the nearest point of X . Diggle (1981) used this function in an ecological context to study the spatial pattern of heather (*Calluna vulgaris* (L.) HULL). The potential of this characteristic for forestry applications is not yet exhausted.

A notable and successful random set model is the Boolean model, where X is the set-theoretic union of small sets randomly scattered in the plane with the possibility that the sets may overlap. For this model formulae are known for all summary characteristics mentioned above. It was applied to the pattern of lichens on a stone in Stoyan *et al.* (1995, p. 86f.).

A typical field of application of set-theoretic methods are those problems which are analysed by *landscape metrics*. Landscape metrics is a collection of summary characteristics for describing the pattern of landscape elements, such as forests and forest patches, which are assumed to be correlated with ecological processes. While these statistics were originally devised to deal with scales of kilometers or more, they can be applied also to plant mosaics at smaller scales, like the pattern of canopy gaps of a forest stand (Dale, 1999; Stoyan *et al.*, 1995).

Another potential field of application of random set statistics includes the analysis of digital images and data obtained from laser scanning techniques (e.g. Moffiet *et al.*, 2005; Riaño *et al.*, 2003; Næsset and Økland, 2002). These do not produce point process data, but data similar to those from high-resolution digital cameras.

¹The spherical contact function is also known as *empty-space function* or *the law of first contact* (Stoyan *et al.*, 1995).

3 Concepts of point process statistics in forestry

The Habilitationsschrift concentrates on two major methods of point process statistics that play an important role in forest science, *nearest neighbour summary statistics* (section 3.2) and *second-order characteristics* (section 3.3). Common themes and edge effects relating to the two methods and other concepts in spatial statistics are addressed in sections 3.1 and 3.4.

3.1 Common themes

There are a number of common themes which apply generally to the descriptive spatial statistics reviewed in sections 3.2 and 3.3 and their identification is an important consideration in this Habilitationsschrift.

3.1.1 Stationarity and isotropy

In the statistical analysis of point patterns it is often assumed that the pattern belongs to an underlying point process which is *homogeneous* (or *stationary*) and *isotropic*, i.e. its probability distribution is invariant to translations and rotations (Diggle, 2003; Illian *et al.*, 2008, p. 42 and p. 35ff., respectively). In contrast, inhomogeneous point processes can, for example, have point densities that vary systematically.

Although methods have also been developed for inhomogeneous point processes (see e.g. Møller and Waagepetersen, 2007), patterns are preferred in the analysis for which the stationarity and isotropy assumptions hold. These markedly simplify the methodology and allow a focused analysis of interactions between trees by ruling out additional factors such as, for example, varying site conditions. The stationary and isotropic case therefore currently forms the main domain of the application of point process methods. In this context the choice of size and location of an observation window² is crucial, because

²An observation window defines the spatial subset or sample included in the analysis. In some text books it is referred to as study area, region or domain.

stationarity and isotropy are scale dependent concepts (Dungan *et al.*, 2002). Both concepts are also often used in the other two fields of spatial statistics.

3.1.2 Test location and point related summary characteristics

Two different types of summary characteristics can be defined from the type of points from which they are measured: *test location* or *point related* summary characteristics³ (Illian *et al.*, 2008, p. 195).

Test location related summary characteristics like the spherical contact distribution function mentioned in section 2.3 are estimated with reference to deterministic test locations or sample points, which are chosen independently of the points of the point process. They describe the relationship between test locations and the points of the spatial pattern under study. Test locations can be based on a systematic grid or randomly placed throughout the observation window and are used particularly in geostatistics and random set statistics. In geostatistics the sample points, y_i , where the sample data were gathered, are the test locations.

In contrast, point related summary characteristics relate only to the process points, i.e. they yield information on the i^{th} point in the observation window. The majority of summary characteristics in point process statistics are point related.

These two concepts of summary statistics were also discussed in relation to structural indices by Gadow and Hui (2002) and Staupendahl and Zucchini (2006). These authors used the terms *point-based* and *tree-based* which originate from *distance methods* used in forest inventory in which Loetsch *et al.* (1973, p. 369f.) distinguish between point-tree and tree-tree sampling methods. Point-tree (= test location related) sampling methods are more common in forest inventory since tree-tree methods (= point related) can lead to biased estimations (Diggle, 2003; Krebs, 1999, p. 34 and p. 170, respectively).

³Diggle (2003, p. 13ff.) uses a slightly different terminology and distinguishes between *point-to-nearest-event* and *inter-event* distances.

3.1.3 Defining local neighbourhood

Quantifying microstructural information from a localised environment of points is another common theme in spatial statistics, which finds applications in all three areas of spatial statistics (see for example Chilès and Delfiner, 1999; Wackernagel, 2003; Illian *et al.*, 2008). The estimation of measures of competition (see section 3.2.1) and structure (see section 3.2.2) requires the selection of a set of neighbouring trees before the principles of constructing marks of section 3.1.4 can be applied.

Local neighbourhood can be defined in many ways and the most common of these are given below.

(1) Distances. Most frequently measures of distance (Euclidean, Euclidean squared, Euclidean weighted, etc.) are used, i.e. neighbouring trees are defined by their distances to the reference tree or point. Sometimes these distances are weighted by the original marks (e.g. diameter, height) of the trees involved (see for example the Hegyi competition index in Table 1).

(2) Basal area factors. Basal area factors (BAF) have been identified as good criteria for selecting nearest neighbours in conjunction with measures of competition (Pretzsch, 2002). This method is based on Bitterlich's angle count or relascope method according to which the probability that a tree is selected as a neighbour is related to its basal area (Bitterlich, 1984). Formula 2 formalises the principle of this neighbour selection method: Neighbouring trees j (represented by P_j) with a distance to the reference tree i (represented by P_i) shorter than a critical distance are included in the local neighbourhood of reference tree i , otherwise they are rejected.

$$\|P_i - P_j\| < \frac{50}{\sqrt{BAF}} \cdot DBH_j \quad (2)$$

(3) Dirichlet cells. In Dirichlet tessellations or two-dimensional Voronoi tessellations (Stoyan *et al.*, 1995; Dale, 1999) each tree holds a territory consisting of the

part of the observation window which is closer to that particular tree location than to any other tree location. This method recruits neighbours whose Dirichlet domains share a common boundary with the reference tree and are therefore likely to have some kind of interaction with the tree under study. Spatial proximity is less important in this approach and direction is given more emphasis. The method ensures that neighbours are selected from all possible directions around the reference tree.

(4) Influence zones. According to this method a two- or three-dimensional influence zone around the reference tree is defined and every other tree which directly intersects this influence zone or whose own influence zone overlaps it is considered to be a neighbour (Bella, 1971; Biging and Dobbertin, 1992, 1995; Berger and Hildenbrandt, 2000). Usually influence zones are defined as more or less simple geometric structures, e.g. circles, cylinders, inverted cones and can have their size fixed or varied depending on an attribute of the reference tree, e.g. the breast height diameter.

With measures of competition (see section 3.2.1) research has shown that the inverted cone and the horizontal basal area factor methods are successful methods of defining nearest neighbours (see Pretzsch, 2002). The number of neighbours selected randomly varies from reference tree to reference tree depending on the neighbourhood selection method and the point pattern.

In contrast, with measures of structure (see section 3.2.2) the number of neighbours is usually fixed for all trees and are determined arbitrarily. Little research has been done to identify an optimal number of nearest neighbours, see for example Pommerening (2006).

Different types of qualitative marks, e.g. the species attributes or health classes, add to the complexity of defining local neighbourhoods. In for example a mixed oak-beech woodland it is useful to estimate summary characteristics separately for each of the two species groups and for the interaction of trees with different species marks. Statistically this kind of point pattern data may be regarded as a *superposition* or union of single-type point patterns (Stoyan *et*

al., 1995; Diggle, 2003; Illian *et al.*, 2008), where two or more component processes are superimposed, and some authors refer to the corresponding analysis of such patterns as *intra*- and *intertype* analysis (Lotwick and Silverman, 1982; Diggle, 2003; Moeur, 1993; Goreaud and Pélissier, 2003). This is analogous to intra- and interspecific competition as defined in forest ecology and has important implications in sampling and analysis of structural indices (section 3.2.2) and second-order summary statistics (section 3.3). Neighbours of points belonging to one type should either always be recruited from the same type set (intratype analysis: e.g. reference point of type oak - neighbour of type oak, reference point of type beech - neighbour of type beech) or from a specified other type to investigate for example the interaction of oak type and beech type points (intertype analysis). With s different qualitative marks, e.g. tree species, there are $\binom{s+1}{2}$ combinations of defining nearest neighbours. Two tree species would lead to $\binom{3}{2} = 3$ and three tree species to $\binom{4}{2} = 6$ combinations of defining nearest neighbours.

3.1.4 Principles for constructing marks

In point process statistics marks, m_i , are attributes of points and the analysis of marked point patterns can provide greater insight into the processes underlying the pattern than an analysis of the points alone (Illian *et al.*, 2008). Marks can be *qualitative*, e.g. tree species, or *quantitative*, e.g. diameter at breast height. It is very important to realise that not only *original* marks may be used but that *constructed* marks play an important role in point process statistics. Original or natural marks are inherent attributes of the point and are usually measured in situ, e.g. tree diameters and tree heights. Constructed marks involve additional information relating to other points such as the nearest neighbours. All traditional spatial structural and competition indices developed in forestry can be interpreted as constructed marks. For these marks it is possible to identify four general construction principles, which are given in Table 1 with examples.

Table 1. Construction principles for marks m_i used for estimating nearest neighbour summary characteristics. i denotes the reference tree and j a NN (nearest neighbour). The principles are explained in the text.

Construction principle	Example(s)
Mark ratio	$m_i = \sum_{j=1}^n \frac{DBH_j}{DBH_i} \cdot \frac{1}{dist_{ij}}$ Hegyi (1974) $m_i = 1 - \frac{1}{n} \sum_{j=1}^n \frac{\min(DBH_i, DBH_j)}{\max(DBH_i, DBH_j)}$ Füldner (1995)
Mark comparison	$m_i = \frac{1}{n} \sum_{j=1}^n 1(DBH_i > DBH_j)$ Hui <i>et al.</i> (1998) $m_i = \frac{1}{n} \sum_{j=1}^n 1(species_i \neq species_j)$ Füldner (1995)
Mark difference	$m_i = \frac{1}{2}(DBH_i - DBH_j)^2$ Stoyan and Stoyan (1994)
Mark product	$m_i = \frac{DBH_i \sum_{j=1}^n DBH_j}{n \overline{DBH}^2}$ Davies and Pommerening (2008)

where

m_i	mark of the reference tree i
n	number of NN
DBH_i	breast height diameter of the reference tree i
DBH_j	breast height diameter of the NN, $j = 1, \dots, n$
$dist_{ij}$	Euclidean distance between the reference tree i and neighbour j
$species_i$	species of the reference tree i
$species_j$	species of the NN, $j = 1, \dots, n$
$\overline{DBH}$	arithmetic mean breast height diameter of all trees in the observation window
$1(A) = 1$	if A is true, otherwise $1(A) = 0$

Representatives of the *mark ratio* principle are the Hegyi competition index (He-

gyi, 1974) and the diameter differentiation index (Füldner, 1995) in which the marks of the reference point and neighbouring points are divided and summed. The ratios are sometimes multiplied by a weight such as the reciprocal distance in the case of the Hegyi index.

The *mark comparison* principle is used in the diameter dominance (Hui *et al.*, 1998) and the species mingling index (Füldner, 1995). This principle involves a qualitative comparison of marks and the result is of a boolean type, 0 or 1. The diameter dominance index is defined as the proportion of the n nearest neighbours that are smaller than reference tree i . The species mingling index is defined as the proportion of the n nearest neighbours that are different species to the reference tree i .

The principle of *mark difference* is demonstrated by the diameter variogram index (Stoyan and Stoyan, 1994) in which half the squared difference of the marks of neighbouring trees is related to the test function used in mark variograms. These are related to geostatistical variograms (see section 2.1) and the diameter variogram index characterises the variability of the diameters of trees at close proximity.

The diameter correlation index is an example of the *mark product* principle. Here the mean product of the diameters of the reference tree and the n nearest neighbours is compared with the squared arithmetic mean diameter of all trees in an observation window. The squared arithmetic mean diameter reflects stochastic independence.

These mark construction principles are not limited to NNSS like the examples in Table 1 but also find an application in test functions used with second-order summary characteristics for marked point processes (see section 3.3).

3.1.5 Functional summary statistics

Functional summary characteristics are preferred in modern point process statistics rather than indices, because the former provide more information on a tree population than simple indices which are usually arithmetic means, ratios or

sums (Stoyan and Penttinen, 2000). Such provision was also an important consideration in the development of summary characteristics for quantifying spatial woodland structure by the Göttingen research group (see Pommerening, 2006, and section 3.2.2) and is a fundamental element of second-order characteristics (see section 3.3), where they are, for example, functions of distance r .

For estimating functional summary statistics so-called *kernel estimators* are usually preferred. These are non-parametric methods which are not based on model assumptions (Diggle, 2003; Illian *et al.*, 2008). Silverman (1986) and Illian *et al.* (2008, p. 481f.) give useful details on the principles of kernel estimation methods. For nearest neighbour summary statistics (see section 3.2) the *Epanechnikov kernel* is recommended while the *box kernel* is a very suitable choice for second-order functions (see section 3.3).

3.2 Nearest neighbour summary statistics

Nearest neighbour summary statistics (NNSS) describe relationships between a test location or a point of the point pattern and the n nearest neighbours and statistically NNSS can be interpreted as means and density functions of constructed marks. They are often called short-sighted methods because they quantify only the relationship between a point and its n nearest neighbours and ignore what is beyond the nearest neighbours (Pommerening, 2002). Also, the distances between nearest neighbours depend on the point pattern and can therefore vary considerably from reference tree to reference tree.

In forestry and ecology these summary characteristics include *competition indices* (e.g. Biging and Dobbertin, 1992, 1995, see section 3.2.1) and *structural or diversity indices* (e.g. Krebs, 1999; Gadow and Hui, 2002; Pommerening, 2002, see section 3.2.2). The two types of indices share a common estimation process, firstly the definition of the local neighbourhood (section 3.1.3), i.e. the nearest neighbours, and secondly the construction of the mark (section 3.1.4).

3.2.1 Measures of competition

Measures of competition or competition indices⁴ have a long tradition in forest and plant sciences and the literature on this subject is extensive (see for example Bella, 1971; Biging and Dobbertin, 1992, 1995; Moravie *et al.*, 1999; Berger and Hildenbrandt, 2000; Barot and Gignoux, 2003). After Begon *et al.* (1996), competition is defined as an interaction between individuals, brought about by a shared requirement for a resource in limited supply, and leading to a reduction of the performance (e.g. survival, growth, reproduction) of at least some of the competing individuals. This interaction effect is particularly strong among neighbouring trees, but is not necessarily always antagonistic as implied by the term competition and can also involve positive aspects of coexistence (Gadow and Hui, 1999, p. 144).

Measures of competition are key elements of forest growth models used to estimate the *response variable*⁵ growth increment, e.g. diameter or basal area increment, and quantify the *competition load* or *competition intensity* on individual trees (Weigelt and Jolliffe, 2003). Competition indices are also an important instrument for generalising growth models so that growth functions are applicable across a wide range of stand densities and species compositions in a larger geographic area.

Measures of competition can be used in growth estimations in two different ways, **1.** in *direct estimations* and **2.** as part of the so-called *potential-modifier approach* (see Table 2) (Hasenauer, 2006, p. 6f.).

In both concepts measures of competition are used as independent variables (regressors, explanatory variable) along with a number of other variables (indicated by Var_n in Table 2 and the plural in modifier functions) such as for example site factors. Additionally in some models the change in competition index over time is included as a separate independent variable. In the modifier

⁴Non-spatial measures of competition are not considered here.

⁵A response variable can be best defined as the property of a system in a structure/property relationship as described in the introduction. Response variables correspond with *dependent variable* and *regressand* in regression terminology.

Table 2. The two basic concepts of using measures of competition in growth estimations. $a_0, \dots, a_n$ are regression coefficients, Var_n indicates other possible independent variables.

$$1. \text{ Expected growth} = a_0 + a_1 \cdot \text{Competition index} + \dots + a_n \cdot Var_n$$

$$2. \text{ Expected growth} = \text{Potential growth} \cdot \text{Modifier function}(s)$$

function the competition index is usually incorporated in the exponent of an exponential function and the potential is derived from trees that have faced no or hardly any competition, e.g. open-grown trees or predominant forest trees.

The components of competition indices, i.e. the definition of the local neighbourhood and the construction of the mark, are usually optimised through correlation and regression analyses. The optimisation is facilitated by a clearly defined response variables (see footnote 5) which in forestry most frequently is tree growth but Weigelt and Jolliffe (2003) also refer to other response variables in the general plant and vegetation science, e.g. reproduction and physiological performance.

With measures of competition in forest science there is usually little interest in summary characteristics, e.g. a mean competition index at forest stand level or a competition density distribution. Weigelt and Jolliffe (2003), however, also report the use of competition summary characteristics in the general plant and vegetation science literature. Despite obvious similarities to measures of structure, the main focus of measures of competition is on the individual marks, i.e. the index value of each tree in the observation window.

3.2.2 Measures of structure

Spatial measures of structure⁶ have been developed for mainly two purposes, 1. for the quantitative description of the spatial arrangement of trees and their

⁶Non-spatial measures of structure are not considered here.

attributes and 2. for providing spatial measures of biodiversity. There are many papers with overviews and example applications (see Neumann and Starlinger, 2001; Gadow and Hui, 2002; Pommerening, 2002; McElhinny *et al.*, 2005; Pommerening, 2006).

For analysing the variability of spatial aspects of plant populations the first summary characteristics were proposed in the 1950s and 1960s, e.g. the aggregation index of Clark and Evans (1954) and Pielou's coefficient of segregation (Pielou, 1977). These and similar early approaches reflected the state of the computer and mensuration technology of the time and expressed plant structure using a single number. Krebs (1999) and Dale (1999) give a good overview. Later the advantages of estimating histograms and functional summary characteristics (see section 3.1.5) were recognised and these provided more detailed information about stand structure (Gadow and Hui, 2002; Pommerening, 2006).

More recently a good quantitative description of spatial forest structure has also been recognised as the key to the ability of simulating it (Lewandowski and Gadow, 1997; Pommerening, 2006; Pommerening and Stoyan, 2008).

Since the sustainability concept has been extended beyond the traditional concept of simple timber sustainability (Pommerening and Murphy, 2004) woodland managers are increasingly required to consider and improve biodiversity, conservation and habitat functions within the forest. There is an urgent need for practical, meaningful and quantitative criteria and measures of these functions. Because of their correlation with habitat functions and population dynamics of endangered mammal and bird species, structural indices have been proposed as surrogates for more direct measures of biodiversity (Pommerening, 2002). To this end Gadow (1999) and Pommerening (2002) subdivide α -diversity into the *diversity of tree positions*, *tree species diversity* and the *diversity of tree dimensions*. At a small scale, the diversity of tree positions reflects the pattern of tree locations: are they regular⁷, clumped (clustered), random or some combination of these? Tree species diversity is concerned with the spatial arrangement of species while diversity of tree dimensions involves the spatial arrangement of, for example,

⁷Dale (1999) refers to regular patterns as over-dispersed.

diameters or heights (Pommerening, 2006). Structural indices are particularly effective in forest inventories with many small replicated observation windows (Pommerening and Stoyan, 2008).

Defining a response variable (see footnote 5 on page 15) is not as straightforward as with measures of competition. Possibilities include the abundance of an endangered animal species, tree (population) growth and the degree to which allometric models can be improved. Especially in the first case, data are often difficult to gather. The difficulties in data acquisition and the absence of clearly defined response variables explain why there is hardly any study concerned with this problem.

Pretzsch (1995) for example investigates the relationship between the diversity of tree positions and stand basal area increment through simulation. Using the aggregation index of Clark and Evans (1954) he could identify that the maximum stand basal area increment was achieved with regularly distributed tree positions, that 95% of the maximum increment was realised when the same trees were randomly distributed and a minimum increment of 50% of the maximum increment in the case of strongly clustered tree positions. Bravo and Guerra (2002) found relationships between stand and tree based structural indices and diameter growth for maritime pine (*Pinus pinaster* AIT.) in Spain and Davies and Pommerening (2008) have recently used measures of spatial structure for improving allometric models in Wales. The authors could show that spatial indices can significantly improve crown models for Sitka spruce (*Picea sitchensis* (BONG.) CARR.) and birch (*Betula* spp.) especially in forests with poor thinning history.

In materials science *reconstruction* (Torquato, 2002, p. 294, see section 4.2) is used as a way of evaluating summary statistics: The extent to which the observed realisation of forest structure can be reconstructed with a given summary characteristic reveals the level of information embodied in it. Pommerening (2006) has independently developed the same idea when evaluating structural indices by reversing forest structural analysis.

In contrast to measures of competition, with measures of structure there

is usually little interest in individual marks but rather in summary characteristics, e.g. a mean structural index at forest stand level or a probability density distribution.

3.3 Second-order summary characteristics

Second-order summary characteristics were developed within the theoretical framework of mathematical statistics and then applied in various fields of natural sciences including forestry (Illian *et al.*, 2008; Møller and Waagepetersen, 2007). They describe the variability and correlations in marked and non-marked point processes.

Functional second-order characteristics depend on a distance variable r and quantify correlations between all pairs of points with a distance of approximately r between them. This allows them to be related to various ecological scales and also, to a certain degree, to account for long-range point interactions (Pommerening, 2002). Several examples are given in this section but more can be found in the literature provided.

The most popular second-order characteristic is Ripley's K -function (Ripley, 1977). $\lambda K(r)$ denotes the mean number of points in a disc of radius r centred at every point i (which is not counted) of the point pattern where λ is the intensity. This function is a cumulative function. Besag (1977) suggested transforming the K -function by dividing it by π and by taking the square root of the quotient, which yields the L -function with both statistical and graphical advantages over the K -function. The L -function is often used for testing the CSR⁸ hypothesis, e.g. in Ripley's L -test (Stoyan and Stoyan, 1994).

Cumulative functions are not always easy to interpret and for detailed structural analysis functions of the nature of derivations are preferred. One of these is the pair correlation function $g(r)$, which is related to the first derivative of the K -function according to the interpoint distance r (formula 3) (Illian *et al.*, 2008),

⁸Complete spatial randomness.

$$g(r) = \frac{K'(r)}{2\pi r}. \quad (3)$$

For a heuristic definition consider $P(r)$ as the probability of one point of the point process being at r and another one at the origin o . Let dF denote the area of infinitesimally small circles around o and r and λdF the probability that there is a point of the point process in a circle of area dF . Then

$$P(r) = \lambda dF \lambda dF g(r).$$

In this approach, the pair correlation function $g(r)$ acts as a correction factor. For Poisson processes and for large distances when the point distributions are stochastically independent $g(r) = 1$. In the case of cluster processes $g(r) > 1$ (indicating attraction of points) and for regular patterns $g(r) < 1$ (indicating inhibition between points).

The partial pair correlation function $g_{ij}(r)$, which by analogy to $K_{ij}(r)$ (Lotwick and Silverman, 1982) can also be referred to as the intertype pair correlation function, is a tool for investigating multivariate point patterns (Penttinen *et al.*, 1992; Stoyan and Penttinen, 2000; Illian *et al.*, 2008). Its interpretation is similar to that of $g(r)$. Namely, consider $P_{ij}(r)$ as the probability of finding a point of type i (e.g. i = species 1) in an infinitesimally small circle of area dF and a point of type j (e.g. j = species 2) in another one of area dF , where the distance between the two circle centres is r . Then

$$P_{ij}(r) = \lambda_i dF \lambda_j dF g_{ij}(r).$$

λ_i and λ_j is the intensity of the points of type i and of type j , respectively.

In the case of quantitative marks other diagnostic tools are available to assess the similarity or dissimilarity of marks. The mark correlation function (Penttinen *et al.*, 1992; Stoyan and Penttinen, 2000; Pommerening, 2002; Illian *et al.*, 2008) uses a *test function*, f , which depends on two marks m_i and m_j , with m_i at the origin o and m_j at r . The test function of the mark correlation function is $f(m_i, m_j) = m_i \cdot m_j$, which is related to the mark product construction principle of Table 1. The concept of the mark correlation index given as an example in

section 3.1.4 is based on the idea of the test function of the mark correlation function.

3.4 Edge effects

Edge effects arise in spatial analysis when the observation window W , in which the pattern is observed, is part of a larger region. The essential problem is that unobserved objects outside W may interact with observed objects within W , but cannot be included in the estimation of important summary characteristics (Diggle, 2003). Ignoring these effects is understood to result in biased statistical estimations.

In point process statistics the precise mathematical formulation of methods to address edge-bias are dependent on the summary characteristic to be estimated and there are different edge-correction methods for nearest neighbour summary statistics and second-order characteristics. Edge-bias issues of competition indices and second-order characteristics have been frequently discussed in the statistical literature (Monserud and Ek, 1974; Martin *et al.*, 1977; Radtke and Burkhart, 1998; Stoyan *et al.*, 1995; Illian *et al.*, 2008). Comparatively little attention has been paid to structural indices, but Pommerening and Stoyan (2006) have addressed this gap in the literature.

There are two main categories of edge-correction: *plus-sampling* and *minus-sampling* (Stoyan *et al.*, 1995; Illian *et al.*, 2008, p. 133 and p. 183ff., respectively). Plus-sampling makes full use of all data within the observation window either by recording objects outside the observation window or by simulating them, if needed, in the summary characteristic estimators. Minus-sampling (also referred to as border method) only makes use of an inner sub-set of the observation window in the summary statistic estimators, i.e. those objects that have no interaction with objects outside the observation window. Table 3 summarises general edge-correction methods according to the two main categories.

Table 3. Common general plus- and minus-sampling methods. Explanations are given in the text.

Plus-Sampling	Minus-Sampling
Translation (= periodic boundary conditions, toroidal wrapping)	Internal buffer
Reflection	– NN1
External buffer	– NN2
– Additional measurements	
– Conditional simulation	

Plus-sampling can be realised by translation and reflection, which are traditional methods used in many spatial approaches. Both methods extrapolate the spatial structure from within W to an infinite plane and join parts of the spatial pattern that do not occur so close together in nature. This can lead to unrealistic spatial patterns near the boundaries of the observation window. Reflection usually performs particularly badly and therefore cannot be recommended (see Pommerening and Stoyan, 2006, for a detailed description and discussion).

The external buffer method can include the measurement of additional objects outside the observation window, which might interact with those inside. Determining the optimal width of the buffer is difficult; if it is too small residual edge effects will remain; if it is too large unnecessary sampling effort is applied. However, with nearest neighbour summary statistics in forest inventories, it is often sufficient to sample the nearest off-plot neighbours of trees near the plot boundaries.

An alternative to measuring additional objects and another variant of the external buffer method is conditional simulation outside W . This method involves a structural analysis of the spatial pattern inside W and a simulation of this pattern outside W based on the results of the structural analysis. The reconstruction method mentioned in section 4.2 can be used for this purpose. “Conditional” means that the simulation does not modify the spatial pattern within the observation window but only generates new patterns outside W which are

statistically similar to those inside and have the statistically correct ties to the inner objects (see Biber, 1999; Illian *et al.*, 2008).

The internal buffer method is a minus-sampling method. Like all minus-sampling methods the buffer method only uses a subset of the objects in the observation window W . Again the choice of the buffer width is difficult (see Diggle, 2003, p. 5) and refined versions have been developed which create a flexible buffer by addressing each object's potential relation to off-plot objects individually. These methods are known as *nearest-neighbour edge correction methods* and have been proposed and investigated as NN1 and NN2 methods in Pommerening and Stoyan (2006), where a detailed description can be found. The NN1 method was identified as a very reliable and robust option. Minus-sampling methods are not suitable for measures of competition (see section 3.2.1), because an unbiased mark must be constructed for every tree position in the observation window and not only for a subset. Numerical edge-correction methods like linear expansion (Martin *et al.*, 1977) and plus-sampling offer alternatives.

4 Spatial modelling approaches

Modelling in spatial statistics can serve two purposes, **1.** to confirm and further explore the data analysis and **2.** for simulation. In the first case models aid in the interpretation and understanding of empirical patterns and their summary characteristics by showing theoretical forms and information on statistical variation. The process of modelling often reveals further information that leads to a better understanding of the evolution of certain structures. Simulation of spatial patterns is based on models and can serve the following purposes (Illian *et al.*, 2008, p. 52f.) which are briefly explained with examples in the paragraphs below:

- Calculation of summary characteristics of more complicated spatial models for which explicit analytical formulae are unavailable.
- Statistical testing.
- Investigation of the performance of statistical methods.
- Visualisation of spatial models.
- Reconstruction, see section 4.2.

For example Pommerening and Stoyan (2006) used simulation to estimate summary characteristics of newly developed, complicated point process models.

In order to test the performance of a new reconstruction method Pommerening and Stoyan (2008) constructed confidence intervals for second-order characteristics from 1,000 simulations. A similar way of testing through simulation was employed by Degenhardt and Pofahl (2000).

When new summary statistics are developed modelling can help to evaluate them by answering the question of how much they contribute to a synthesis of forest structure (Pommerening, 2006). In this example simulation is used as a way of investigating the performance of statistical methods.

Spatially explicit growth models, sampling simulators and landscape visualisation tools require spatial data for their initialisation (Pommerening, 2000). Despite advances in surveying technology spatial data are still expensive to gather. In Pommerening and Stoyan (2006) examples are given for the visualisation of newly developed point processes which were obtained by simulation.

Reconstruction of spatial woodland structure (see section 4.2) offers the opportunity to re-establish patterns within large observation windows where no samples have been taken, for example outside smaller subwindows, such as those used in forest surveys (Pommerening and Stoyan, 2008). In the same way reconstruction can also be used as a plus-sampling edge-correction method (see section 3.4). Such methods can extrapolate potential tree locations and other attributes for missing trees outside the observation window through simulation.

Approaches to modelling spatial structure can be subdivided into *parametric* and *non-parametric* methods (Pommerening and Stoyan, 2008).

4.1 Parametric modelling methods

A number of point process models with only few parameters have been developed for modelling spatial patterns. They are based on model assumptions with local or regional parameters. Two main classes of point processes can be distinguished, Cox process models (see Stoyan and Penttinen, 2000) and Gibbs process models (e.g. Tomppo, 1986; Degenhardt and Pofahl, 2000). Comas and Mateu (2007) provide an overview.

Pretzsch (1997) for example developed an approach which is based on an empirical function, estimating the distance to the nearest neighbour, and a set of probability functions (a combination of an inhomogeneous Poisson process and a hard-core process) to model the spatial pattern of mixed beech - larch forests in Lower Saxony (Germany). In a case study in tropical forests Batista and Maguire (1998) model the distribution of regeneration trees in former clear cut areas with a homogeneous Neyman-Scott process and the spatial pattern of ingrowth trees under canopy with an inhomogeneous Poisson process. Mateu *et al.* (1998) apply marked Gibbs point process models to two Mediterranean forest types. In another study Stoyan and Stoyan (1998) combine an inhomogeneous Poisson process and a Gibbs process to model a hickory woodland in North Carolina. Degenhardt (1999) models the time series of a mixed lime-ash research plot in North-East Germany by a series of Gibbs processes with time dependent parameters. Finally, Shimatani and Kubota (2004) developed a new non-stationary approach, which is a combination of an inhomogeneous Poisson process and a Neyman-Scott process, and applied it to an *Abies sachalinensis* (FR. SCHMIDT) MAST. population in the boreal region of northern Japan.

An interesting combination of methods from point process and random set statistics was published by Penttinen and Niemi (2007). The authors developed a new random set marked Cox process, as part of which the establishment of

tree seedlings in patches was modelled using random set statistics. This idea is based on the observation that tree seedlings often occur in dense clusters or patches which can be the result of better light conditions in gaps or of randomly exposed mineral soil by harvesting machinery. The forest area is accordingly subdivided into two phases, a phase of high seedling density, where the canopy gaps or patches of exposed mineral soil are located, and another phase with much lower seedling density outside the random set. In both phases a Poisson process is simulated with two different intensities.

Apart from the fact that the underlying mathematical methodology can be complicated in parametric approaches, especially with increasing numbers of tree species, point process models are limited to the special geographic cases from which the parameters have been derived.

4.2 Non-parametric modelling methods

Other authors have tried to decrease the role of model assumptions and empirical parameters: Lewandowski and Gadow (1997) constructed a three-phase model approach using nearest neighbour summary statistics (NNSS). Elaborating on Pretzsch (1997), Pommerening (2000) generalised the empirical functions by deducing spatial information from forest survey data. Pommerening (2006) presented an approach using NNSS and cellular automata. The model was used to appraise the performance of NNSS based on the idea that meaningful summary characteristics should allow the analysis to be reversed and enable the simulation of spatial patterns from estimated summary statistics (see also section 3.2.2). Tscheschel and Stoyan (2006) applied a non-parametric reconstruction method to point processes that had previously only been used in physics and materials science.

Pommerening and Stoyan (2008) published a new method of reconstructing spatial tree point patterns from NNSS sampled in small circular subwindows, which uses a stochastic optimisation technique based on a variant of simulated annealing (Kirkpatrick *et al.*, 1983; Torquato, 2002; Chen and Gadow,

2002; Tscheschel and Stoyan, 2006) and conditional simulation (see page 22). The simulated annealing variant used was the so-called *improvements-only* or *relaxation algorithm*, which has proved to be efficient and faster than the original simulated annealing algorithm though the latter has better convergence properties (Tscheschel and Stoyan, 2006).

An interesting by-product of this study is a method for estimating Ripley's K -function and other second-order characteristics from standard forest inventories.

5 Conclusions and outlook

Many different spatial approaches have been developed in forest science and most of them can be related to three areas of spatial statistics, i.e. geostatistics, point process statistics and random sets. Their theoretical concepts provide useful terminology and a framework for future research into the spatial nature of forestry data. In particular the similarities and common themes (see sections 3.1.1-3.1.5) of the different areas of spatial statistics provide direction for further investigations.

For example the construction of marks (section 3.1.4) is such a common theme. Concepts originally developed for second-order characteristics can be applied to spatial indices (like the diameter correlation index in Table 1) and vice versa. In the same way some concepts explicitly developed for quantifying competition pressure on single trees can be used to construct summary characteristics for the spatial structure of woodlands. Elements of these different concepts are interchangeable which potentially opens the way to a whole series of new developments. Other common themes include definition of local neighbourhood (section 3.1.3) and the use and estimation of functional summary characteristics (section 3.1.5).

It can be anticipated that competition and structural indices will remain important concepts for spatial growth projection and for the quantification of

spatial forest structure. However, recently more sophisticated functional concepts have been developed such as competition or neighbourhood fields (Adler, 1996; Berger and Hildenbrandt, 2000; Bauer *et al.*, 2004) and second-order summary characteristics (Møller and Waagepetersen, 2007, and section 3.3). These approaches are likely to gain more importance in the future.

The study by Pommerening and Stoyan (2008) confirmed the value and the power of NNSS in situations where survey data from small observation windows have to be used and second-order statistics cannot develop their full potential. Second-order characteristics are more powerful in analysing spatial patterns if mapped data in larger observation windows are available. To sample such data in forests still requires a considerable effort. It is, however, conceivable that the constantly improving surveying and computing technologies will soon reduce this effort to a minimum.

In many spatial forest growth models only a comparatively small part is really spatially explicit, i.e. the estimation of growth increment by means of competition indices as described in section 3.2.1. Considering the encouraging results in Davies and Pommerening (2008), mixed modelling techniques using spatial autocorrelation and spatio-temporal growth processes (Zhang *et al.*, 2005; Zhang and Gove, 2005) could potentially improve regional growth projections and lead to a whole generation of new growth models.

Mason *et al.* (2007) investigated the structure of semi-natural pinewoods in Scotland. One objective of the study was to explore whether the comparison between patterns reflecting different development stages and management regimes using structural indices and second-order characteristics could help to derive management guidelines for the restoration of native woodland communities. The description of the impacts of human management on woodland structure by means of structural indices has also been an important topic of the Göttingen research group (see e.g. Daume *et al.*, 1997). Mason *et al.* (2007) revealed valuable information in the pattern analysis but concluded that the development of detailed management guidelines from spatial pattern analysis requires more research. There are still too few long-term time series studies

of woodland development with detailed spatial records suited to this type of research.

Most studies in forestry limit the application of second-order summary statistics to tree positions. However, Gavrikov and Stoyan (1995) for example show that an analysis using tree top positions instead of the usual stem centre positions in spatial time-series data revealed more about the competition process in an even-aged Scots pine forest. The authors attributed this finding to the fact that tree trunks are naturally rather conservative and changes in stem centre positions are only possible through mortality and regeneration processes. In contrast, tree crowns can react to changes in local growth conditions and therefore potentially reveal more information on competition processes in forest ecosystems. This example suggests that there is scope for reviewing the definition of tree positions in order to better identify interaction processes.

Random set statistics (see section 2.3) are an interesting field of spatial statistics which could be used more systematically to analyse patterns in forestry such as the distribution of canopy gaps in a forest, the spatial shrub species distribution in hedgerows, forest fragmentation and the distribution of woody plants in coppice woodlands to give only a few examples.

Finally, a method widely used in geostatistics for structural analysis, the geostatistical variogram (Cressie, 1993; Chilès and Delfiner, 1999), is often applied to point patterns as the empirical mark variogram and is another example of how interchangeable methods from different fields of spatial statistics can be. But sometimes the empirical mark variogram is interpreted like a geostatistical variogram, which is incorrect (see Wälder and Stoyan, 1996; Stoyan and Wälder, 2000; Schlather *et al.*, 2004). Further research into the use and limitations of empirical mark variograms would help to make this powerful statistical tool more effective in forest research.

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Appendix: Included publications I-VI

Six publications by the author have been included in the habilitation submission and are summarised in the table on the following page. The last column indicates to which sections of the Rahmenschrift (p. 1 - 36) the papers relate. Exact bibliographic details are given in the references on page 30ff. Reprints of these papers can be found on the following pages with kind permission of the three scientific journals.

In the two cases where Pommerening published with Stoyan (publications III and V), Pommerening was the first and corresponding author. As such he developed the concept for the study ideas and the experimental design. Pommerening was also responsible for implementing the experimental design in a higher programming language, for conducting the computer experiments and the statistical analysis and for the interpretation of the results. Pommerening also wrote the majority of the text in both cases. Stoyan provided statistical advice and contributed to the text.

In the work published with Mason, Connolly and Edwards (publication IV), Pommerening led the statistical analysis and aided the interpretation of the results. Connolly did general data processing but was advised and guided by Pommerening in this process. Mason was concerned with the editorial work on this paper and Edwards contributed local site knowledge. Pommerening also provided the analysis software CRANCOD which has been developed by him since 2004 and significantly contributed to the text.

The idea for publication VI followed on from provisional calculations included in a PhD thesis by Davies, that was supervised by Pommerening. The conceptual framework and the experimental design were developed by Pommerening; Davies was responsible for data collection and processing in close consultation with Pommerening. The authors jointly developed the text of the manuscript.

No	Reference	Title	Journal	Rahmenschrift
I	Pommerening (2002)	Approaches to quantifying forest structures	Forestry	2.2, 3.2.2, 3.3
II	Pommerening (2006)	Evaluating structural indices by reversing forest structural analysis	For. Ecol. Manage.	2.2, 3.1.3, 3.2.2, 3.4, 4.2
III	Pommerening and Stoyan (2006)	Edge-correction needs in estimating indices of spatial forest structure	Can. J. For. Res.	2.2, 3.2.2, 3.4
IV	Mason <i>et al.</i> (2007)	Spatial structure of semi-natural and plantation stands of Scots pine (<i>Pinus sylvestris</i> L.) in northern Scotland	Forestry	2.2, 3.2.2, 3.3, 3.4
V	Pommerening and Stoyan (2008)	Reconstructing spatial tree point patterns from nearest neighbour summary statistics measured in small subwindows	Can. J. For. Res.	2.2, 3.2.2, 3.3, 3.4, 4.2
VI	Davies and Pommerening (2008)	The contribution of structural indices to the modelling of Sitka spruce (<i>Picea sitchensis</i>) and birch (<i>Betula</i> spp.) crowns	For. Ecol. Manage.	2.2, 3.1.3, 3.1.4, 3.2.1, 3.2.2, 4.1

I

Approaches to Quantifying Forest Structures

Approaches to quantifying forest structures

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Summary

For some time, structure indices – quantifying spatial stand structure – have been integrated into forest research and are used to provide a measure of biodiversity. In addition, correlation functions – developed initially for problems outside forestry – enable analysis and characterization of forest stand structures, generating more accessible information. This paper outlines a classification of structural indices measuring alpha diversity and examines typical representatives of the classification groups such as the Shannon index, the aggregation index of Clark and Evans, the contagion index, the coefficient of segregation of Pielou, the mingling index, the diameter differentiation index, the pair correlation and the mark correlation function. These can be used to measure differences between forests in time and space, to generate forest structures, to analyse the differences between observed and expected structures and to characterize modifications of forest structure resulting from selective harvesting. These algorithms are the keys for assessing complex forest structures, which can be the result of continuous cover forestry methods. Continuous cover forests with selective harvesting are being promoted in the new forest policies of Britain. Case studies have shown that from given spatial forest structures one can possibly conclude the suitability for habitats, a hypothesis which has yet to be proved by further appropriate analysis. The equations for the quantification of stand structure presented in this paper have the advantage that they are easier to survey during forest inventory than the more direct measures of ecological variety.

Introduction

Continuous cover forestry has become a common term over the last two decades throughout Europe, although its history is much older (Helliwell, 1997). New forest policies have included several alternatives to clearcutting which are summarized by this term and which are expected to have a major impact on the forests of Great Britain in the future (Forestry Commission, 1998, 2000, 2001; Kerr, 1999). Part of the reason for adopting this new management type is to provide

improved habitats for wildlife dependent on continuity of woodland conditions and stands of diverse structure. Therefore measures are needed, firstly to distinguish between stands of different structure, and additionally to provide surrogate indices of habitat quality.

Spatial stand structure is an important factor in determining habitat and species diversity. Increasing heterogeneity of horizontal and vertical stand structure is linked to a higher number of species and stands with greater ecological stability. Silvicultural options can modify the stand structure

and therefore have an important potential role in securing stand diversity and ecological stability (Pretzsch, 1998; Humphrey *et al.*, 2000).

Ecosystem diversity on a spatial and areal scale is subdivided into alpha, beta, gamma and delta diversity (MacArthur, 1965; Whittaker, 1972). In forest ecosystems, alpha diversity operates within forest stands; beta diversity refers to the variation between forest stands; gamma and delta diversity operate on larger scales (Lähde *et al.*, 1999).

There is an increasing demand for information on alpha diversity, in particular on the spatial distribution of trees and their attributes (Mason and Quine, 1995; Ferris and Humphrey, 1999). Therefore, structural indices have been developed which describe, as mean values or distributions, certain horizontal aspects of forest stand structure (Upton and Fingleton, 1985, 1989).

Since the 1970s, statisticians have been developing functions which not only express forest stand

structures as mean values or as an empirical distribution, but are also able to describe spatial structure on a continuous basis (Ripley, 1977). This paper demonstrates how these functions add to the traditional concepts of structure indices.

The objective of this study is to discuss the methodology of some of the latest developments relating to variables and functions which characterize forest stand structure as a part of alpha diversity. A classification of these measures is presented in order to provide a better understanding of the different concepts (Figure 1). The equations of typical examples of the different concepts are explained and applied to three experimental stands. The results are discussed with the help of simulated references and suggestions made as to how these methods could be applied in British forests. The focus is primarily at the forest and stand scale rather than at a landscape or regional perspective.

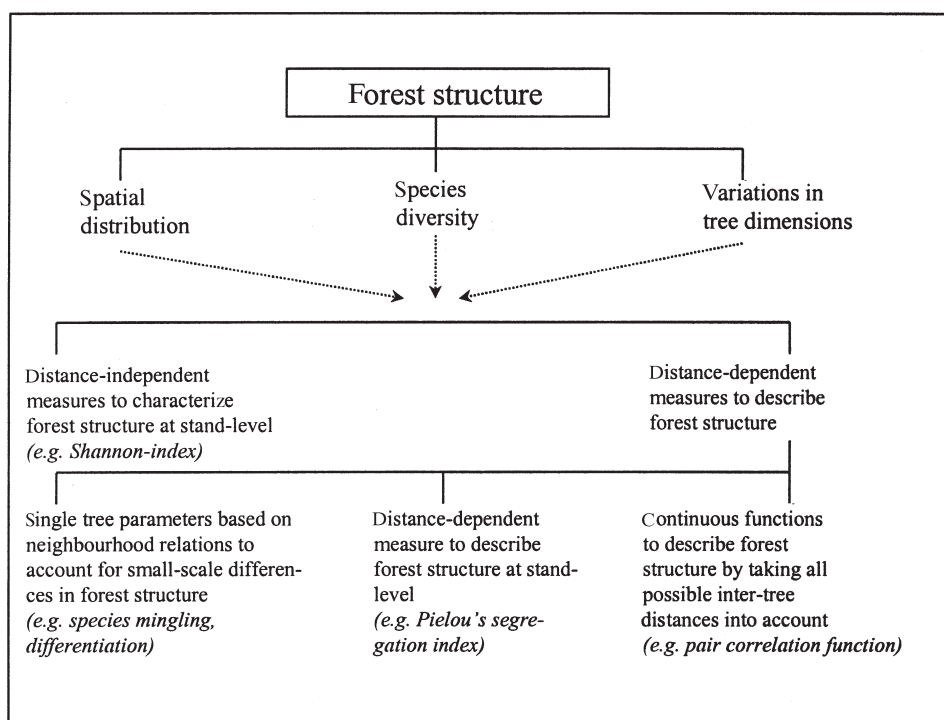


Figure 1. Overview of the three major characteristics of forest structure and the groups of variables by which forest structure is assessed (modified from Albert, 1999).

Brief review of stand structure algorithms

In managed forests, as well as in old-growth ecosystems, biodiversity is a key element for evaluating the stability of the system (Kimmins, 1997, p. 391). Kimmins (1997, p. 352) defines the structure of a plant community in terms of vertical and horizontal spatial organization.

In mathematical terms the majority of indices quantifying forest structure can be divided into two major groups: distance-independent and distance-dependent measures. While the first group evaluates stand structure without any spatial reference, the latter group can be subdivided into: (1) individual or single tree parameters based on neighbourhood relations, accounting for small-scale differences in biodiversity; (2) distance-dependent measures to describe forest stand structure at stand level; and (3) continuous functions. Figure 1 gives an overview of the three major aspects of forest structure and the groups of measures. The following paragraphs present typical examples of these groups.

Distance-dependent variables for characterizing stand structure

Aggregation index of Clark and Evans In the 1950s and 1960s numerical variables were developed to describe aspects of variability of tree locations in forest stands by a single value. One example is the aggregation index of Clark and Evans (1954). It is defined as:

$$R = \frac{\bar{r}_{observed}}{E(r)} \quad \text{where} \quad E(r) = \frac{1}{2 \cdot \sqrt{\frac{N}{A}}}; \quad (1)$$

$$R \in [0, 2.1491]$$

where $\bar{r}_{observed}$ stands for the mean of the distances from the trees to their nearest neighbours in a given forest stand, while $E(r)$ is the mean nearest neighbour distance in a stand with completely random tree locations ('Poisson forest') of intensity $\lambda = N/A$ with A = area of the forest stand and N = number of trees. Usually, the interpretation of R values is as follows: $R > 1$ if the pattern has a tendency to regularity, $R = 1$ if it is completely random (Poisson process), and $R < 1$ if there is clustering in the pattern.

Contagion index As a single-tree-based alternative to the aggregation index, Gadow *et al.* (1998) developed the 'contagion' variable or 'neighbourhood pattern', W_i , to define the degree of regularity of the spatial distribution of tree positions in a forest. Unlike the index of Clark and Evans, this variable is a single-tree parameter. Assuming complete regularity of the positions of the n nearest neighbours around a reference tree i , the expected standard angle α_0 between two neighbours would be equal to $360^\circ/n$. For example, $\alpha_0 = 90^\circ$ in a constellation involving four neighbours. Each pair of neighbours shares two angles, α and β , with $\alpha + \beta = 360$ and $\alpha \leq \beta$. Contagion is defined as the proportion of α angles which are smaller than the standard angle α_0 (see also Figure 2):

$$W_i = \frac{1}{n} \sum_{j=1}^n w_{ij} \quad \text{where} \quad w_{ij} = \begin{cases} 1, & \text{the } \alpha\text{-angle } j \text{ is smaller than } \alpha_0 \\ 0, & \text{otherwise} \end{cases} \quad W_i \in [0, 1] \quad (2)$$

In the 'structural group of four' W_i can be visually assessed in the field by comparing the actual angle α with an angle of 90° . Thus a quick decision can be made in the forest on whether α is smaller than 90° or not. If tree positions are recorded, W_i can be derived from the trees' coordinates. In a constellation involving four neighbours, W_i can assume five possible values (0, 0.25, 0.5, 0.75 and 1.0). The distribution of the W_i allows evaluation of the point pattern of tree positions in a forest. The average contagion ($\bar{W}$) may be used to classify the point pattern into the categories 'regular', 'random' and 'clumped' (Gadow *et al.*, 1998). In order to carry out a sensitivity analysis for the average contagion ($\bar{W}$), Albert (1999, p. 67) simulated 10 random, clumped and regular forest stands. The results indicate that stands of trees with a mean contagion value greater than 0.6 can be considered as clumped, those with values between 0 and 0.5 indicate regular tree distributions and between 0.5 and 0.6 are random. However, Albert (1999, p. 67) did point out that these distinctions may not be sharp.

Shannon index The Shannon index (Shannon and Weaver, 1949) is an example of a distance-

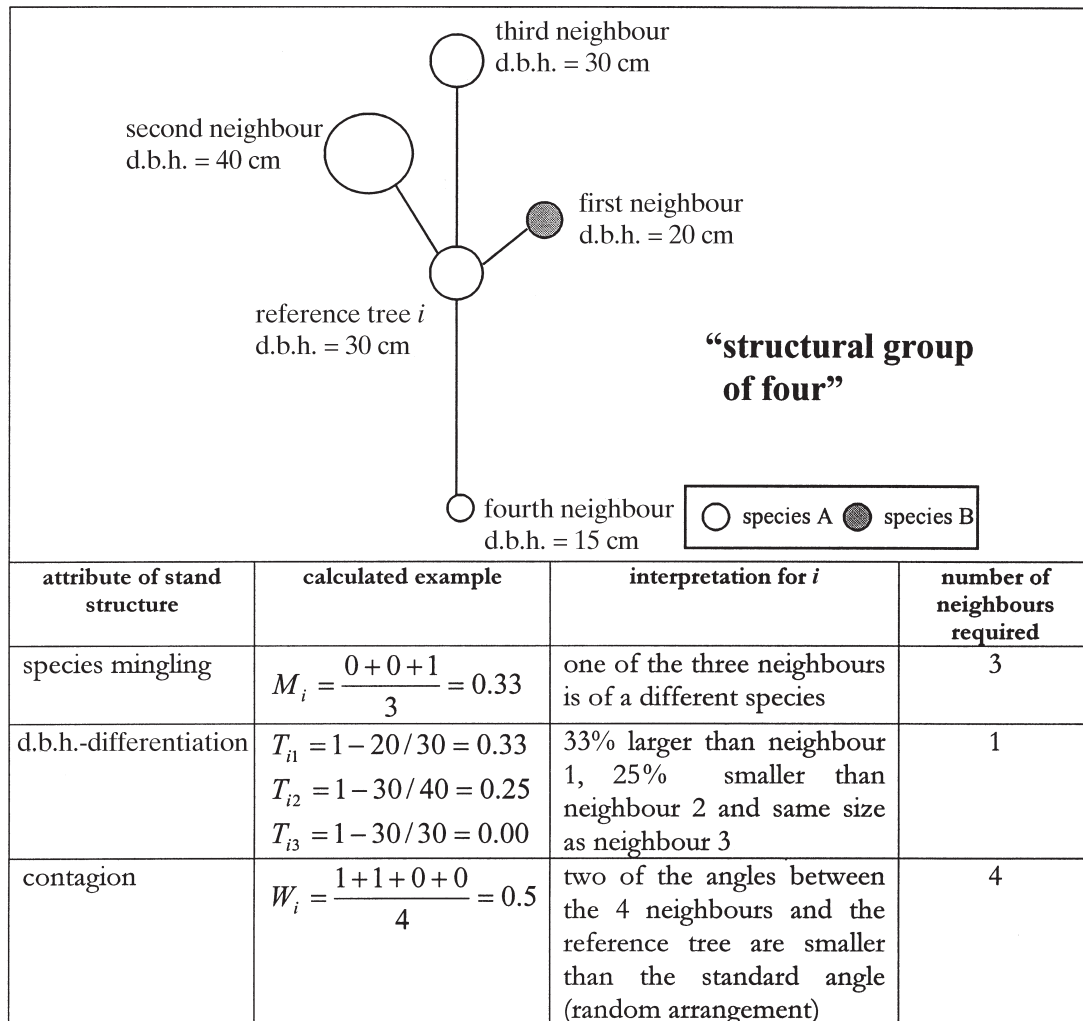


Figure 2. Structural variables (species mingling, d.b.h.-differentiation and contagion) for reference tree *i* and its nearest neighbours. Note that for calculating the species mingling, three neighbours are required, for the d.b.h.-differentiation one neighbour is needed as a minimum and for the contagion, four neighbours are required (modified after Albert and Gadow, 1998).

independent algorithm (see Figure 1) describing species mingling. It is defined by:

$$H'(p_1, p_2, \dots, p_n) = - \sum_{j=1}^n p_j \cdot \ln(p_j) \quad (3)$$

where p_j = probability of a randomly selected tree belonging to tree species j ; n = number of tree species in the forest.

A similar index, also working with probabilities, was developed by Simpson (1949).

Pielou's coefficient of segregation A further index is the coefficient of segregation, S , of Pielou (1977). It describes the degree of mixing of trees of two species A and B in a forest, and like the aggregation index, R , it is based on the nearest

neighbour tree distances. In Pielou's notation, S is given as follows:

Let N be the number of all pairs of trees (reference tree – nearest neighbour tree), let m and n be the numbers of trees of species A and B, respectively, and let r and s be the number of times trees of species A and B are found as the nearest neighbours of a reference tree. These numbers can be set out in a simple form in a 2×2 table as follows.

		Species of the nearest neighbour		
Species of reference tree	A	A	B	Total
	B	a	b	m
	Total	c	d	n
		r	s	N

The coefficient of segregation is defined by

$$S = 1 - \frac{N \cdot (b + c)}{m \cdot s + n \cdot r}; \quad S \in [-1, 1] \quad (4)$$

If the nearest neighbours are always of the same species as the reference trees, then $S = 1$. If all neighbours are of different species $S = -1$. In the case of complete randomness of species distribution, one can expect values around 0.

Mingling index The corresponding single-tree variable to Pielou's coefficient of segregation, known as *mingling* (M_i), gives the proportion of the $n = 3$ nearest neighbours j ($j = 1 \dots n$) of the i th reference tree which do not belong to the same species as the reference tree i (Figure 2):

$$M_i = \frac{1}{n} \sum_{j=1}^n v_{ij}; \quad M_i \in [0, 1] \quad (5)$$

$$\left\{ \begin{array}{l} \text{where } v_{ij} = \\ \left\{ \begin{array}{l} 1, \text{ reference tree } i \text{ and neighbour } j \text{ are of different tree species} \\ 0, \text{ otherwise} \end{array} \right. \end{array} \right.$$

The current state of a forest may be described very effectively using the distribution of the mingling variable (see, for example, Fuldner, 1995; Pommerening, 1997; Albert, 1999). In a constellation involving three neighbours ('structural group of four') M_i can assume four possible values (0.00; 0.33; 0.67; 1.00). For the calculation of the mingling variable for a whole stand,

all M_i values are added up and divided by the number of trees. The bigger the mean mingling M , the more the different tree species are intermingled. Small values indicate large groups of only one tree species and therefore segregation (Pommerening, 1997).

Diameter differentiation index The single tree diameter differentiation variable, T_{ij} , gives the size difference of neighbouring trees on a continuous scale and describes the spatial distribution of tree sizes. For the i th reference tree and its $n = 3$ nearest neighbour j ($j = 1 \dots n$) the diameter differentiation T_{ij} is defined as:

$$T_{ij} = 1 - \frac{\min(DBH_i, DBH_j)}{\max(DBH_i, DBH_j)}; \quad T_{ij} \in [0, 1] \quad (6)$$

where DBH = breast height diameter (d.b.h. in cm).

The value of T_{ij} increases with increasing average size difference between neighbouring trees. $T_{ij} = 0$ means that neighbouring trees have an equal size. This index is based on pairs of reference trees – first, second or third nearest neighbour tree. For the calculation of the diameter differentiation of a whole forest stand, all T_{ij} values are summed and divided by the number of trees, so that three single numbers T_1 , T_2 and T_3 describe the size difference of neighbouring trees, taking the first, second and third nearest neighbours, respectively. Diameter differentiation values can be interpreted as follows:

- 1 small differentiation:** comprises the classes $0.0 \leq T < 0.3$. The tree with the smallest d.b.h. is 70 per cent or more of the neighbouring tree's size
- 2 average differentiation:** comprises the classes $0.3 \leq T < 0.5$. The tree with the smallest d.b.h. is 50–70 per cent of the neighbouring tree's size
- 3 big differentiation:** comprises the classes $0.5 \leq T < 0.7$. The tree with the smallest d.b.h. is 30–50 per cent of the neighbouring tree's size
- 4 very big differentiation:** comprises the classes $0.7 \leq T \leq 1.0$. The tree with the smallest d.b.h. is less than 30 per cent of the neighbouring tree's size.

Diameter differentiation, mingling and contagion indices may be established for the stand as a whole or for a given sub-population.

Figure 2 shows a hypothetical sampling group of four and the corresponding values of the structural attributes of species mingling, d.b.h.-differentiation and contagion for the reference tree i . All single-tree parameters can also be calculated as frequency distributions which will be demonstrated later in this paper. Fuldner (1995) used species-specific indices as well as variables relating to the dominant height.

The single-tree structure variables have the additional advantage that it is possible to assess them easily from common forest inventories. Pommerening (1997), Pommerening and Schmidt (1998) and Pommerening and Gadow (2000) evaluated the performance of the sampling method 'structural group of four' and the standard fixed-area plot commonly used in forest inventories concerning these variables and found that both methods are useful and imply only a small sampling error. This offers the possibility of incorporating structural indicators in forest inventories and monitoring.

There are a considerable number of other indices for describing spatial structure, which are based on similar concepts. They cannot be discussed in detail here, as this would be beyond the scope of this paper. For further information, see Gadow and Hui (1999), Gadow *et al.* (1998), Gleichmar and Gerold (1998), Smaltschinski (1998), Upton and Fingleton (1985, 1989), Lähde *et al.* (1999), Latham *et al.* (1998), Neumann and Starlinger (2001), O'Hara *et al.* (1996), Spies (1998), and Zenner and Hibbs (2000).

Pair correlation function Modern point process statistics use functions instead of indices or empirical distributions. These functions depend on the inter-tree distance r .

Unlike the aggregation index, R , the pair correlation function $g(r)$ does not result in one number, but in a function, which can be plotted as a graph. It characterizes the variability of the pattern of tree locations. This function is based on the mathematical theory of marked point processes. In the marked point process model, the points are tree positions given with respect to a Cartesian coordinate system. The marks are qualitative or quantitative tree characteristics; for example, tree species, diameter at breast height or total tree height (Penttinen *et al.*, 1992). Assuming that the observed forest stand can be described by a

homogeneous point process (i.e. there are no systematic fluctuations of point density and no preferred directions in the point pattern), the tree density or intensity λ is defined as the mean number of trees per area. The familiar forest variable 'stems per hectare' can be used as intensity. Intensity λ has the following interpretation. Consider an infinitesimally small circle of area dF , then the probability of finding one tree in it is λdF , because the area is too small to find two or more trees. When describing variability and correlations in tree stands we have to consider pairs of trees. Let us consider two infinitesimally small circles of areas dF_1 and dF_2 of inter-centre distance r . Let $P(r)$ denote the probability that both circles each contain a point of the point process, therefore:

$$P(r) = \lambda^2 \cdot g(r) \cdot dF_1 \cdot dF_2 \quad (7)$$

The function $g(r)$ is called the pair correlation function and is a function of the inter-point distance r . It answers similar questions to the aggregation index of Clark and Evans (for interpretation examples, see Tomppo, 1986; Penttinen *et al.*, 1992; Pommerening *et al.*, 2000; Shimatani, 2001). Figure 3 shows the pair correlation functions for two fundamental types of point processes, namely for a cluster process (positive correlation of tree locations, mutual attraction) and for a process with mutual inhibition between the trees (negative correlation of tree locations). For a forest with trees distributed at random, $g(r) = 1$, which means that the tree locations are spatially uncorrelated. The dotted graph in Figure 3, which runs parallel to the abscissa at a value of 1, illustrates this. In this case, according to the product formula of the theory of probabilities, $P(r) = \lambda \cdot dF_1 \cdot \lambda \cdot dF_2$.

If the tree locations show a tendency towards regularity, e.g. in very young stands planted in rows or in very old stands (where there are large distances between the trees, contradicting the assumption of random tree locations), then for small values of r , $g(r)$ takes the form $g(r) = 0$, because pairs of trees at these distances do not exist (Figure 3). This is the case at the so-called hardcore distance r_0 , which is the smallest observed inter-tree distance. For bigger values of r , $g(r) > 0$ and $g(r)$ approaches the value of 1. Young forest stands of natural origin very often have trees arranged in clusters where the values of

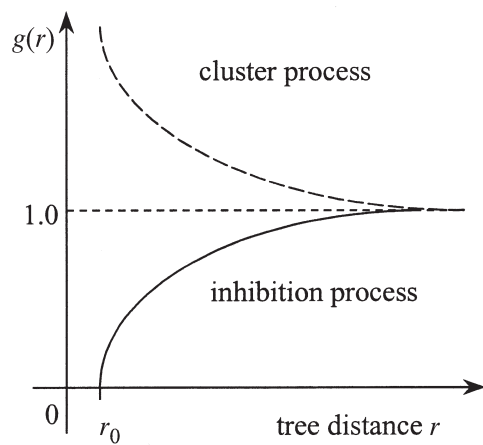


Figure 3. Schematic diagram of the pair correlation function $g(r)$ for analysing forest structures. The hardcore distance r_0 is the smallest possible inter-tree distance of a forest stand. Values of $g(r) > 1$ indicate that interpoint distances around r are relatively more frequent than those in a forest with random tree locations. Small values of r typically mean clustering. Conversely, values of $g(r) < 1$ indicate that the corresponding inter-tree distances are rare in the forest stand under study, which may indicate inhibition of trees caused by competition.

the pair correlation function tend to be larger than 1 at small inter-tree distances r (indicated by the dotted graph called 'cluster process' in Figure 3). Values of the pair correlation function $g(r)$ larger than 1 indicate that the inter-tree distances around r are relatively more frequent compared to those in a completely random point process. If this is the case for small values of r , typically there is clustering. Conversely, values of $g(r)$ smaller than 1 indicate that the corresponding distances are rare, which may indicate inhibition (Penttinen *et al.*, 1992). Estimated pair correlation functions reflect the behaviour often reported in forestry literature, that natural forests often begin with cluster-process-like patterns. Competition between trees causes (self) thinning, and at the final stage the forest is similar to a Poisson process with random tree locations or even more regular (Stoyan and Penttinen, 1998). Unlike the aggregation index, the pair correlation function depends on the inter-tree distance and allows spatially explicit investigations of the interaction between trees (e.g. type

of interaction, determination of the range of interaction and estimation of its strength). Shimatani (2001) suggested an interesting combination of the Simpson index (Simpson, 1949) and the pair correlation function.

Mark correlation function Clearly, not only the number of trees in a stand is of interest but also their diameters, heights, damage class, tree species and other so-called 'marks'. In the theory of point processes a 'mark' is a value, which is assigned to a point. If the 'point' is a tree location, the 'mark' is a tree attribute (see Penttinen *et al.*, 1992). As in the pair correlation function, a pair of trees is regarded as having the inter-tree distance r . From the marks m_1 (e.g. the d.b.h. of the first tree) and m_2 (e.g. the d.b.h. of the second tree) a value is calculated, which assesses the dissimilarity or similarity of the trees' marks. This relationship is quantified by $f(m_1, m_2)$, where f is a suitable 'test' function. Here, f is defined as $f(m_1, m_2) = m_1 \cdot m_2$. From these single values a mean value is calculated while the inter-tree distance r remains constant. Thus we conceive a function $\kappa_m(r)$, which is dependent on r . It is advisable to divide the function $\kappa_m(r)$ by the square of the mean value m of the observed tree parameters in order to make interpretation easier (Penttinen *et al.*, 1992; Stoyan and Penttinen, 1998). The mark correlation function, $k_m(r) = \kappa_m(r)/m^2$ is very suitable for analysing the mutual influence of trees as far as it affects the observed tree parameter. If d.b.h. is used, then it is often the case that for small inter-tree distance r values, $k_m(r)$ is < 1 . This indicates a tendency that at small inter-tree distances both trees of a pair have smaller diameters than the average of the stand. This behaviour may indicate that the price which trees have to pay for being close together is that their diameters tend to be smaller than the mean in the whole forest. However, small values of $k_m(r)$ can also be the consequence of clusters with young and thin trees (Stoyan and Penttinen, 1998). Penttinen *et al.* (1992), Gavrikov and Stoyan (1995) and Pommerening *et al.* (2000) give some examples.

Reference values and edge effects

Before applying the presented parameters and functions to trial stands in order to find out how

they work, there are two important issues which need to be addressed.

One question often asked is what an observed number means exactly. For example should an observed mingling be regarded as desirable or not and, if so, to what extent? This question is about a *reference* which can be used to compare with observed values. Fuldner (1995) calculated structural variables for different forest compartments and for the main tree species in a stand within each compartment before and after thinning. He made comparisons between compartments, between the two main tree species within each compartment and assessed changes in structure caused by thinning. This approach could be used to select particular forest compartments as references in that they reflect a typical or desired spatial structure. Research within nature reserves also uses the concept of reference plots to monitor natural forest development. The spatial structure of such 'untouched' forest compartments could be compared with that of managed stands.

The disadvantage of this method is that it assumes uniformity in natural stand structures: no forest compartment is like another, and it may not be justified to use a limited number of research plots as references. Until now no study has made an attempt to derive generally applicable values of structure indices for particular tree species mixtures and management types. Moreover, Sprugel (1991) concluded that naturalness is difficult, if not impossible, to define. A great variety of forest communities might only be described as natural for a given site and a given time. Shape, dimension, altitude, relief, slight differences in species compositions and site conditions, to name only a few factors, can make the suitability of comparison questionable.

To overcome this disadvantage Pommerening (1997) suggests using computer simulation to make comparisons between observed structural variables and those expected in a random forest. Using random permutations, it is possible to calculate a test criterion which can be used to evaluate differences between observed and expected forest structures. The advantage of this method is that the tree species composition and other tree attributes from the stand under study are used to derive the reference rather than those from a different forest compartment. One can criticize

this approach by arguing that the random arrangement of tree attributes which is used as a reference may not correspond to a particular biological meaning. However, Stoyan and Penttinen (1998) investigated a number of different stand development phases and came to the conclusion that old-growth stands do often tend to have random tree locations. Randomness as a reference is very often used in statistics, and is easy to establish. This approach has been adopted in a number of studies of stand structure (e.g. Clark and Evans, 1954; Zenner and Hibbs, 2000). By using randomness as a substitute for naturalness it is possible to quantify how much the value of a structural variable in a given forest compartment differs from this theoretical reference and to suggest the reason for this.

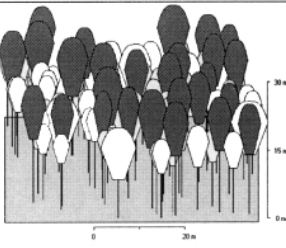
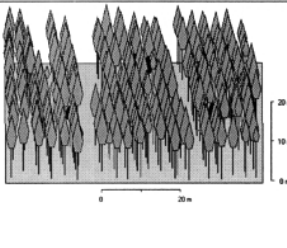
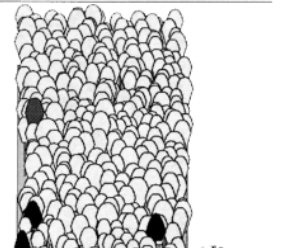
In this paper, randomness is used as a reference and the results of the random simulations are mean values of 5000 independent single simulations per forest compartment.

Another important issue is that of edge effects, which need to be addressed if distance-dependent structural parameters are applied. The problem is that neighbourhood relationships are not truly represented at the stand or plot boundary when potential neighbours lie outside the research plot. This edge bias becomes bigger the smaller the research plot is and the fewer trees are involved. This problem is shared with distance-dependent competition indices (Biging and Dobbertin, 1992). To overcome this effect, several methods have been developed (Monserud and Ek, 1974; Martin *et al.*, 1976; Radtke and Burkhardt, 1998). All of these methods, however, have considerable disadvantages and there is no one method which can be applied to all distance-dependent structure variables and plot shapes. There is still considerable need for research in this area. Due to the fact that most of the boundaries of the three trial stands used in this paper are real forest boundaries (forest and public roads), no edge correction has been applied to the structure variables. The correlation functions, however, have internal edge corrections integrated in their algorithms.

Study sites

Data from three forest stands are used to demonstrate the reviewed indices (see Table 1). The three

Table 1: Forest yield data of the stands Manderscheid 198, Manderscheid 187 and Ammeloe 14e. In the schematic pictures beech trees appear white, oaks dark grey, Douglas firs light grey and Scots pine black. *G* = basal area per hectare ($\text{m}^2 \text{ha}^{-1}$), *N* = number of trees per hectare, *V* = volume per hectare ($\text{m}^3 \text{ha}^{-1}$)

compartment	Manderscheid 198				Manderscheid 187				Ammeloe 14e			
schematic picture												
survey	1996				1996				1996			
natural production	Oak	G 16.31	N 112	V 293.86	Doug. Fir	G 27.30	N 660	V 246.86	Beech	G 18.76	N 546	V 211.14
	Beech	G 10.95	N 253	V 140.64	Scots pine	G 0.53	N 53	V 2.86	Oak	G 0.21	N 3	V 3.03
	-	-	-	-	Other	G 0.20	N 23	V 0.69	Scots Pine	G 1.07	N 6	V 13.72
	Total	G 27.26	N 365	V 434.5	Total	G 28.03	N 736	V 250.41	Total	G 20.04	N 555	V 227.89
age	118 years				24 years				53 years			
area	0.24 ha				0.36 ha				0.65 ha			

stands were especially selected to reflect rather different spatial structure. Comparable data from Britain were not available when this study started. However, current research projects at the University of Wales, Bangor and the Forestry Commission's Northern Research Station aim at filling this gap.

The stands Manderscheid 198 and 187 are situated in the forest district of Manderscheid in the German federal state Rhineland-Palatinate. The compartment Ammeloe 14e is situated west of Münster in the federal state Northrhine-Westphalia.

Compartment Manderscheid 198 covers an area of 0.24 ha and is a part of a 11.9 ha forest stand. It is stocked with 120-year-old oak (*Quercus petraea* Lieth), intermingled with beech (*Fagus sylvatica* L.) occurring mostly in small groups. This compartment was included in this study because it reflects a mature stand with an intimate species mixture.

Compartment Manderscheid 187 covers an

area of 0.36 ha and is stocked with 24-year-old planted Douglas fir trees (*Pseudotsuga menziesii* (Mirb.) Franco). The compartment was originally established as a pure stand and the planting rows are still distinctly visible, but the area now includes a few isolated, naturally regenerated beech (*Fagus sylvatica* L.), pine (*Pinus sylvestris* L.), spruce (*Picea abies* (L.) Karst.), birch (*Betula pendula* Roth.) and larch (*Larix decidua* Mill.) trees. This compartment was selected because it is a young plantation with regular tree positions and a rather low degree of species mingling.

Compartment Ammeloe 14e covers an area of 0.65 ha and is stocked mainly with beech (*Fagus sylvatica* L.), apart from some pines (*Pinus sylvestris* L.) and oaks (*Quercus robur* L.). The trees are 53 years old. Ammeloe 14e reflects typical more or less mono-species beech forests of Central Europe. In terms of overall stand age it is between Manderscheid 187 and Manderscheid 198.

The following parameters were recorded

within each compartment: diameter at breast height (d.b.h.), tree species and the Cartesian coordinates of the tree locations. The data collection was conducted by the Institute of Forest Management and Forest Yield Sciences (University of Göttingen, Germany) in 1996 as part of a pilot project with the state forest services of Rhineland-Palatinate and Northrhine-Westphalia. The three plots selected for this study were part of a larger network of reference plots where new methods of forest management were tested. These new methods aimed at combining conventional surveying of yield and stocking information with a detailed quantitative description of stand structure.

Results and discussion

Spatial distribution

The aggregation index R (Clark and Evans, 1954) and the 'contagion' or 'neighbourhood pattern' index W both describe the horizontal spatial arrangement of tree positions. According to the aggregation index, compartment Manderscheid 187 shows the most regular tree patterns (Table 2), as might be expected from a young Douglas fir plantation. The Ammeloe 14e beech stand also appears to be quite regular while the mixed oak-beech compartment of Manderscheid 198 reflects random tree positions. The observed contagion distribution (which also describes the horizontal arrangement of tree locations) can be compared to the simulated contagion distribution (assuming that the given trees of each of the three forest compartments are arranged randomly). The comparison in Figure 4 shows that each of the three stands tend to have only small deviations from the corresponding random distribution of tree locations. The deviation between observed and simulated distributions can be quantified by the absolute discrepancy algorithm AD (Gregorius, 1974; Pommerening, 1997, p. 77):

$$AD = \frac{1}{2} \cdot \sum_{i=1}^k |\hat{\theta}_i - \theta_i|; \quad AD \in [0, 1] \quad (8)$$

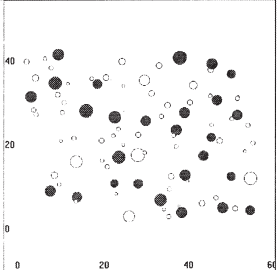
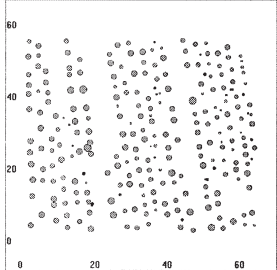
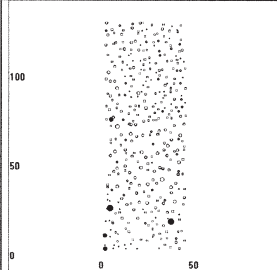
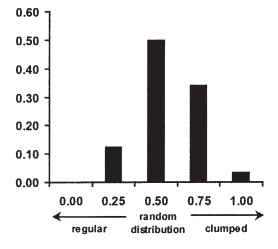
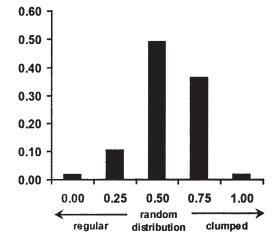
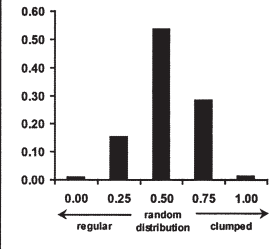
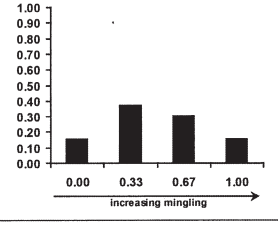
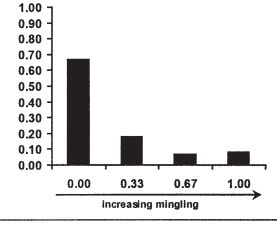
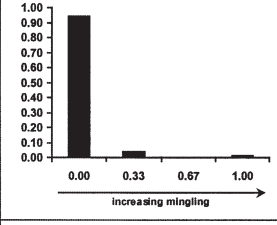
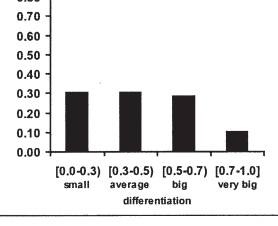
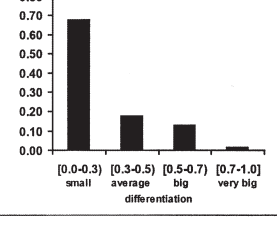
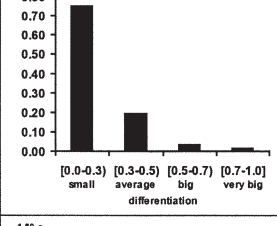
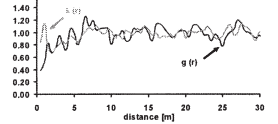
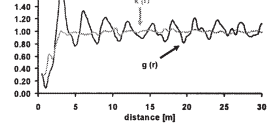
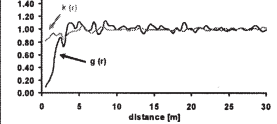
with $\hat{\theta}_i$ simulated relative frequency in the distribution classes i to k , and θ_i observed relative frequency in the distribution classes i to k .

Using this algorithm we find that $AD = 0.068$ for compartment Manderscheid 198, $AD = 0.106$ for compartment Manderscheid 187 and $AD = 0.067$ for compartment Ammeloe 14e. This simply means that approximately 7 per cent of some of the five classes of the observed contagion distributions in Manderscheid 198 and in Ammeloe 14e would need to be swapped over to other classes to obtain a contagion distribution under random conditions. The analysis of Manderscheid 187 shows that 10 per cent need to be swapped. Therefore the observed arrangement of tree locations in Manderscheid 187 is slightly less random than the other two compartments. The reason for this can be found in the regular arrangement of tree locations in the Douglas fir plantation. The comparison of the observed relative frequencies to the simulated frequencies is what the Clark and Evans index does internally. Comparing the observed value to the random value, when the aggregation index $R = 1$, yields similar information. Due to different algorithms (as explained in the review) the results calculated by the index of Clark and Evans and contagion are slightly different: Manderscheid 198 and Ammeloe 14e show a similar deviation from randomness in terms of contagion. Ammeloe 14e, however, seems to be slightly closer to randomness, although the index of Clark and Evans hints that Manderscheid 198 is closer to random conditions.

Figure 5 shows how the contagion distribution of the two main tree species of one forest compartment can be compared with each other. Oaks in Manderscheid 198 are more randomly distributed, while beech shows a tendency towards a clumped arrangement. This is referred to later when the species mingling of this stand is discussed.

Even more information on the spatial arrangement of tree locations is available by applying the pair correlation function $g(r)$. The pair correlation function (see Table 2) is also associated with an internal reference. A parallel to the abscissa through the value 1 defines a complete random arrangement of tree locations. Up to an inter-tree distance of 6 m, the pair correlation function in the compartment Manderscheid 198 is located below the value 1.0 which means that at these distances there are fewer trees observed than would be expected under random

Table 2: A quantitative description of the structure of the three sample stands by means of variables and correlation functions. The label of the ordinates of contagion, mingling and T1 distributions is relative frequency. The label of the ordinates of the correlation functions is correlation. In the square plots, beech trees appear white, oaks dark grey, Douglas firs light grey and Scots pine black.

compartment	Manderscheid 198	Manderscheid 187	Ammeloe 14e
square plots			
Clark & Evans	1.0073	1.3508	1.2481
contagion	0.5710	0.5646	0.5342
Shannon	0.6165	0.4249	0.0997
mingling	0.4886	0.1888	0.0307
Pielou	-0.2818	0.1141	-0.0138
T1	0.4173	0.2495	0.2145
contagion distribution			
mingling distribution			
T1 distribution			
correlation functions			

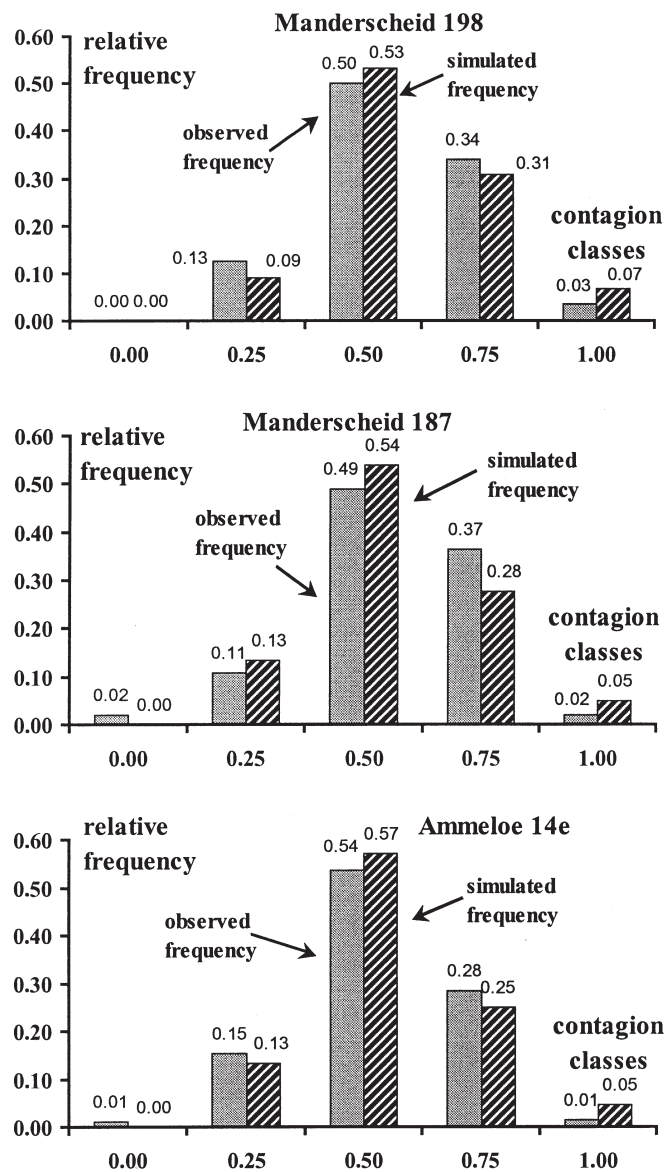


Figure 4. Observed and simulated values of the contagion distribution for the three forest compartments.

conditions. At an inter-tree distance of 6.5 m the pair correlation function has a value of 1.26, which means that there are more trees having between them a distance of about 6.5 m than would be expected. The pair correlation function behaves similarly in the compartments

Manderscheid 198 and Ammeloe 14e. Since the curve for $g(r)$ reaches 1 at $r \approx 8$ m in Manderscheid 198 (if random fluctuations around 1 are ignored) one can conclude that the inter-tree interaction does not go further than 8 m. Trees with an inter-tree distance less than 8 m seem to

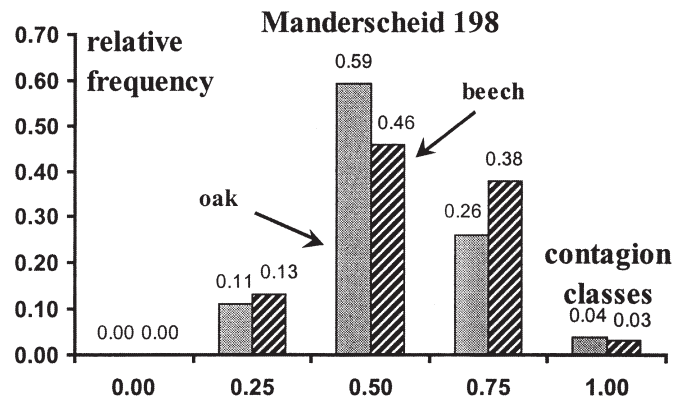


Figure 5. Differences in the contagion index distribution between the two main tree species recorded within the Manderscheid 198 compartment.

be rare, so that there must be an inhibition caused by competition and/or thinning. The corresponding critical distance in Ammeloe 14e is 5 m. These results are very plausible because both stands are older than Manderscheid 187 and their trees have greater diameters; large trees require big inter-tree distances. The pair correlation thus offers information about the interaction radius, which is the maximum radius of direct tree interaction. Penttinen *et al.* (1992) observed similar patterns in an 80-year-old Norway spruce forest in Saxony (Germany) and a 50-year-old mixed stand in Northern Finland consisting of Scots pine and silver birch. A very different picture is drawn by the pair correlation function in the compartment Manderscheid 187. The regular pattern of fluctuations very much reflects a current spacing of 3×3 m. The maxima of the function indicate that trees were preferably located at the corresponding distances, while values below 1 leading to a minimum show that trees can be found at these distances with increasing difficulty. In this case the interaction radius of 3 m can be more easily identified in the curve of the mark correlation function. Pommerening *et al.* (2000) investigated a selection forest in the Bavarian forest and, in contrast to the results of this study, found that the pair correlation function always had values above 1.0 up to a distance of 4 m. This was caused by clumps of regeneration trees which can be typically found in selection forests. The authors also derived species specific inter-tree interaction radii for that stand between 6.5 and 9 m.

Species diversity

The distance-independent Shannon index suggests that species mingling is highest in compartment Manderscheid 198, lower in Manderscheid 187 and that there is almost no species diversity in Ammeloe 14e (Table 2). The Shannon index gives a higher value the more species there are in a forest compartment and the more equally they are represented in terms of relative abundances. Shannon index values are higher in compartment Manderscheid 198, because oak and beech are quite equally represented, rather than in compartment Manderscheid 187, where there are more tree species but Douglas fir is dominant.

The distance-dependent measure of segregation S of Pielou (Table 2) indicates aggregation of different tree species in Manderscheid 198, species segregation in Manderscheid 187 and more or less random species distribution in Ammeloe 14e.

This picture can be completed by examining the mingling variable M (see Figure 6). Using equation (8) we find that $AD = 0.072$ for compartment Manderscheid 198, $AD = 0.052$ for compartment Manderscheid 187, and $AD = 0.007$ for compartment Ammeloe 14e. This means that in the two Manderscheid compartments between 5 and 7 per cent of some of the four classes of the observed species, mingling distributions need to be swapped over to other classes to obtain random conditions. In Ammeloe 14e this is only 0.7 per cent. In contrast to the

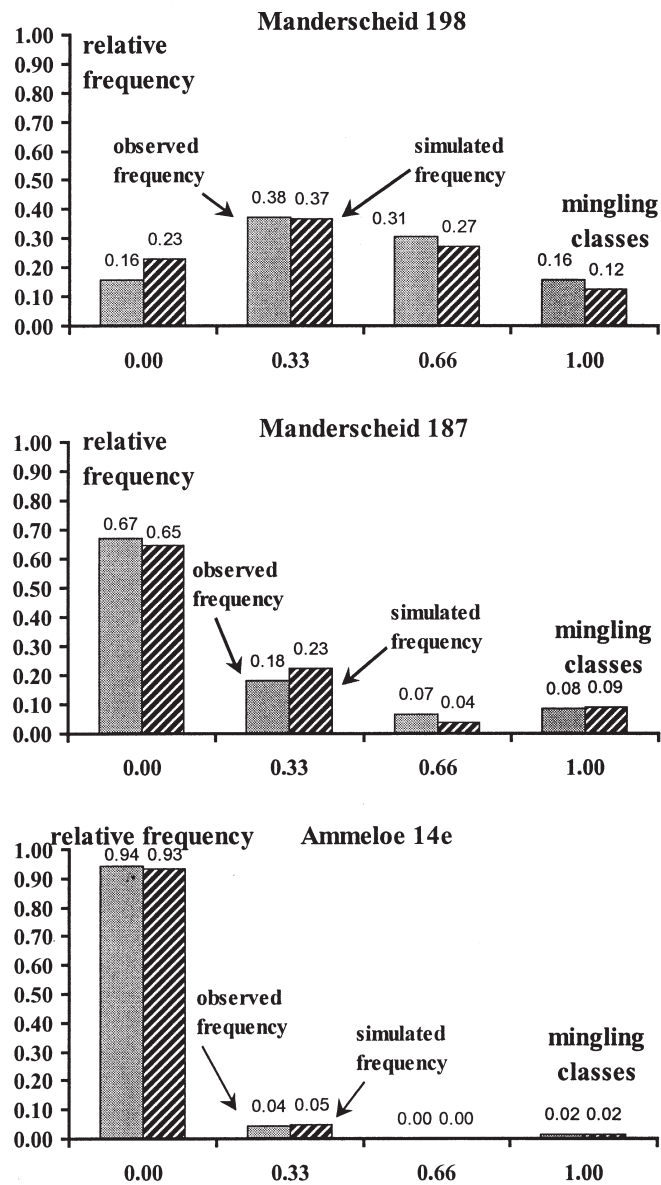


Figure 6. Observed and simulated values of the species mingling distribution for the three forest compartments.

contagion simulations, the tree positions were regarded as constant while the attribute 'tree species' was assigned at random to these positions. In the three sample forest stands we can detect only very small deviations from a random distribution.

The general tendencies of the values of the three measures of species diversity are therefore similar. Figure 7 shows how the species mingling distribution of the two main tree species of compartment Manderscheid 198 can be compared to each other without using a theoretical reference.

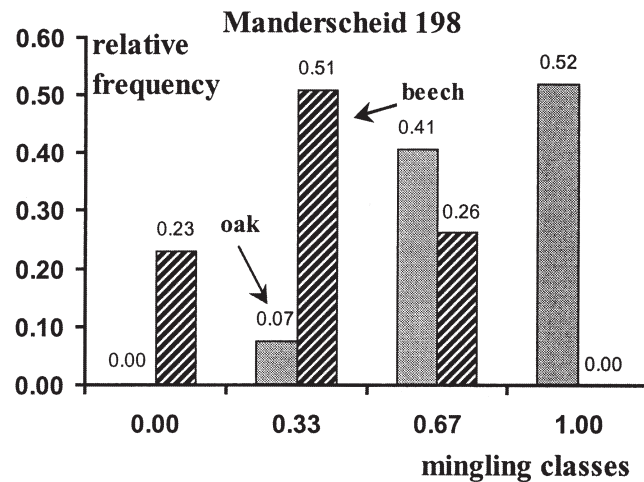


Figure 7. Differences in mingling of the two main tree species within the Manderscheid 198 compartment.

The compartment seems to have been managed in such a way that oaks are closely intermingled with beeches, while the latter are arranged in clumps. We come to this conclusion because of the fact that most of the oaks belong to the higher mingling classes, while most of the beeches are located in the smaller mingling classes. When the species-specific contagion distribution of Manderscheid 198 was discussed (see above), it was clear that oak was more randomly distributed than beech, with the latter appearing more clumped. Obviously the mingling and contagion variables are in many ways related to each other. Figure 7 underpins the visual impression gained from the schematic picture in Table 1 or from a field visit to the forest with a quantitative description. The spatial pattern of the stand is set by a special management of this woodland type which promotes oak and keeps beech as an admixture serving the oak. Földner (1995, p. 77) investigated mixed beech–ash stands with admixtures of sycamore and maple in the Bovenden forest district near Göttingen, and came to a very similar result with regard to the mingling of the main species beech and ash.

Variations in tree dimensions

The diameter differentiation $T1$ (Table 2) is the spatially explicit counterpart to the distance-independent diameter distribution in that it

allows the difference between immediate tree neighbours to be measured.

The mean $T1$ values indicate a quite high diameter differentiation in Manderscheid 198 but a very low one in both the other compartments. The distribution (Table 2 and Figure 8) shows that most of the trees within compartments Manderscheid 187 and Ammeloe 14e belong to the first two differentiation classes, which means that their immediate neighbours have a diameter from 50 per cent and predominantly more than 70 per cent of their own diameter. By contrast, in Manderscheid 198, the trees seem to be almost equally distributed over all the four classes. These differences are not only a function of age and stand development stage but are very much due to the special character of the management of mixed oak–beech stands, where oak is systematically promoted while beech is managed to serve the oaks and improve their timber quality. This results in bigger diameters of oak and smaller diameters of beech.

Using equation (8), we find that $AD = 0.205$ for compartment Manderscheid 198, $AD = 0.108$ for compartment Manderscheid 187 and $AD = 0.151$ for compartment Ammeloe 14e (Figure 8). These results show marked deviations from a random distribution of tree diameters. As with the mingling simulations, the tree positions were regarded as constant while the tree attribute ‘diameter’ was assigned randomly. The small

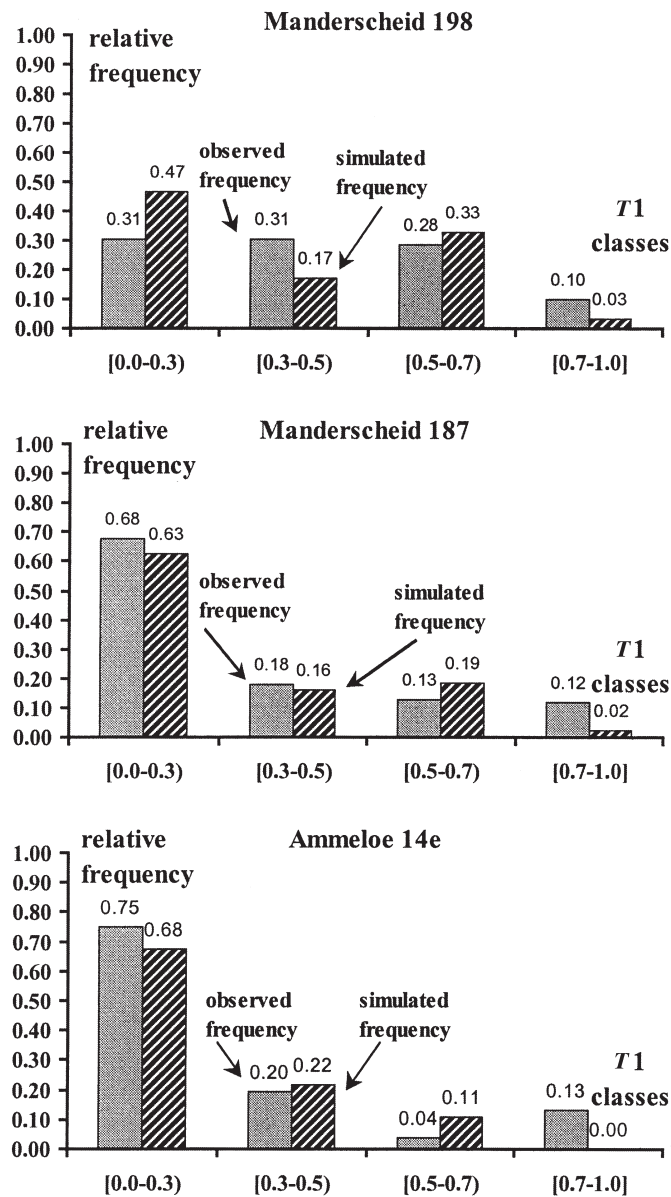


Figure 8. Observed and simulated values of the diameter differentiation T1 distribution for the three forest compartments.

deviation in Manderscheid 187 might be due to the fact that the dimensions of young plantation trees are very similar to each other anyway so that a random assignment presents no great difference in terms of diameter differentiation. In Mander-

scheid 198 the differences are especially large in the first two classes, indicating small and average differentiation. This might indicate that special management has taken place to protect the light-demanding oaks from the climax tree species

beech, perhaps by removal of some of the latter. Although randomness as a reference is no substitute for naturalness, deviation from randomness is a suitable indicator that in a particular forest compartment something special has been detected.

Applying the mark correlation function $k(r)$ as another measure to characterize the spatial arrangement of tree dimensions, we find that there are correlations between the diameters of trees only at small distances between trees. In Manderscheid 187 the diameters of trees located more closely to each other tend to be negatively correlated, because at distances of up to 3 m between trees the values of $k(r)$ are below 1. This indicates a trend that at small inter-tree distances both trees of a pair have smaller diameters, which is the price that they have to pay for being close together. At distances greater than 3 m, there are only random fluctuations around 1, indicating that there is no longer any correlation between diameters. In Ammeloe 14e we see a similar tendency; however, the value of 1 is reached later, at an inter-tree distance of 5.5 m. The speed with which the function approaches the value 1 depends on the extent of interaction between trees based on their dimensions. Obviously the older compartment Ammeloe 14e has larger tree interaction radii than the comparatively young plantation Manderscheid 187. Similar results were found by Pommerening *et al.* (2000) in a selection stand of the Bavarian forest. The value of $k(r)$ has an interesting peak at an inter-tree distance of 1 m in compartment Manderscheid 198. Occasionally, where two or more dominant trees are located closely to each other, they still appear to be growing satisfactorily, perhaps as a result of local management and/or good site conditions. After this peak, $k(r)$ shows quite 'normal behaviour' and approaches the value of 1 again at an inter-tree distance of 6 m.

Conclusions

Forestry in Britain as well as in Europe has to demonstrate that its management is sustainable in terms of providing wildlife habitat. Landscape and forest compartment structures determine to a large extent the occurrence and population dynamics of a range of species. From given spatial

forest structures one can identify the suitability of habitats or the population development of important wildlife and plant species (McKelvey *et al.*, 1993; Pitkanen, 1997; Letcher *et al.*, 1998; Wiegand, 1998). The equations for the quantification of stand structure presented in this paper have the advantage that they are easier to survey by means of common forest inventories than more direct measures of diversity. The latter can be surveyed only selectively at certain points and after considerable effort. At least with some of the variables discussed in this paper the more direct measures of diversity seem to correlate reasonably well (Spanuth, 1998; Neumann and Starlinger, 2001), but the correlation still needs to be verified by further appropriate analysis. While there are many studies comparing structural indices which focus on methodology, only a few studies exist on the relationship between stand structure and direct measures of diversity (Neumann and Starlinger, 2001).

Apart from using these algorithms and concepts to relate forest stand structure to habitat functions, they can also be used to inform managers about the consequences of silvicultural activities. As continuous cover forestry in Britain will certainly give rise to more diverse forests, it might be sensible to define 'ecological' management objectives. The indices and functions discussed can be used to determine quantitative critical values which need to be exceeded to ensure a minimal amount of biodiversity. As these methods are relatively new, and researchers are currently only at the early stages of understanding spatial stand structure, further follow-up studies are needed in order to be able to relate quantities like the ones presented in this paper to habitat functions or to derive critical values for forest management. There are only a few studies comparing different indices over a wide range of ecological conditions (Neumann and Starlinger, 2001). As a consequence of new forest policies in Britain (e.g. Forestry Commission, 2001), increased research in irregular forest management now includes investigations on the dependence of tree growth on tree location. This will provide a number of forest stands with accurately mapped tree positions and their development over time, which can be used for studying this in greater detail.

The process of analysing and quantifying forest

structures can also be seen in reverse. Starting with a quantitative statement and quantifying suitable variables describing a desired forest structure, the spatial arrangement of trees can be reproduced on the computer to meet these objectives (Lewandowski and Gadow, 1997; Pretzsch, 1997). Using these approaches it is possible to develop a forest structure or habitat generator, which is able to simulate the habitats preferred by certain animal species (Wiegand, 1998). These computer simulations can then be translated into management guidelines or recommendations to be used by woodland managers and conservationists. The quoted papers present first steps in this direction and still need to be followed up in greater detail. However, this work undoubtedly offers great potential for conservation and ecological woodland management.

Over the last 20 years, new methods have been developed for describing complex forest structures and their possible changes resulting from growth, natural mortality and selective harvests. With these tools, scientific forest management will be in a much better position to establish, manage and maintain a great variety of complex structures without jeopardizing the traditional emphasis on sustainable harvests. One of the keys to achieving sustainable management in any type of forest is having spatial variables which can be assessed in the field at low cost and which can be used to describe, compare and evaluate forest structures and their modification by harvesting activities (Gadow and Pogoda, 2000). The sampling method 'structural group of four' (Figure 2), especially designed to assess structural information in forest stands, does not require time-consuming distance measurements and takes into account neighbourhood relationships. The Shannon index (Shannon and Weaver, 1949), the contagion index (Gadow *et al.*, 1998), the coefficient of segregation of Pielou (1977), the mingling index (Füldner, 1995) and the diameter differentiation index (Füldner, 1995) can be easily combined with the 'structural group of four' or traditional circular sample plots. When presented as a frequency distribution, the individual values of these neighbourhood-based parameters also indicate small-scale differences in the structure of forests. Algorithms such as the correlation functions and other indices, which require the surveying of coordinates, can be measured in research

plots where normally more detailed research work is carried out. However, Tomppo (1986) presented a method which allows the estimation of correlation functions from traditional circular sample plots used in forest inventories.

A disadvantage of the indices and variables is that they are based on small-scale data and thus can only give information on the variability within this range. The nearest neighbour of a reference tree, however, may be located some distance from the reference tree and thus this inter-tree distance can often encompass various ecological scales. If distances between trees are large, it is possible that there is only little or sometimes no interaction between them, while small distances between reference trees and nearest neighbours imply strong interactions. Measures based on the nearest-neighbour concept make no distinction between these two cases. They mix together the influence of ecological patterns on various scales. However, these indices are of particular value in terms of practicability when applied *in situ* in a forest, when only the nearest neighbour distances are measured or estimated and not tree locations. Usually short distances or counting in sampling units of small area are practical. If the tree locations are recorded, correlation functions, which avoid the mixing of different ecological scales, can provide more information on forest structure and tree interaction. In particular, the additional information on tree interaction radii can lead to a better understanding of competition effects in mixed stands.

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II

Evaluating Structural Indices
by Reversing Forest Structural
Analysis



Evaluating structural indices by reversing forest structural analysis

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Abstract

It is widely acknowledged that spatial forest structure is a driving factor behind growth processes and that forest growth, in return, influences the structural composition of woodlands. Also any impact on forests is primarily a change of spatial forest structure. In the last few decades an impressive number of structural indices have been developed to quantify spatial forest structure and it has also been suggested that they can be used as surrogate measures for quantifying biodiversity [Pommerening, A., 2002. Approaches to quantifying forest structures. *Forestry* 75, 305–324]. Of particular interest in this regard is the development of a family of individual tree neighbourhood-based indices, which are measures of small-scale variations in tree positions, species and dimensions, developed by Gadow and Hui [Gadow, K.v., Hui, G., 2002. Characterising forest spatial structure and diversity. In: Bjoerk, L. (Ed.), *Proceedings of the IUFRO International workshop 'Sustainable forestry in temperate regions'*, Lund, Sweden, pp 20–30]. Especially when expressed as frequency distributions these indices offer valuable information on spatial woodland structure. An important element of appraising the merits of such indices is a detailed evaluation of their performance for a specified purpose. One possible evaluation path is based on the idea that a successful quantification of spatial forest structure should allow the analysis to be reversed and enable the synthesis of forest structure from the indices derived. This idea is investigated here with a simulation model that uses the concept of cellular automata combined with further development of an approach by Lewandowski and Gadow [Lewandowski, A., Gadow, K.v., 1997. Ein heuristischer Ansatz zur Reproduktion von Waldbeständen (A heuristic method for reproducing forest stands). *Allg. Forst- u. J.-Zeitung* 168, 170–174]. The rules according to which the spatial pattern of tree positions “grows” in the stand matrix are deduced directly from the distributions of the structural indices of the input data. Different combinations of indices are used to assess and simulate the structure of four sample stands. The results show that simulations using species specific distributions of indices and a limit to the number of neighbours used for index calculation to three or four neighbours are most successful at reconstructing the original stand structure. The specific sequence of simultaneous distributions of structural indices was not significantly superior to the use of marginal distributions. Contrary to the suggestion in Hui et al. [Hui, G.Y., Albert, M., Gadow, K.v., 1998. Das Umgebungsmaß als Parameter zur Nachbildung von Bestandesstrukturen (The diameter dominance as a parameter for simulating forest structure). *Forstwiss. Centralbl.* 117, 258–266] no significant trend could be detected with regards to the use of the diameter dominance (formula 5) versus the diameter differentiation (formula 4).

The artificial synthesis of forest structure is of particular importance to conservationists who wish to develop forest landscapes to create a particular habitat pattern in order to support or re-introduce rare animal species. The topic is also important for modellers who require individual tree coordinates as input data for simulation runs or visualisations, which are hard to obtain in forest practice.

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Keywords: Spatial stand structure; Structural indices; Evaluation; Neighbourhood-based indices; Simulating spatial stand; Marginal and simultaneous distributions; Cellular automata

1. Introduction

A proper understanding of spatial forest structure is one of the keys to the sustainable management of mixed uneven-aged forests. The growth of trees is a reaction to their spatial context and conversely the growth processes influence the spatial forest structure and all biotic and abiotic, including human, impacts

modify spatial forest structure. A good understanding of these dependencies and their quantification is crucial for the management of woodlands for economic as well as environmental purposes.

The simulation or synthesis of spatial forest structure is an important aspect of environmental planning. For example, if there is a strong correlation between a particular spatial forest structure and the abundance of a particular animal species it should be possible to synthesize this structure elsewhere or at least to quantify the difference between the existing structure and an ideal structure in order to create new habitats for this

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Table 1

Overview of the most important individual tree indices developed by the Göttingen group since 1992, their formulae and the corresponding publications

No.	Index (reference)	Formula	Where
1	Uniform angle index (Gadow et al., 1998; Hui and Gadow, 2002)	$W_i = \frac{1}{n} \sum_{j=1}^n v_j$	$v_j = \begin{cases} 1, & \alpha_j < \alpha_0 \\ 0, & \text{otherwise} \end{cases}$
2	Species mingling (Füldner, 1995; Aguirre et al., 2003)	$M_i = \frac{1}{n} \sum_{j=1}^n v_j$	$v_j = \begin{cases} 1, & \text{species}_j \neq \text{species}_i \\ 0, & \text{otherwise} \end{cases}$
3	DBH differentiation (1) (Füldner, 1995; Pommerening, 1997, 2002)	$T_{ij} = 1 - \sum_{j=1}^n \frac{\min(\text{DBH}_i, \text{DBH}_j)}{\max(\text{DBH}_i, \text{DBH}_j)}$	j is the 1st neighbour tree
4	DBH differentiation (2) (Füldner, 1995; Gadow, 1999)	$T_i = 1 - \frac{1}{n} \sum_{j=1}^n \frac{\min(\text{DBH}_i, \text{DBH}_j)}{\max(\text{DBH}_i, \text{DBH}_j)}$	$j = 1 \dots n$ neighbour trees
5	DBH dominance (1) (Hui et al., 1998)	$U_i = \frac{1}{n} \sum_{j=1}^n v_j$	$v_j = \begin{cases} 1, & \text{DBH}_j \geq \text{DBH}_i \\ 0, & \text{otherwise} \end{cases}$
6	DBH dominance (2) (Gadow and Hui, 2002; Aguirre et al., 2003)	$U_i = \frac{1}{n} \sum_{j=1}^n v_j$	$v_j = \begin{cases} 1, & \text{DBH}_i \geq \text{DBH}_j \\ 0, & \text{otherwise} \end{cases}$

n is the number of neighbour trees. All index values are distributed between 0 and 1.

species. There is also an increasing demand for spatial tree data sets for certain growth models, sampling simulators and visualisation software (Pommerening, 2000). However, such data are very rare and there is often a need to simulate the necessary tree positions.

Over the past few decades many indices quantifying spatial forest structure have been developed (Pommerening, 2002) though very little work has been undertaken to formally evaluate these new developments (Neumann and Starlinger, 2001). One possible method of evaluation is to examine the correlation between the indices and more direct measures of biodiversity (e.g. the abundance of a particular rare species). An example of such a study using the indices discussed here is Spanuth (1998). However, another approach is to assess their ability to synthesize spatial forest structure. This would provide a closed circle starting with the analysis of spatial forest structure as a system and finishing with the reconstruction of this system. Such an evaluation can be understood as an application of systems analysis in that through the process of data reconstruction it is possible to learn what the advantages and the shortcomings of structural indices are. This is the approach to evaluating structural indices that this paper is concerned with.

2. Methods and data

2.1. Quantifying spatial forest structure

According to Gadow (1999) and Pommerening (2002) α -diversity can be subdivided into the diversity of tree positions (e.g. formula 1 in Table 1), tree species diversity (e.g. formula 2 in Table 1) and the diversity of tree dimensions (e.g. formulae 3–6 in Table 1). The diversity of tree positions reflects, at a small scale, whether the pattern of tree locations is regular, clumped (clustered), random or some combination of these. Tree species diversity is concerned with the spatial arrangement of species while diversity of tree dimensions involves the spatial arrangement of, for example, diameters or heights. The fundamental idea of this study is that a set of indices, which covers all three aspects of α -diversity should be sufficient to quantitatively describe and simulate spatial forest structure. Most structural indices attempt to examine and explain only the horizontal spatial structure and it has to be assumed that vertical aspects are allometrically related to horizontal aspects.

Since 1992 a research group at the Institute of Forest Management of the University of Göttingen (Germany) has developed a family of individual tree indices which are

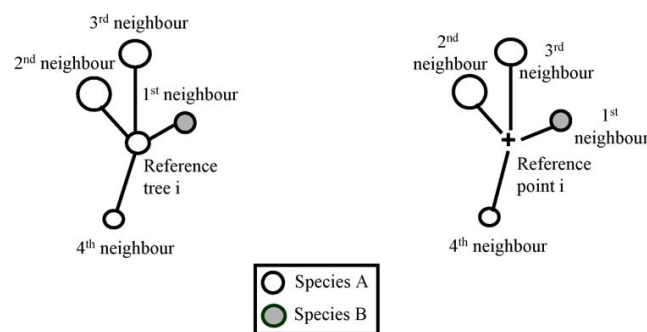


Fig. 1. Examples of a tree-based (left) and point-based (right) structural group involving in this case four neighbour trees. This structural group of reference tree/point and neighbouring trees is the computational unit for the indices listed in Table 1.

neighbourhood-based and which can account for small-scale aspects of the spatial distribution of tree attributes (Gadow and Hui, 2002; Fig. 1; Table 1). Their approach is similar to that of describing the structure of chemical molecules in that they focus on the quantification of the immediate neighbourhood of each tree or discrete points in a forest stand. An individual index value is then assigned to each tree, or to selected points, in the stand. Their algorithmic structure is very similar to that of distance dependent competition indices, which makes this family of indices particularly well suited to the simulation of spatial forest structure. Another advantage is that it is also comparatively easy to collect data for these indices during standard forest inventories (Fuldner, 1995; Pommerening, 1997). The neighbourhood based indices are also very flexible in that the number of neighbour trees, n , to be considered is not fixed, except in the case of the uniform angle index. For this index $n = 4$ and a standard angle $\alpha_0 = 72^\circ$ ($=80$ gon) appear to be ideal (Hui and Gadow, 2002). The indices can be tree- or point-based (Gadow and Hui, 2002; Fig. 1) and, though point-based indices have certain advantages when used with inventories, only tree-based indices have been studied in this investigation because their concept is closer to that of cellular automata.

Tree-based indices result in an index value for each tree of the stand. An important development by the Göttingen research group was to make the distributions of index values the fundamental unit of information rather than the arithmetic mean values as in most of the previous approaches to constructing structural indices. Accordingly the main focus of this study is on the distributions of indices. With uniform angle index (formula 1), species mingling (formula 2) and DBH dominance (formulae 5 and 6) there is only a limited number of values the index can take. For example with $n = 4$ neighbours there are $n + 1 = 5$ possible values: 0.00, 0.25, 0.50, 0.75 and 1.00. Using these scores all trees of the stand can be classified and stand structure summarised by the distribution of values. With the indices mentioned above the number of classes is a function of the number of neighbours. This is not the case with the diameter differentiation (formulae 3 and 4) where a different approach to constructing classes has been chosen (see e.g. Pommerening, 2002).

2.2. A model frame work for evaluating structural indices

A simplified overview of the evaluation process is illustrated in Fig. 2. The object of interest is the forest, which is first measured, analysed and then re-synthesised using the indices described above (Table 1). Most investigations finish after the analysis stage but for evaluating the merit of structural indices a synthesis can be helpful. This would answer the question of how much can the indices contribute to a synthesis of forest structure. The synthesis is driven by a novel point process model, the probability distribution of which depends on the structural indices as kinds of super-parameters.

There are various approaches described in the literature for simulating spatial forest structure, which use classical models of the theory of point processes with a very small number of

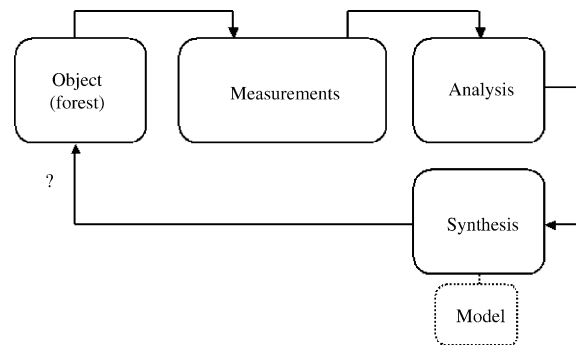


Fig. 2. Schematic overview of the evaluation process used in this study.

parameters; examples are Cox (Stoyan and Penttinen, 2000) or Gibbs processes (Tomppo, 1986). In their pure form most of these models are difficult to apply to forest structures with more than three species and their application is also limited when used with sample data. Therefore, more complicated point process models are necessary. Pretzsch (1997) developed an approach which is based on an empirical function, estimating the distance to the nearest neighbour, and a set of probability functions (a combination of an inhomogeneous Poisson process and a hard-core process) and Biber (1999) developed a method of simulating spatial forest structure in order to avoid edge effects in growth simulations. However, the latter two approaches are only applicable to forests that are located in the area where the data sets were taken for parameterization. Elaborating on Pretzsch's method, Pommerening (2000) generalised the empirical functions by deducing spatial information from forest inventories.

For an evaluation of the potential of the structural indices described in Table 1 for simulating forest structure it is necessary for the model to process the information generated by these indices only. The only approach close to the objectives of this study is that developed by Lewandowski and Gadow (1997) and parts of this are incorporated into the new method described below.

For this study a new class of point process models was developed using the concept of cellular automata. The method can be subdivided into three different phases (Fig. 3), with phase 1 modelling only the point pattern of tree positions, phase 2 modelling qualitative (tree species attributes) and phase 3 modelling quantitative marks (tree diameters). In phase 1 the spatial pattern of tree positions is simulated with the start configuration consisting of one tree being randomly placed on the stand grid. All the other $n - 1$ points are positioned subsequently using a set of rules of the type that drive cellular automata. More details about these techniques are provided in Section 2.3. The end product of phase 1 is a point pattern with no attributes assigned to the points other than coordinates, which are permanently fixed. The next two phases are an adaptation of the modelling approach published by Lewandowski and Gadow (1997). At the beginning of phase 2 the attribute 'tree species' is randomly assigned to all the points. This information can be obtained from a marginal species

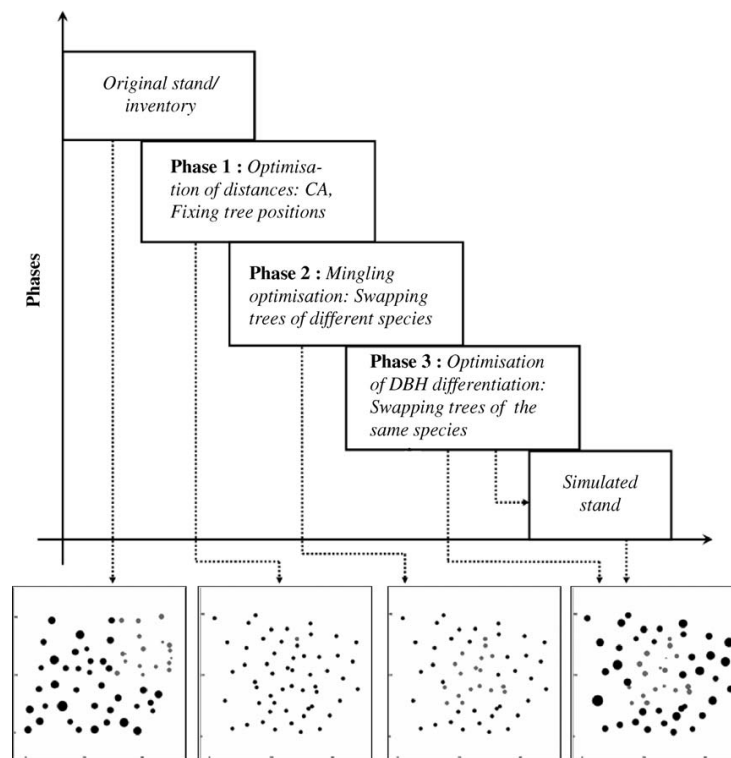


Fig. 3. The model sequence (CA: cellular automata, DBH: diameter at breast height).

distribution (no link with other distributions) or from a simultaneous distribution of species and uniform angle index. This random assignment is repeated 200 times and the configuration, which is closest to the empirical mingling distributions is carried forward to the optimisation part of this phase. During the optimisation of mingling the species attributes of all points with different attributes are swapped if this process of exchange results in a reduction of the deviation from the observed mingling distributions. Swapping trees of the same species would not change the mingling value. In each iteration all possible exchanges are investigated and only the one which results in the greatest reduction of the deviation from the observed mingling distributions is carried forward. When no further improvement is possible then phase 2 is finished. At the beginning of phase 3 diameters are assigned to all the points at random using diameter distributions with 1 cm-classes. The diameters can be assigned from marginal DBH distributions (non-species specific) or from simultaneous distributions of diameters and species. Phase 3 continues in a similar way to phase 2 but this time attributes (species and DBH) of tree locations are swapped only if they have the same species attribute. This restriction is necessary in order to retain the mingling configuration optimized in phase 2. Throughout this process all of the input tree attributes, coordinates, species and DBH are assigned to exactly the same number of trees as in the original stand or sample.

The last two phases of this modelling approach are similar to the idea of the simulated annealing reconstruction as described by Torquato (2002).

The model is not driven by pre-defined statistical functions, which would only be valid in a restricted geographic area. All the rules used in this approach are deduced from the input data which can come from full enumerations as well as inventories. It is, therefore, possible to collect the information required for calculating the structural indices as part of a standard forest inventory. Gadow et al. (2003) developed a method of estimating intertree distance distributions from the stand density and the mean of the uniform index value. This approach can be used to produce the necessary input data for this model if it is not feasible to sample tree distances in the field. However, the quality of the output does depend on the quality of the input and, the more accurate and unbiased the inventory is, the better the simulation result. In this study fully enumerated stand data were used in order to assess simulation errors. The structure and mechanism of the model is deliberately kept as simple as possible in order to retain a transparent process.

The data analysis methods, the model and most of the evaluation algorithms were implemented in an integrated software package called "Crancod" using structured programming techniques in the Borland Delphi environment which is based on the Pascal language. "Crancod" is currently being

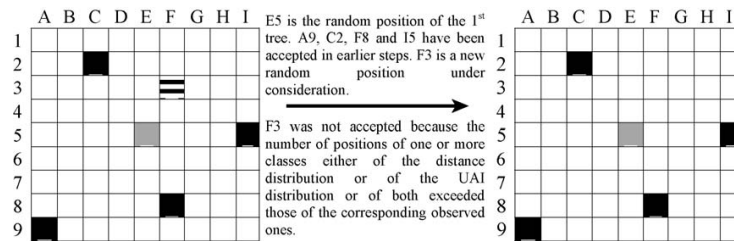


Fig. 4. An example of the working principle of the cellular automata used in this modelling approach.

revised and reorganised in an object-oriented way by using the language Java and the programming environment NetBeans.

2.3. Extended cellular automata

A cellular automaton (CA) is a system composed of cells, which are connected in regular structures. Each cell communicates its present state to its nearest neighbours and computes future states from its own, current, state and that of its neighbours. The concept of cellular automata was initially developed in the early 1950s by von Neumann and Ulam, and is an artificial life approach to simulation modelling. Since then the idea has been further developed by Conway in the 1970s, Wolfram in the 1980s (Wolfram, 2002) and others. CA have been successfully used for modelling biological systems for almost two decades (Ermentrout and Edelstein-Keshet, 1993; Ganguly et al., 2003) and can help to explain them especially where differential equations become too complex or do not

In a similar way the indices listed in Table 1 are based on small-scale neighbourhood relationships and therefore an application of cellular automata seems to be very appropriate as an approach to simulate tree positions.

The rules of the application of cellular automata of this model approach are derived from the intertree distance distribution to the first neighbour tree and the distribution of the uniform angle index. These two distributions were selected because Gadow et al. (2003) discovered a strong correlation between the uniform angle index (UAI, formula 1) and the distance to the first neighbour tree. The intertree distance distribution to the first neighbour tree is computed in a similar way to the diameter distribution by defining 50 cm distance classes¹. The cells of the simulated forest stand are examined and updated in random order as to whether they can accommodate a tree according to the distance and the UAI distributions. This simulation process is illustrated in Fig. 4.

The rule can be expressed in the following way (obs: observed, sim: simulated):

IF $number_{sim}[class[distance\ 1]] + 1 \leq number_{obs}[class[distance\ 1]]$ **AND**
 $number_{sim}[class[UAI]] + 1 \leq number_{obs}[class[UAI]]$ **THEN**
accept square mid-point as tree position
OTHERWISE
reject square mid-point as tree position

return a definite solution. There have been few applications of CA in forestry and most of these have dealt with the subjects of forest fires (e.g. Li and Magill, 2001; Karafyllidis and Thanailakis, 1997), land use and remote sensing (e.g. Soares-Filho et al., 2002).

Despite their simple construction cellular automata are able to simulate very complex behaviour and their application is very useful when studying systems consisting of many similar or identical units whose behaviour is influenced by local interactions. Characteristic of cellular automata is their ability for self-organisation. The computation laws for the new state are given by localized neighbourhood rules of occupancy (Zhang et al., 2004) and the behaviour of a CA can be specified by giving the structure of the interconnection grid, the neighbourhood of each cell, the boundary conditions, the rules for each cell in the grid and the initial conditions (Wolfram, 2002).

The distance distribution ensures that the simulated intertree distances do not fall below observed minimum distances and the UAI distribution ensures the correct geometric arrangement of tree positions. Unlike other approaches that simulate tree positions (e.g. Pretzsch, 1997) this method does not attempt to accommodate an array of trees with given attributes on the area of a forest stand. The simulation process starts with an array of available coordinates and subsequently examines whether they can host a tree or not.

The particular type of cellular automata used can be referred to as extended cellular automata. They deviate from standard CA in that the state of the two dimensional grid cells is updated

¹ The distance D1 between each tree of the forest stand and its first neighbour tree is allocated to a class of the distance distribution following the principle: $0\text{ m} \leq D1 < 0.5\text{ m}$, $0.5\text{ m} \leq D1 < 1\text{ m}$, $1\text{ m} \leq D1 < 1.5\text{ m}$, etc.

sequentially or asynchronously (Adachi et al., 2004). This basically means that every change in the state of a cell has an immediate effect on the state of all the other cells in the same time step. The simulation of the spatial tree pattern of a stand can therefore be completed in a single time step. The cells are examined and updated in random order using uniform random numbers either with or without replacement. Also the immediately adjoining cells do not define the neighbourhood as in earlier cellular automata applications (Wolfram, 2002). Rather the cells in which the tree neighbours are actually positioned (see Fig. 4) are used, and as these can vary the neighbourhood can be described as dynamic. The model and the algorithm for calculating the structural indices uses an edge-correction, which compensates for biased estimations caused by off-plot trees. In two-dimensional cellular automata the use of a so-called torus or toroidal wrapping (Diggle, 2003) is very common, in which the edges of the finite grid are pasted together resulting in a three-dimensional solid (Hegselmann and Flache, 1998; Diggle, 2003; Ripley, 1981). In forestry related modelling this method is also frequently referred to as “translation” (Monserud and Ek, 1974).

In this study the width of the interconnection grid in which the cellular automata operate is 5 cm.

2.4. Structural indices used in the different phases

The distributions of structural indices can be depicted either as marginal (univariate) or simultaneous (bivariate) and the model can work with both. Figs. 5 and 6 illustrate these different types of distributions.

The left diagram in Fig. 5 shows a marginal distribution of mingling (formula 2) regardless of species or any other index

while the right hand diagram offers us species-specific mingling distributions. The first two diagrams in Fig. 6 show marginal distributions of mingling (formula 2) and dominance (formula 5). The third diagram gives an impression of the simultaneous or bivariate distribution of mingling and dominance incorporating the information of the first two diagrams of Fig. 6.

Additionally, there is the option to work regardless of species or species specifically which can lead to simultaneous distributions of indices and species (i.e. a combination of the right hand diagrams in Figs. 5 and 6).

When developing rules for the structural synthesis the procedure varies depending on whether marginal or simultaneous distributions of the structural indices are used. For marginal distributions the procedure is as follows:

- Phase 1: Distance and UAI distributions (formula 1) sequentially
- Phase 2: Mingling distribution (formula 2)
- Phase 3: DBH differentiation (formula 3) or dominance distribution (formula 5)

In the case of simultaneous distributions of indices the model is organised in such a way that one index is always retained from one phase to the next to provide continuity throughout the simulation:

- Phase 1: Distance and UAI distributions simultaneously
- Phase 2: UAI and mingling distribution
- Phase 3: Mingling and DBH differentiation or dominance distribution

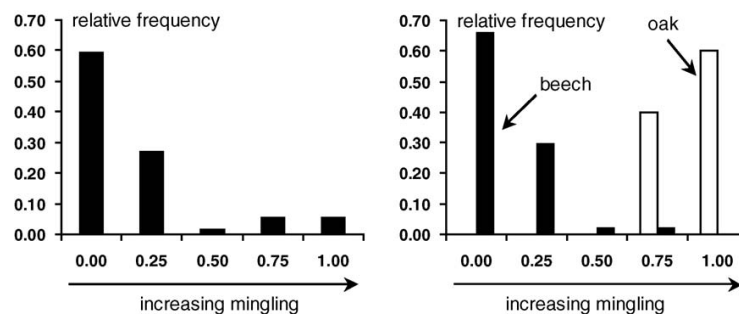


Fig. 5. The marginal mingling distribution (formula 2, left) and the simultaneous or bivariate distribution of mingling and species (right) for the German sample stand.

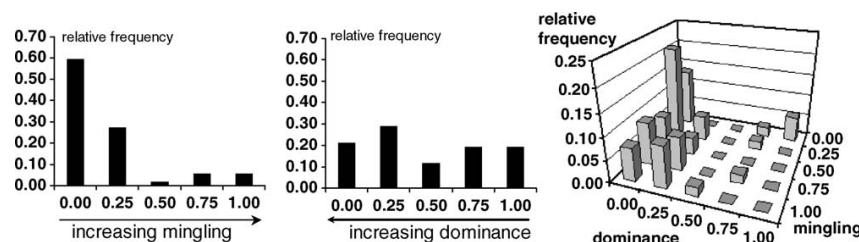


Fig. 6. The separate marginal distributions of mingling (formula 2) and dominance (formula 5, left) and the simultaneous or bivariate distribution of mingling and dominance (right) for the German sample stand.

By doing this it is hoped to retain the correlation between the structural indices throughout the simulation process.

The deviation between the observed and the simulated distributions of structural indices is measured with formula (7) using the absolute number of points/trees. The number of points/trees in each class of a distribution should be smaller or equal to the number of the corresponding observed one.

$$\text{Dev} = \sum_{i=1}^n |s_i - s'_i| \quad (7)$$

where n : number of classes, s_i : number of trees in class i of the observed distribution, s'_i : number of trees in class i of the simulated distribution.

The model is designed in such a way that each simulation run yields the exact basal area, mean squared diameter, stems per hectare and DBH distribution of the original forest.

2.5. Experimental plan and evaluation

The model is a tool for a multidimensional evaluation of structural indices and could use any number of factors to do this. However, because of the exponential increase of simulation runs required with increasing numbers of factors the experiment was restricted to five main factor groups:

1. The number of neighbouring trees: 3, 4, 5 and a mixed arrangement (four neighbours for the UAI distribution, three neighbours for the mingling distribution and three neighbours for the distributions of diameter differentiation/dominance).
2. DBH differentiation (formula 3) versus DBH dominance (formula 5) in phase 3.
3. Ordering of cells with/without replacement.
4. Marginal versus simultaneous distributions of structural indices.
5. Non-/species specific distributions of structural indices (phases 2 and 3 only).

This results in 64 possible different combinations of factors, each of which was simulated with five replications, giving a total of 320 model runs.

Three different groups of indices/functions have been used to evaluate the quality of the simulations. The first two groups comprise the indices which have been part of the simulation rules (formulae 1–3, 5, distance distribution) plus the additional index T (formula 4) and the unused index of the pair T1 (formula 3)/U (formula 5). The difference between group 1 and group 2 is that indices in the latter are species specific and in the former they are not. For group 2 the structural indices of the less abundant of the two main species was always selected. In the third group correlation functions were used for evaluation, i.e. pair-, mark correlation functions and mark connection functions (Stoyan and Penttinen, 2000; Pommerening, 2002). The pair correlation function only investigates the point pattern regardless of any other tree attributes while the mark correlation

function in this application also takes the tree diameters into account. The mark connection function calculates the probabilities that the tree species of a forest are associated with each other at particular intertree distances. In this study the intertree distances up to 10 m with a step width of 0.5 m were investigated. These continuous functions were used rather than indices like Clark and Evans (1954) or Pielou (1977) because they are more precise and deliver more information about the long-range structure of the simulated forests (Stoyan and Penttinen, 2000).

The main criterion to quantify the disparity between observed and simulated distributions in this application is the absolute discrepancy (AD; Gregorius, 1974; Pommerening, 1997; Aguirre et al., 2003).

$$\text{AD} = \frac{1}{2} \sum_{i=1}^n |s_i - s'_i| \quad \text{AD} \in [0, 1] \quad (8)$$

where n : number of classes, s_i : relative frequency of trees in class i of the observed distribution, s'_i : relative frequency of trees in class i of the simulated distribution.

It is defined as the relative proportion of individuals, which must be exchanged between the classes if the simulated distribution is to be transformed into the observed. Correspondingly, $1 - \text{AD}$ is the proportion common to both distributions. A value of $\text{AD} = 1$ means that both distributions have no common class, whereas $\text{AD} = 0$ signifies that the distributions are absolutely identical. AD is applied to all structural indices and to the mark connection function. The classes i are those of the indices as described in Section 2.1. As the mark connection function calculates probabilities for each species combination of a forest stand the probabilities at each discrete distance add up to 1 and can be interpreted as a class. The difference between observed and simulated pair and mark correlation functions are calculated as absolute differences similar to formula (7) because formula (8) is not applicable.

Because of the comparatively large number of individual simulations (320) and the evaluation criteria involved the process of identifying the best combination of factors had to be automated and simplified. For each of the three groups of evaluation criteria as described above arithmetic mean values were produced. In the case of the correlation functions this first required the calculation of an arithmetic mean over all intertree distances (from 0.5 to 10 m) for each individual simulation. Finally a total arithmetic mean was calculated by averaging the results of the three evaluation groups. The combination of factors with the lowest final mean value was identified as the best option. Additionally, the five factors as listed above were investigated individually.

2.6. Sample forest stand data

Four sample forest stands were selected to evaluate the performance of the structural indices. In order to underpin the universal applicability of the model the sample data sets were

Table 2

Basic stand data of the four sample forest stands used in this study (B/A: basal area, Dg: mean squared diameter, Clark & Evans: aggregation index of Clark and Evans (1954), Pielou: Pielou's (1977) coefficient of segregation)

Species	B/A (m ²)	Trees (per ha)	Dg (cm)
Germany (50 m × 50 m, 52 tree; oak 151, beech 65 years)			
Beech	21.4	752	19.0
Oak	21.3	80	58.3
Total	42.7	832	25.6
Clark & Evans	1.123	Pielou	−0.078
Greece (40 m × 30 m, 139 trees, fir 102, beech 166 years)			
Beech	34.2	758	24.0
Fir	8.3	400	16.2
Total	42.5	1158	21.6
Clark & Evans	0.798	Pielou	0.045
	B/A (m ²)	Trees (per ha)	D (cm)
Wales 1 (50 m × 40 m, 56 trees, both species 51 years)			
Sitka spruce	22.7	190	39.0
Lodgepole pine	3.9	90	23.5
–	–	–	–
Total	26.6	280	34.8
Clark & Evans	1.370	Pielou	0.836
Wales 2 (22 m × 12 m, 67 tree, all species 33 years)			
Sitka spruce	18.4	833	16.8
Birch	6.0	1288	7.7
Goat willow	1.5	417	6.7
Total	25.9	2538	11.4
Clark & Evans	0.989	Pielou	0.071

taken from very different European regions involving different species, environmental conditions, intervention history, etc.

Basic stand data including plot size, age, yield data and total number of trees are given in Table 2.

The stands are a mixed oak-beech stand (*Quercus robur* L. and *Fagus sylvatica* L.) from West Germany (Boeselager estate in the Sauerland region; Bölsing, 1996), a mixed beech-fir stand (*Fagus sylvatica* L./*Fagus moesiaca* Czeczott and *Abies borisii-regis* Mattf.) from North Greece (Northern Pindos mountains; Beinsen, 1996) and two stands from North Wales (mixed Sitka spruce [*Picea sitchensis* (BONG.) CARR.]–lodgepole pine [*Pinus contorta* DOUGL. ex LOUD.] from Cloacaenog forest and a mixed Sitka spruce–birch [*Betula pendula* ROTH]–goat willow [*Salix caprea* L.] site from Coed y Brenin).

The data cover a wide range of age and tree dimensions. According to the aggregation index of Clark and Evans (1954) the pattern of tree positions ranges from slightly clumped in

“Greece” to quite regular in “Wales 1”. “Wales 2” exhibits an almost random arrangement of tree positions. The pattern of tree species aggregation/segregation according to Pielou's (1977) coefficient of segregation ranges from close to randomness of species distribution in “Germany”, “Greece” and “Wales 2” to a segregation of species in “Wales 1”.

3. Results

Tables 3 and 4 show the combination of factors leading to the best and worst results from the four sample stands. The five factors are in the same order as listed in Section 2.5. The value in the column “deviation” is the mean of the absolute discrepancies (formula 8) of all evaluated indices/functions with the exception of the pair and the mark correlation functions where as stated earlier the absolute deviations have been used.

Although there is no overall consistency in the combinations that lead to the best or worst results it is possible to identify some trends on which hypotheses can be based. The worst results are generally linked to combinations where more than three or four neighbours are used for the calculation of structural indices. There does not seem to be any conclusive evidence as to whether the DBH differentiation (formula 1) or the DBH dominance (formula 5) is associated with better or worse results. The majority of best results are associated with an ordering of cells with replacement while the majority of worst results are associated with an ordering of cells without replacement. There is some uncertainty regarding the use of marginal or simultaneous distributions although all worst results are associated with simultaneous distributions. A clearer idea is conveyed with regard to non-species/species specific simulations. All the best options have been simulated with species specific distributions in phases 2 and 3 and all the worst ones with non-species specific ones apart from “Greece”, which seems to be an anomaly as will be demonstrated later. The differences between lowest and largest deviations are comparatively small which suggests that the indices show a similar simulation behaviour.

To further test the best choice of factors within groups 2–5, these were investigated using a paired *t*-test. For this purpose all the 64 results of the individual simulations were grouped in eight blocks with specific combinations of neighbours and DBH differentiation/dominance. However, for the investigation of DBH differentiation/dominance this was impractical and four blocks were identified. For each test all factors were kept constant except for the one under investigation. The

Table 3

The combination of factors leading to the best results (lowest deviation) with regard to the four sample stands

Sample stand	Neighbour combination	Differentiation or dominance	With or without replacement	Marginal or simultaneous	Non-species or species specific	Deviation
Germany	4 + 3 + 3	Dominance	Without	Simultaneous	Specific	0.1282
Greece	4 + 3 + 3	Differentiation	With	Simultaneous	Specific	0.0816
Wales 1	4 + 3 + 3	Differentiation	With	Marginal	Specific	0.1006
Wales 2	3 + 3 + 3	Differentiation	With	Marginal	Specific	0.0906

Table 4

The combination of factors leading to the worst results (highest deviation) with regard to the four sample stands

Sample stand	Neighbour combination	Differentiation or dominance	With or without replacement	Marginal or simultaneous	Non-species or species specific	Deviation
Germany	5 + 5 + 5	Differentiation	Without	Simultaneous	Non-specific	0.3026
Greece	3 + 3 + 3	Dominance	Without	Simultaneous	Specific	0.3059
Wales 1	5 + 5 + 5	Differentiation	Without	Simultaneous	Non-specific	0.2139
Wales 2	4 + 4 + 4	Dominance	With	Simultaneous	Non-specific	0.1908

hypothesized difference between the deviation means of alternative factors was set to zero. The results of the test show that there are only a few cases of significant difference between the use of factor options. Obviously there can be no definite statement as to whether the use of formula 3 is superior to the use of formula 5 or vice versa. The same is true for the question of whether an ordering of cells with replacement is advantageous in comparison with an ordering of cells without replacement. With regard to the superiority of marginal or simultaneous distributions of structural indices there is only one significant case in “Wales 1” where the use of simultaneous distributions is more advantageous. This is the stand with a segregation of the two species, which are correlated with very different diameter ranges. With the majority of sample stands the use of species-specific distributions of structural indices in phases 2 and 3 is significantly superior to the use of non-species specific ones. As detected in Tables 4 and 5 there is a somewhat different simulation behaviour when using the Greek sample data, with results generally not following the pattern of the other three.

In order to investigate the performance of different combinations of neighbours a single factor ANOVA test was carried out and the results are shown in Table 6. It is apparent that in only two cases are at least two of the four neighbour combinations significantly different at the 0.05 level.

A consequential pairwise analysis of the significant cases in Table 6 using the method described in Bortz (1999, p. 252f) reveals that with “Wales 1” simulations using three neighbours (3 + 3 + 3) are superior to those using four neighbours (4 + 4 + 4). The analysis also shows that the mixed scenario (4 + 3 + 3) is significantly superior to the use of five (5 + 5 + 5) and four neighbours (4 + 4 + 4). With the “Greece” data set the relation between different sets of numbers of neighbours is to the contrary. Using four neighbours (4 + 4 + 4) is superior to using three neighbours (3 + 3 + 3), the mixed scenario (4 + 3 + 3) and five neighbours (5 + 5 + 5). Other comparisons are not significant.

Table 5

Results of the paired *t*-test for the factors 2–5

Sample stand	Differentiation vs. dominance $t_{(3;0.05)} = 3.18$	With vs without replacement $t_{(7;0.05)} = 2.36$	Marginal vs. simultaneous $t_{(7;0.05)} = 2.36$	Non-species vs. species specific $t_{(7;0.05)} = 2.36$
Germany	1.6115	–1.8340	–1.3870	8.7286 ^a
Greece	1.8748	–0.4574	–1.0936	0.8779
Wales 1	–1.7160	1.4215	–4.8292 ^a	3.9985 ^a
Wales 2	–1.1990	0.4242	–1.7377	12.6984 ^a

^a Deviation means are significantly different at the 0.05 level [2-tailed]).

To illustrate the results a visualisation of the best and the worst individual simulation results for “Wales 1” (see Tables 3 and 4) in relation to the original stand structure is shown in Fig. 7. In both cases a visual impression suggests that the species segregation has been simulated reasonably well. As the model uses an edge correction all the lodgepole pines in the top right corner of the original stand are surrounded by Sitka spruces in the two simulations. However, in the worst simulation lodgepole pine trees are given much larger diameters than in the original stand. This is a typical effect of non-species specific distributions of structural indices in phases 2 and 3 because information regarding species and

Table 6

Results of the single factor ANOVA test for the factor 1

Source of variation	SS	d.f.	MS	<i>F</i>	<i>F</i> _{crit}
Germany					
Between groups	0.0134	3	0.0045	2.75	2.95
Within groups	0.0455	28	0.0016		
Total	0.0589	31			
Greece					
Between groups	0.0770	3	0.0257	13.68 ^a	2.95
Within groups	0.0525	28	0.0019		
Total	0.1296	31			
Wales 1					
Between groups	0.0067	3	0.0022	5.32 ^a	2.95
Within groups	0.0117	28	0.0004		
Total	0.0183	31			
Wales 2					
Between groups	0.0015	3	0.0005	0.63	2.95
Within groups	0.0220	28	0.0008		
Total	0.0235	31			

SS: sum of squares, d.f.: degrees of freedom, MS: mean square, *F*: empirical *F*-value, *F*_{crit}: critical *F*-value.

^a Deviation means are significantly different at the 0.05 level [two-tailed].

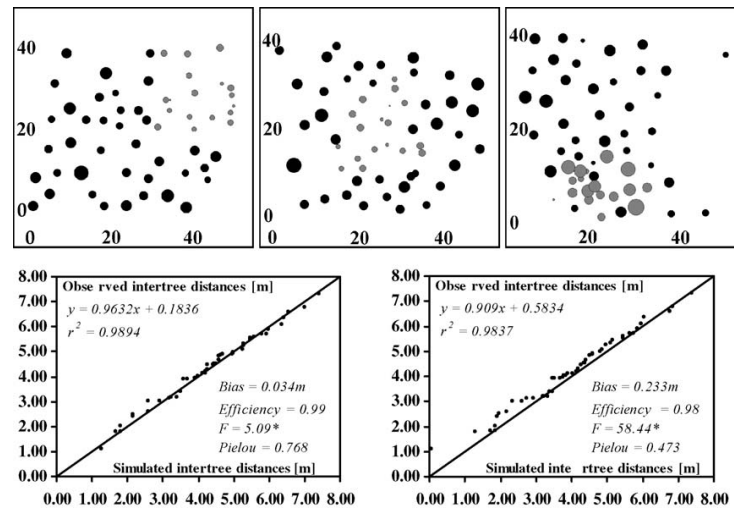


Fig. 7. Tree location maps and evaluation graphs of the original data set (top left), the first replication of the optimal simulation (top centre) and of the worst simulation (top right) for the sample stand “Wales 1” (Black: Sitka spruce, Grey: Lodgepole pine). The graphs showing the statistics bias, efficiency and simultaneous F -test were calculated according to Sterba et al. (2001) and refer to the best and the worst simulation.

dimension is separated. The statistical figures and the graphs suggest that the point patterns of all three data sets are not so different which is the result of the cellular automata approach. However, the bias of the worst simulation is much larger than that of the best one. While the Pielou (1977) value of the best simulation is reasonably close to that of the original stand (Table 1), the cluster effect in the worst simulation is exaggerated.

4. Discussion and conclusions

This multidimensional approach to evaluating structural indices has highlighted a number of valuable aspects as to how structural indices work and the degree to which they are able to quantify spatial forest structure. The simulation results support Wolfram's (2002) argument that simple rules such as distributions of spatial indices can lead to quite complex point patterns like the spatial arrangement of two-dimensional tree positions.

Although there do not seem to be any overall trends in the five factors investigated there are a few aspects that deserve further study in the future. There appears to be an advantage in using species-specific distributions especially in stands with segregated species. The use of more than five neighbours and in some cases also four neighbours for calculating and simulating the three aspects of α -diversity are not optimal. This finding is helpful with regard to sampling and quantifying the indices of Table 1 as part of forest inventories based on circular sample plots where a requirement for a larger number of neighbours can lead to a significant bias due to edge effects (Pommerening and Gadow, 2000). Hui et al. (1998) state that the use of the diameter dominance is superior to the use of the diameter differentiation when simulating spatial forest structure but this could not be confirmed in this study. The fact that there is no trend concerning the use of marginal versus simultaneous

distributions could be explained by a lack of understanding of the correlations between structural indices. These correlations seem to be specific to different forest stands or at least to groups of different forest stands. Instead of utilising a rigid chain of simultaneous distributions as in this paper (see Section 2.4) it might be better to identify the specific correlations for each stand and then develop the chain of simultaneous distributions accordingly. The comparatively good performance of marginal distributions can also be explained by the fact that these have fewer classes that are not so specific.

The point-based versions of the indices (see right hand diagram in Fig. 1; Gadow and Hui, 2002) used in this study could, theoretically, also be used with the cellular automata approach although the tree-based concept is closer to the original idea of CA. The total population of point-based indices, however, is based on all possible points within the stand boundaries, which is infinite. The total population of tree-based indices consists only of those points within the stand boundaries, which are tree positions. The rule of the CA would therefore need to be adapted. With tree-based indices each newly accepted point represents a reference tree and potentially a neighbour of other reference trees. With point-based indices each newly accepted point represents a neighbour tree of points only. An application of cellular automata to simulate the pattern of tree positions could be achieved by defining the mid points of all the cells as the total population of points to be examined in the analysis and the synthesis.

The only other indices, which are similar to those used in this study are distance dependent competition indices (Biging and Dobbertin, 1992). Although their main concern is to quantify competition pressure for each tree of a forest stand it might be possible to test the merits of some competition indices, or to merge some aspects of their concepts, with those of the structural indices of Table 1. The similarity between structural and competition indices is particularly obvious with the

dominance index (formulae 5 and 6 in Table 1). It would also be interesting to explore whether the inclusion of a vertical dimension could improve the simulation of the two-dimensional patterns of tree positions.

As mentioned in Section 2.2 the use of cellular automata is only one possible approach to simulating spatial patterns and can be understood as a special case of modelling spatial correlation. In fact, this method is not so different from the idea of Gibbs processes.

As a positive by-product of this evaluation study it has proved possible to optimise a model for simulating spatial forest structure which is not based on pre-defined statistical functions and can be readily applied anywhere in the world without the need for local adaptation. Although this approach also uses quite a few empirical parameters, i.e. the structural indices, these are individually deduced from the input data for each simulation. Given the demand for spatially explicit individual tree data such a model could become an important tool for other investigations that need such data and the modelling process itself will help to develop a better understanding of spatial forest structure. Similar index approaches at landscape level could allow the simulation of larger entities.

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III

Edge-Correction Needs in
Estimating Indices of
Spatial Forest Structure

Edge-correction needs in estimating indices of spatial forest structure

Arne Pommerening and Dietrich Stoyan

Abstract: Indices quantifying spatial forest structure are frequently used to monitor spatial aspects of tree attributes including biodiversity in research plots of limited size. The treatment of edge trees, which are close to the plot boundaries, can affect the estimation of such indices that include neighbour effects, since some of their neighbours are likely to fall outside the plot. This paper investigates whether and under what circumstances edge-correction methods are necessary and evaluates the performance of six different approaches: no edge correction, translation, reflection, buffer zone, and two new nearest-neighbour methods. The performance of edge-correction methods depends strongly on the algorithmic structure of the indices and the spatial pattern of tree positions involved. Some edge-correction methods introduce more error than ignoring edge bias altogether. For indices accounting for the diversity of tree positions and especially for those computing angles, translation or buffer zone methods reduce the estimation error regardless of the sample size. The use of the reflection method is associated with large bias values. One of the new nearest-neighbour edge-correction methods proves to be capable of reducing the bias considerably. The results confirm the need for sufficiently large monitoring plots to avoid bias from edge effects. Where this is impossible, neighbours beyond the plot boundary need to be included in the survey, thus providing unbiased estimates but at the cost of extra measurements. Sensitivity analysis is required for newly introduced indices prior to their first application.

Résumé : Des indices qui permettent de quantifier la structure spatiale de la forêt sont fréquemment utilisés pour faire le suivi de l'aspect spatial des attributs des arbres, incluant la biodiversité dans les placettes expérimentales à superficie restreinte. Le traitement des arbres situés en bordure de la placette peut affecter l'estimation de ces indices puisque certains de leurs voisins peuvent vraisemblablement se retrouver à l'extérieur de la placette. Cet article examine si des méthodes de correction des effets de bordure sont nécessaires et dans quelles circonstances. Les auteurs analysent également la performance de six approches différentes : aucune correction de l'effet de bordure, translation, réflexion, zone tampon et deux nouvelles méthodes du plus proche voisin. La performance des méthodes de correction est fortement dépendante de la structure de l'algorithme des indices et de la configuration spatiale de la position des arbres concernés. Certaines méthodes de correction introduisent une erreur plus importante que le fait de simplement ignorer les biais dus à l'effet de bordure. Pour le groupe d'indices qui tiennent compte de la diversité dans la position des arbres et spécialement pour ceux qui servent au calcul des angles, les méthodes de translation et de la zone tampon réduisent l'erreur d'estimation sans égard à la taille de l'échantillon. La méthode par réflexion génère des biais élevés. L'une des deux nouvelles méthodes du plus proche voisin a pu considérablement réduire le biais. Les résultats confirment la nécessité que les placettes aient une superficie suffisamment grande pour éviter les biais dus aux effets de bordure. Lorsque cela est possible, les voisins situés à l'extérieur des limites de la placette devraient être inclus dans l'inventaire pour obtenir des estimations non biaisées au prix, cependant, de mesures supplémentaires. Il est nécessaire de procéder à l'analyse de sensibilité des nouveaux indices avant de les utiliser pour la première fois.

[Traduit par la Rédaction]

Introduction

The quantitative characterization of spatial structure is an important part of the study of ecological processes in forests (Upton and Fingleton 1985, 1989; Pretzsch 1998; Barot and Gignoux 2003; Bauer et al. 2004). For this purpose many statistical summary characteristics have been developed that characterize forest structure and aspects of biodiversity using numerical values or functions. Examples of such functions are Ripley's K function, $K(r)$, or the nearest-neighbour dis-

tance distribution function, $G(r)$ (Stoyan and Stoyan 1994), while typical examples of numerical characteristics are structural indices such as the aggregation index (Clark and Evans 1954) and the species segregation index (Pielou 1977).

Usually it is assumed that the two-dimensional arrangement of tree positions in a forest can be described by a point process (Penttinen et al. 1992). In a point process each individual tree, i , can be understood as a point or event defined by its position in the monitoring plot or observation window W using two-dimensional Cartesian coordinates $\{x_i, y_i\}$. The

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observation window is usually a rectangular or circular area in the forest that is assumed to yield representative information on the forest as a whole. Marks $\{m_{il}, \dots, m_{in}\}$ are often attached to the points, which are sets of qualitative (e.g., species) and (or) quantitative (e.g., diameter at breast height, DBH) values. Frequently, these marks are constructed or computed, for example, competition indices describing the influence of neighbour trees on the reference tree. These indices are usually calculated by means of mathematical formulae using intertree distances and original tree marks such as DBH or species.

In statistical analysis it is assumed that the point process studied is spatially homogeneous (or, in terms of point process theory, stationary). Under the spatial homogeneity assumption it is meaningful to characterize the tree point pattern of the underlying process, in the research plot, using global characteristics. The simplest of these are perhaps mean number of stems per hectare (SPH), mean basal area, or mean volume per hectare. In ecological research other characteristics are used, such as the mean values of computed individual tree characteristics, for example, mean competition indices (Barot and Gignoux 2003; Bauer et al. 2004). Also, $K(r)$ and $G(r)$ are characteristics that have meaning only in the homogeneous case.

When calculating spatial characteristics, edge effects are likely to play an important role. Ignoring these effects is usually believed to result in biased statistical estimations. This is because some of the immediate neighbours of trees near the boundary of W are outside W and therefore not recorded; however, these trees are needed to calculate the correct values of statistical quantities such as competition or nearest neighbour indices. While edge-bias issues of correlation functions and competition indices have been frequently discussed in the statistical literature (e.g., Monserud and Ek 1974; Martin et al. 1977; Stoyan and Stoyan 1994; Stoyan et al. 1995; Radtke and Burkhart 1998; Gignoux et al. 1999), comparatively little attention has been paid to structural indices (Donnelly 1978). As the algorithms of nearest-neighbour structural indices are very similar to those of competition indices, it is likely that careless treatment of edge trees may also lead to a bias in the estimation of such indices especially when the window of observation is small and (or) tree numbers are low. Such biased estimation of structural indices can lead to incorrect conclusions and interpretations of the spatial biodiversity of a woodland. It should be noted that the situation is different with statistical tests, for example, concerning the hypothesis of complete spatial randomness, as in Gignoux et al. (1999). These authors compared summary characteristics of the Poisson process with those from its samples and found that tests without edge correction have a higher power for small sample sizes than those with correction.

The attention paid to edge effects has varied, and the early publications of Clark and Evans (1954) and Pielou (1977) simply ignored the problem. Later, Donnelly (1978) investigated the effect of edge bias on the aggregation index of Clark and Evans (1954) and developed an empirical edge correction specifically for this index. Since then various forms of edge correction have been developed. A very simple one, used by many researchers (e.g., Neumann and Starlinger 2001; Aguirre et al. 2003), is the use of a bound-

ary strip or "buffer" zone, of width d , inside the monitoring plot W . This method is also referred to as the border or guard method (Ripley 1981). In the statistical analysis only the trees in a reduced window of observation are used as reference trees, namely those that have a distance larger than d from the window boundary. The width of the buffer zone (e.g., $d = 5$ m) should be large enough that all relevant interaction between trees in the reduced window is accounted for within W . Determining the optimal width of the buffer zone is difficult; if it is too small residual edge effects will remain; if it is too large valuable data are discarded unnecessarily (Diggle 2003).

Other edge-correction methods can be considered as being only speculative (Pretzsch 2002; Diggle 2003), as they extrapolate the spatial structure from within W to an infinite plane and join parts of the point pattern that do not occur so close together in nature. Two examples are translation (also referred to as torus or periodic boundary conditions; Diggle 2003) and reflection (e.g., Radtke and Burkhart 1998; see Fig. 1).

It is clear that these two methods do not easily work with circular sample plots or with plots of irregular shape (Windhager 1997). Furthermore, translation and reflection are believed to result in unrealistic periodicities of the spatial point pattern and therefore neighbourhood situations that do not naturally occur (Pretzsch 2002; Diggle 2003).

This paper explores the need for edge correction of five spatial structural indices involving nearest neighbours. The performance of six edge-correction methods for these indices is investigated through simulations. Included in the study are two recently developed nearest-neighbour edge-correction concepts, NN1 and NN2 (Hanisch 1984).

Materials and methods

Indices used in this study

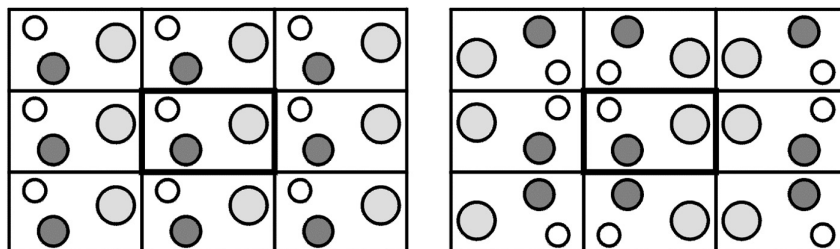
Tables 1 and 2 give an overview of the nearest-neighbour indices used in this study.

The integer variable n in Table 1 is the number of nearest neighbours included in the determination of the index. In this study, n is set to 4 for indices 1, 2, and 3, while the DBH differentiation, Clark and Evans, and Pielou indices generally use only one neighbour. The uniform angle index is derived from the angles subtended by adjacent neighbours at the reference trees. The six indices were selected so that similar aspects of spatial structure are accounted for by two indices (1 and 5, 2 and 6, 3 and 4; Table 1).

The two new edge-correction concepts, NN1 and NN2 (Hanisch 1984), can be applied to any structural index that is based on the nearest-neighbour principle, for example, those of Clark and Evans (1954) and Pielou (1977). The mingling index, an indicator of spatial species diversity (Füldner 1995; Fig. 2), will be used to explain and demonstrate NN1 and NN2. It is representative of a group of neighbourhood-based indices describing the spatial distribution of tree attributes.

The mingling index (M_i) gives, for each individual tree i , the proportion of its n nearest neighbours that do not belong to the same species as the reference tree i . Figure 2 illustrates the index for the case $n = 4$.

Fig. 1. Illustration of the translation (left) and reflection (right) edge-correction methods used in this study. The observation window, W , is in the centre with boundaries in bold.



The mingling index can only take one of a limited number of values. For example, when $n = 4$ neighbours there are $n + 1 = 5$ possible values of M_i , calculated as k/n , with $k = 0, 1, \dots, n$: 0.00, 0.25, 0.50, 0.75, and 1.00 (Fig. 2). Using these scores, all trees of the stand can be individually classified, and the overall stand structure can be characterized by the distribution of the values of M_i . The mean population mingling, M , can be expressed as

$$[1] \quad M = \sum_{k=1}^n \frac{k}{n} m_k = \frac{1}{N} \sum_{i=1}^N M_i$$

where m_k is the proportion of trees that have a mingling index of k/n and N is the number of trees in the population considered. Naturally, the full distribution of m_k provides more information than M .

The number n is fixed for all trees of an observation window, but can be chosen by the forestry statistician, determined by the objectives of individual studies on species diversity, the size of the window of observation and the density of points. For example, Fuldner (1995) recommended $n = 3$ neighbours, while more recent publications (Aguirre et al. 2003) have suggested that $n = 4$ neighbours is to be preferred. For more information on how to use and interpret the mingling index and the other indices listed in Table 1, see Aguirre et al. (2003) and Pommerening (2002).

NN1 and NN2 estimators

In addition to the existing edge-correction methods, which will be described in Table 3, the authors have considered two new concepts, the NN1 and NN2 estimators. Stoyan (2005) demonstrated that NN1 in particular, which has its origin in Hanisch (1984), is a very successful estimator of the nearest-neighbour distance distribution function, $G(r)$, and therefore it seemed appropriate to study its performance in the estimation of indices. The estimator NN1 is ratio-unbiased and includes more points than the unbiased buffer zone estimator, which, as noted earlier, has the additional disadvantage that buffer zone width is statistically difficult to determine. NN1 and the closely related NN2 are specifically designed for the statistical estimation of characteristics related to nearest neighbours and so are referred to as nearest-neighbour or NN-correction estimators. The principle is explained in Stoyan and Stoyan (1994, p. 297) for the case of $G(r)$. The idea is to use variable buffer zones for the trees close to the edge of the observation window. NN1 and NN2 incorporate a buffer but, in contrast to the traditional buffer method with a constant d , the decision to exclude trees close

to the window's edge is made individually, based on the spatial arrangement of the trees.

In the statistical estimation of nearest-neighbour indices, tree i is used as a reference tree only if its distance s_i to the boundary of the window W is further than or equal to its distance c_i from its n th nearest neighbour in W . In this case all nearest neighbours are known to be within W and all necessary information for the index estimation of tree i is provided by the window. However, if the distance from the boundary is shorter than that from the n th neighbour, then it is possible that nearer trees to tree i occur outside W . In this case it is uncertain whether the window provides all the necessary information for index estimation, and therefore these trees are rejected in the analysis.

An indicator variable n_i is introduced, where $n_i = 1$ if tree i is to be accepted for analysis and $n_i = 0$ if it is to be rejected. The principal is illustrated in Fig. 3. Points P_1 and P_2 have the same distance $s_1 = s_2$ to the boundary of the window of observation W . Points P_{1n} and P_{2n} are the corresponding n th nearest neighbours in W . Because $c_1 > s_1$, $n_1 = 0$ and point P_1 is not accepted. Point P_2 , however, is accepted, since $c_2 < s_2$ and $n_2 = 1$.

As in the buffer method, there is an inevitable loss of a number of data points. To compensate for this loss when applying NN approaches, the accepted points are weighted to obtain an unbiased estimator. In statistical theory such weights are called Horvitz–Thompson weights (Horvitz and Thompson 1952), and the idea is to give rare events larger weights than the more common ones. The weights in this case are the inverses of the so-called reduced areas, F_i , of W . The reduced area, F_i , is a function of c_i and W . For the two most important cases it is

$$[2] \quad F_i = \begin{cases} (a - 2c_i)(b - 2c_i), & \text{rectangular plot} \\ (r - c_i)^2 \pi, & \text{circular plot} \end{cases}$$

where c_i is the distance between tree i and its n th nearest neighbour; a, b are the lengths of the sides of a rectangular sample plot; and r is the radius of a circular sample plot.

If c_i is small it is probable that $c_i < s_i$, F_i is large, and tree i receives only a small weight; for larger c_i , it is increasingly unlikely that $c_i < s_i$, F_i is small, and therefore larger weights are given. Theoretically, F_i can also be defined for windows of irregular shape.

In the estimation of M (eq. 1) by means of the NN1 estimator a further point process parameter plays an important role, the intensity or point density, that is, the mean number of points per unit area. Intensity is simply another way of

Table 1. Overview of individual tree indices used in this study, their formulae and the corresponding publications.

No.	Index (reference)	Biodiversity of	Formula	Variable definition
1	Uniform angle index (Hui and Gadaw 2002)	Tree positions	$W_i = \frac{1}{n} \sum_{j=1}^n w_j$	$w_j = \begin{cases} 1, \alpha_j < \alpha_0 \\ 0, \text{otherwise} \end{cases}$ α_j Angle pointing away from the reference tree i to neighbours j
2	Species mingling (Füldner 1995; Aguirre et al. 2003)	Tree species	$M_i = \frac{1}{n} \sum_{j=1}^n m_j$	$\alpha_0 = \frac{360^\circ}{n+1}$ $m_j = \begin{cases} 1, \text{species}_j \neq \text{species}_i \\ 0, \text{otherwise} \end{cases}$
3	DBH dominance (Gadow and Hui 2002; Aguirre et al. 2003)	Tree dimensions	$U_i = \frac{1}{n} \sum_{j=1}^n u_j$	$u_j = \begin{cases} 1, \text{DBH}_i > \text{DBH}_j \\ 0, \text{otherwise} \end{cases}$
4	DBH differentiation (Füldner 1995; Pommerening 2002)	Tree dimensions	$TI_i = 1 - \frac{\min(\text{DBH}_i, \text{DBH}_j)}{\max(\text{DBH}_i, \text{DBH}_j)}$	j is the first neighbour tree of i
5	Aggregation index (Clark and Evans 1954)	Tree positions	$R = \frac{\bar{r}_{\text{observed}}}{E(r)}$	Arithmetic mean of distances between reference trees and neighbours j
6	Coefficient of segregation (Pielou 1977)	Tree species	$S = 1 - \frac{N(b+c)}{ms+nr}$	Area of the observation window No. of trees, $R \in [0, 2.1491]$ Species No. of trees No. of reference trees No. of times A and B serve as NN

See Table 2. $S \in [-1, 1]$ **Note:** $n = 4$ is the number of neighbour trees used with indices 1–3. Values of indices 1–4 are distributed between zero and one.

Fig. 2. Illustration of the mingling index for $n = 4$ neighbours. The neighbours are numbered according to increasing distance from the reference tree.

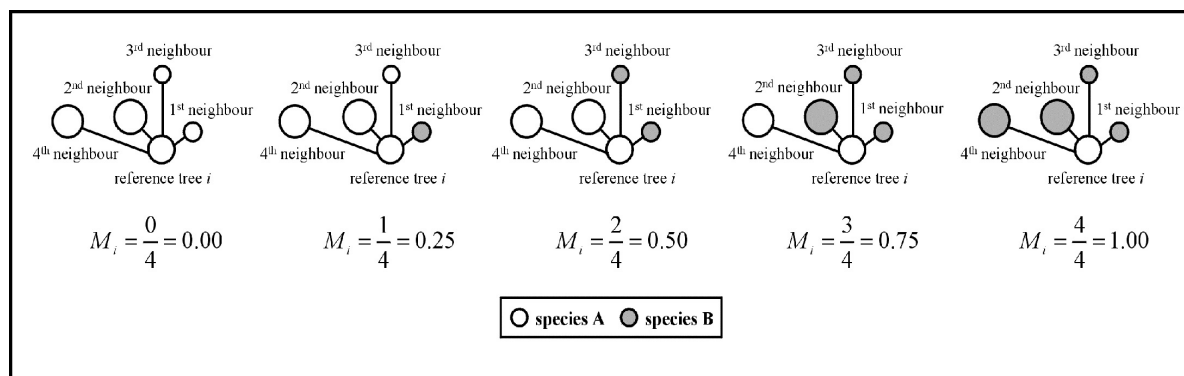


Table 2. A 2×2 table for the calculation of the coefficient of segregation (Pielou 1977).

Species of reference tree i	Species of nearest neighbour		
	A	B	
A	a	b	m
B	c	d	n
	r	s	N

expressing the forestry variable stems per hectare (SPH). It characterizes the density of trees in a certain area regardless of their marks or attributes such as their diameters.

The NN1 estimators of the mingling index class k and the mingling mean value are

$$[3] \quad \hat{m}_k = \frac{1}{\hat{\lambda}} \sum_{i=1}^N \frac{1_i(k)n_i}{F_i} = \frac{\overline{\lambda m_k}}{\hat{\lambda}}$$

and

$$[4] \quad \bar{M} = \frac{1}{\hat{\lambda}} \sum_{i=1}^N \frac{M_i n_i}{F_i} = \frac{\overline{\lambda M}}{\hat{\lambda}}$$

where

- $\hat{\lambda}$ estimated intensity of the point pattern (see eq. 6);
- N number of trees of the forest stand or monitoring plot (window of observation) under consideration;
- M_i mingling index of tree i ;
- $\bar{M}$ estimator of the mean mingling index;
- F_i reduced area of the window of observation (see eq. 2);
- n_i indicator variable defining whether a tree's index is affected by plot edge (0) or not (1);
- $1_i(k)$ indicator function of mingling index class k assuming the value of one, if the neighbouring tree is of a different species, and zero otherwise.

It can be seen that M and m_k are not directly estimated but only $\lambda \bar{M}$ and λm_k , which are then divided by an intensity estimator $\hat{\lambda}$. Thus, the estimation is not unbiased but only ratio-unbiased.

The standard intensity estimator $\tilde{\lambda}$ is given by eq. 5.

$$[5] \quad \tilde{\lambda} = \sum_{i=1}^N \frac{N}{F_i}$$

where $\tilde{\lambda}$ is the number of points in W divided by the area F of W .

However, in this study a less precise estimator $\hat{\lambda}$ (eq. 6) was used. This is because $\hat{\lambda}$ has the same structure as $\lambda \bar{M}$ and it can be expected that fluctuations of $\lambda \bar{M}$ are evened out by fluctuations of $\hat{\lambda}$. This effect is weaker with $\tilde{\lambda}$:

$$[6] \quad \hat{\lambda} = \sum_{i=1}^N \frac{n_i}{F_i}$$

NN2 is closely related to NN1, having the advantage of perhaps being intuitively more appealing and easier to compute. However, it is not ratio-unbiased.

In this estimator the sum of all M_i of the accepted points (those with $n_i = 1$) is simply divided by the number of accepted points:

$$[7] \quad \tilde{m}_k = \sum_{i=1}^N \frac{1_i(k)n_i}{\sum_{i=1}^N n_i}$$

and

$$[8] \quad \tilde{M} = \sum_{i=1}^N \frac{M_i n_i}{\sum_{i=1}^N n_i}$$

Experimental design

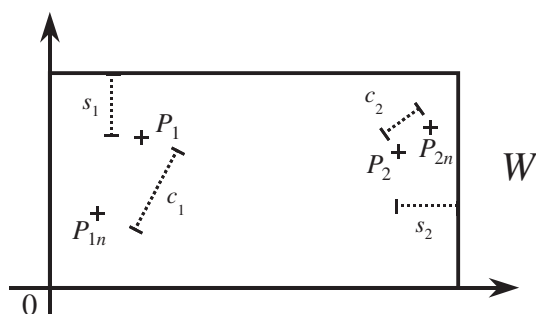
Edge-correction methods

Six different edge-correction methods were considered and compared in this study (Table 3). Estimation without edge-correction (method 0) was used as a control. In the control all trees within the monitoring plot or window of observation were processed as reference trees and their nearest neighbours were taken from within the window, ignoring the existence of outside trees. "Translation" or "periodic boundary conditions" (method 1) means that the plot W containing the tree locations is exactly replicated throughout space to

Table 3. Overview of the edge-bias compensation methods used in this study.

No.	Name	Reference	Comment
0	No edge-bias compensation	—	Control
1	Translation (= torus, periodic boundary conditions)	Torquato 2002; Radtke and Burkhart 1998	Fig. 1
2	Reflection	Pretzsch 2002; Radtke and Burkhart 1998	Fig. 1
3	Buffer zone	Diggle 2003; Gadow et al. 2003; Ripley 1981	5 m
4	NN1	Equations 3–4	—
5	NN2	Equations 7–8	—

Fig. 3. Illustration of the working principle of the indicator variable n_i . For point P_1 , $n_1 = 0$ (since its distance s_1 to the boundary of W is smaller than the distance c_1 to the nearest neighbour P_{1n} in the window), and for point P_2 , $n_2 = 1$.



form an infinite lattice (Torquato 2002). This method is the same as wrapping the window on a torus and joining opposite edges (see Fig. 1), but as the name implies, the additional locations of the trees are obtained by reflection along the plot edges. As in the translation method, all reference and neighbour trees are to be found in the window of observation or its reflected copies.

The reflection method (2) also replicates the pattern throughout space (see Fig. 1), but as the name implies, the additional locations of the trees are obtained by reflection along the plot edges. As in the translation method, all reference and neighbour trees are to be found in the window of observation or its reflected copies.

The buffer zone method (3) uses only the trees in the inner part of W as reference trees; the trees within a buffer strip of 5 m along the plot boundaries are used only in the calculation of the indices. NN1 and NN2 (methods 4 and 5) have already been explained in detail in the previous section. The last three edge corrections belong to the group of minus-sampling methods (Stoyan and Stoyan 1994; Stoyan et al. 1995).

Investigated point processes

Data from both point process models and real forests were used in the comparisons. Point process models guarantee knowledge of the true population values of the mean indices around which the sample values must necessarily fluctuate. Employing these models also enables unlimited repetitions and the statistical assumption of stationarity is guaranteed (Stoyan et al. 1995). To validate the results obtained from the point process models two real forest sites were included in the investigation.

Three different processes were simulated: a Poisson, a Matérn hard-core, and a Matérn cluster process (Stoyan et al. 1995). The second process is a process with some degree of regularity and a positive minimum interpoint distance.

The third is a classical cluster process, where the cluster centres form a Poisson process and the cluster points are randomly scattered in circles of constant radius around the centres. The three different processes and their specific parameters (see Table 4) were chosen to produce rather distinct patterns. The point process models approximate patterns found in real forest situations. In our study the Matérn cluster processes represent young forests of natural origin consisting of trees with heavy seeds or with heterogeneous soil conditions leading to a clustered arrangement of trees. Alternatively, this process may represent a failed re-stocking with species 1 and a moderate secondary invasion of species 2. The Matérn hard-core process can be thought of as representing older forests where it is impossible for trees to be located closer together than a certain minimum distance, perhaps because of inter- and intra-specific competition and (or) forestry thinning activities. The Poisson process is a situation between the young clustered and the more mature hard-core arrangement of tree positions. Such patterns, which are close to randomness, are common in forests (Tomppo 1986; Pommerening 2002).

For the simulation of all three processes, points were marked both dependently and independently with species and diameter attributes. Dependent marking leads to a correlation between tree locations and species and diameter attributes, while independent marking excludes such a correlation. Both cases can be found in real forests. The marks were assigned as follows:

- Dependent marking: Within a circle of radius r_0 around each tree the number m of other trees was counted. If $m > 1$ the tree was assigned the species attribute 2, otherwise it was assigned species 1.
- Independent marking: The qualitative marks species 1 and species 2 were assigned randomly.

In both cases for the assignment of diameters, uniform random numbers were transformed to Weibull-distributed random numbers by applying the method developed by Nagel and Biging (1995). The species-specific scale and form parameters of the two-parameter Weibull function were estimated from a quadratic mean diameter by using linear relationships. In contrast to their original approach, the maximum diameter, which was required for estimating the form parameter, was estimated from the quadratic mean diameter, again by using a linear equation. The diameters assigned to each point are sampled from the corresponding species-specific Weibull functions by using the inverse transformation method. The Weibull distributions were restricted by a minimum diameter of 5 cm. The regression parameters used for the estimation of the parameters of the two-parameter Weibull function were taken from Nagel and Biging's (1995) original publication and are representative of northwest Ger-

many (where species 1 is oak (*Quercus robur* L., *Quercus petraea* (Matt.) Liebl.) and species 2 is beech (*Fagus sylvatica* L.)). Specific quadratic mean diameters were applied to each species (see Table 4) and as a result the diameter ranges of both species are also very different.

The purpose of the dependent marking procedure is to obtain isolated trees of species 1 and clusters of species 2. Such patterns appear, for example, in managed central European oak forests, where each dominant oak tree is surrounded by a group of smaller sized trees of one or more shade-tolerant species, for example, beech (Pommerening 2002).

Tables 4 and 5 summarize the specific processes used in the study. While the point patterns of Table 4 illustrate only 1 replication out of 10 000, the numerical values in the first six columns of Table 5 are the estimated true values, for the simulated models, of the indices for the processes involved. Formulae for these values could be derived mathematically in the case of Poisson processes with independent marks, but not for the more complicated processes. Thus, all values of Table 5 relating to the processes were determined numerically by simulation. For this purpose 100 very large forests (1000 m × 1000 m) were simulated to be sure that edge effects are negligible. The true index values were then determined as the arithmetic means of the 100 replications.

The mean values of the indices clearly reflect the qualitative differences between the processes. For example, for the aggregation index of Clark and Evans, a value larger than 1 for the Matérn hard-core processes indicates the given tendency towards regularity. For the Matérn cluster processes on the other hand the value is, as expected, smaller than 1.

The uniform angle index also behaves as expected, and in the case of the Poisson process a value of 0.5 is obtained exactly in accordance with its method of construction (Hui and Gadow 2002).

Real forest data

To validate the results of the statistically advantageous point processes, data from two monitoring sites in mixed oak-beech woodlands were used. Detailed information on the two monitoring plots is given in Fig. 4 and Table 5 (last two columns).

Manderscheid is a monitoring plot of 80 m × 80 m situated in the forest district of Manderscheid in the German federal state Rhineland-Palatinate (Pommerening 2002). The plot was surveyed in 1996 and consisted of a 118-year-old oak (*Quercus petraea*) stand intermingled with beech (*Fagus sylvatica*).

The second monitoring site, Boeselager, is also a mixed oak-beech stand of *Quercus robur* and *Fagus sylvatica*. The plot of 95 m × 116 m was also surveyed in 1996 and is situated in the Boeselager Estate in the Sauerland region (German federal state Northrhine-Westphalia; Bölsing 1996). At the time of surveying the oak trees were 151 years old and the beech trees 65 years old.

Manderscheid has a structure similar to the simulated model case Poisson with dependent marks, and Boeselager reflects Matérn hard-core with dependent marks.

Simulation procedure

The simulation for quantifying the performance of the six

different edge-correction methods (Table 3) was done in the following way:

- (i) Processes: Simulation of 10 000 forests (Poisson, Matérn hard-core, and Matérn cluster processes) in a window of 250 m × 250 m. Samples with smaller windows of decreasing size from 200 m × 200 m down to 30 m × 30 m were cut out of the centre.
- (ii) Real forest data: Simulation of 100 replications in the original plot windows. Samples with smaller windows of decreasing size from 70 m × 70 m (90 m × 90 m) down to 15 m × 15 m were cut, with a centre point randomly selected for each replication so that the sample window was located fully inside the original plot window.

Evaluation statistics

For the evaluation of the performance of the six edge-bias compensation methods (estimators) the bias and root mean squared error (RMSE) were used.

$$[9] \quad \text{Bias} = M - \hat{M}$$

$$[10] \quad \text{RMSE} = \sqrt{\frac{1}{n-1} \sum_{k=1}^n (\hat{M}_k - M)^2}$$

where

- M “theoretical” mean index value (i.e., “population” value), that is, “theoretical” mean species mingling;
- $\hat{M}$ index value, that is, species mingling, estimated from the 10 000 (100) replications;
- $\hat{M}_k$ index value, that is, species mingling, estimated from replication k ;
- n number of replications; here $n = 10\,000$ (100).

Because of a lack of space the bias is displayed only for the uniform angle, Clark and Evans, and Pielou indices in the next section. Graphs showing the bias of the other indices can be obtained from the authors upon request.

All results were obtained from the authors' own software routines written in Pascal and Java and from the first author's CRANCOD program.

Results and discussion

Figures A1–A12 (Appendix A) show the behaviour of the edge-correction methods expressed by bias (eq. 9) and RMSE (eq. 10) for each of the six indices depending on number of trees and point process model or monitoring plot. To enable comparisons, plot size was converted to number of trees using the known intensities. The Clark and Evans and the uniform angle index only account for the spatial arrangement of tree positions, so the results in Figs. A1, A2, A6, and A7 are the same for both dependent and independent marking. As expected, the RMSEs increase with decreasing numbers of trees.

The reaction of the six edge-correction methods of Table 3 to decreasing numbers of trees differs with the index involved. The speculative methods of no edge correction (0) and translation (1) (Pretzsch 2002; Diggle 2003) perform surprisingly well. This can be explained by the fact that these methods use all points in the window as reference points and accept some incorrect local values of the indices, which perhaps cancel each other out. Edge-bias investiga-

Table 4. Visualization (in a 100 m × 100 m window) of the processes involved in this study (species 1, black; species 2, gray).

	Matérn cluster	Poisson	Matérn hard-core
Parameters	$R = 3.0$ $\lambda = 0.03$ $dg_1 = 36$ cm $dg_2 = 13$ cm	$\lambda = 0.02$ $dg_1 = 60$ cm $dg_2 = 26$ cm	$h = 5.5$ $\lambda = 0.01$ $dg_1 = 75$ cm $dg_2 = 41$ cm
Illustrates	"Young forest"	"Poisson forest"	"Old forest"
Dependent	(a) $r_0 = 2.0$	(b) $r_0 = 5.7$	(c) $r_0 = 7.5$
	(d) $p_1 = 0.270$	(e) $p_1 = 0.267$	(f) $p_1 = 0.542$

Note: "Dependent" and "independent" refer to the dependent and independent ways of marking tree locations with binary species and DBH. The specific parameters of the processes are given in a specific row and below the images. R , cluster diameter (m); λ , intensity; dg_1 , dg_2 : quadratic mean diameter (cm) of species 1 and 2, respectively; h , minimum distance between tree locations; r_0 , tree influence zone radius (m); p_1 , probability of species 1.

tions into competition indices came to similar conclusions (Monserud and Ek 1974; Radtke and Burkhart 1998). The translation method provides good estimators for all indices investigated. No edge correction is a good approach with most indices investigated and generally does not result in bad estimations. Two exceptions are the behaviour of the uniform angle and the Clark and Evans index. With these two indices, applying no edge correction is usually associated with a considerable bias throughout the range of numbers of trees observed. In the case of the uniform angle index, for points close to the window edge the nearest neighbours in the window are positioned towards the interior of the window, resulting in quite unusual angles, which spoil the statistical estimation.

The traditional method of reflection (2) leads, in many cases, to the highest RMSE (e.g., DBH dominance, DBH differentiation, Clark and Evans, and Pielou indices). The bias analysis reveals that the reflection method (2) results in

the largest bias values across all processes and real forest data, the only exception being the uniform angle index, for which reflection provides a good estimator.

Edge-correction methods that reject trees close to the boundaries of the observation window (methods 3–5) usually result in rapidly increasing bias values with decreasing numbers of trees, since the remaining number of trees becomes so small that ratio-unbiased estimators lose their good properties.

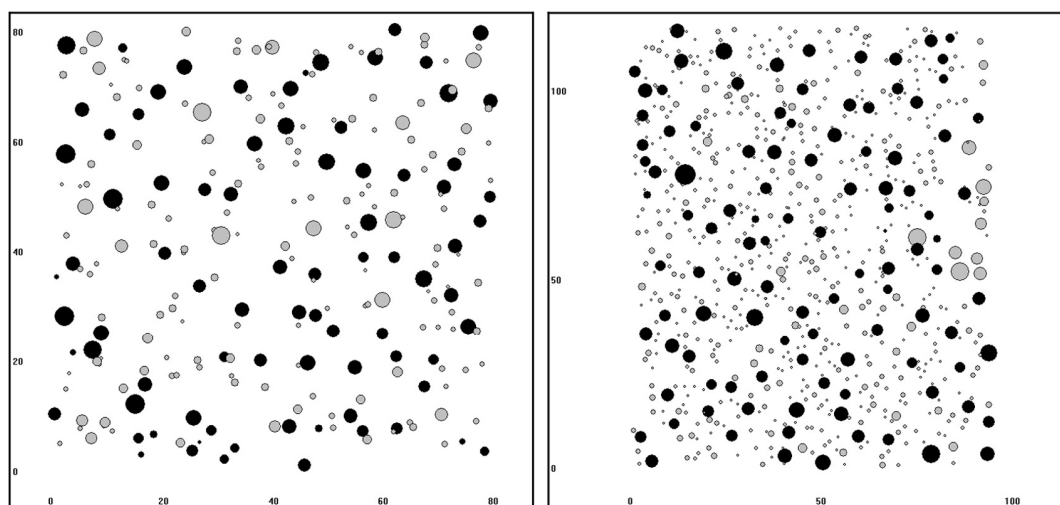
In most cases the buffer method (3) produces reasonable estimators (with the exception of cluster processes and Pielou index for hard-core processes and the mingling, DBH differentiation, Clark and Evans, and Pielou indices for the two real forest data sets). Where it performs well it often leads to similar RMSEs as those produced by the NN1 estimator.

A comparison of the theoretically unbiased methods, NN1 (4) and NN2 (5), shows that throughout the application

Table 5. “Theoretical” arithmetic mean values (mean values of 100 very large forest stands were simulated in a 1000 m × 1000 m window) and population values (of the two real forest sites) of selected forestry characteristics.

Marking	Matérn cluster		Poisson		Matérn hard-core		Real forests	
	Dep.	Indep.	Dep.	Indep.	Dep.	Indep.	Manderscheid	Boeselager
UAI	0.60	0.60	0.50	0.50	0.42	0.42	0.47	0.46
Mingling	0.21	0.39	0.25	0.39	0.38	0.50	0.44	0.26
DBH dominance	0.53	0.50	0.53	0.50	0.52	0.50	0.53	0.51
T1	0.33	0.41	0.35	0.40	0.33	0.34	0.40	0.42
Clark and Evans	0.43	0.43	1.00	1.00	1.44	1.44	1.03	1.18
Pielou	0.44	0.00	0.38	0.00	0.22	0.00	−0.22	−0.15
BA (both species)	13.72	13.49	22.62	22.57	30.07	30.06	28.69	40.59
SPH (both species)	363.90	357.87	200.01	199.98	99.97	99.97	381.25	668.78
dg (both species)	21.33	21.91	37.94	37.91	61.89	61.88	30.95	27.80
BA (species 1)	9.97	9.80	14.97	14.94	24.12	24.11	17.12	23.48
SPH (species 1)	98.11	96.45	53.49	53.41	54.22	54.18	126.56	97.10
dg (species 1)	35.98	35.97	59.70	59.68	75.26	75.27	41.50	55.49
BA (species 2)	3.75	3.68	7.64	7.63	5.95	5.95	11.57	17.11
SPH (species 2)	265.79	261.42	146.52	146.57	45.75	45.79	254.69	571.69
dg (species 2)	13.40	13.40	25.77	25.75	40.71	40.69	24.05	19.52

Note: UAI, uniform angle index (eq. 1 in Table 1); mingling, species mingling (eq. 2 in Table 1); DBH dominance (eq. 3 in Table 1); T1, DBH differentiation (eq. 4 in Table 1); Clark and Evans, Clark and Evans (1954) aggregation index; Pielou, Pielou’s (1977) coefficient of species segregation; BA, basal area (m²); SPH, trees per hectare; dg, quadratic mean diameter (cm). “Dep.” and “Indep.” refer to the dependent and independent ways of marking tree locations with binary species and DBH. In the case of the real forest sites oak is coded as species 1 and beech as species 2.

Fig. 4. Visualization of the real forestry data used in this study (left: Manderscheid, right: Boeselager; oak (species 1), black; beech (species 2), gray).

range of this study both approaches behave similarly, although NN1 leads in a number of cases (uniform angle, DBH dominance, DBH differentiation, Clark and Evans, Pielou indices) to a smaller bias and can therefore be regarded as a safe option. Judging by the RMSE, NN1 provides a good estimator for the Clark and Evans index in particular. Both NN1 and NN2 perform well for DBH dominance, DBH differentiation, Clark and Evans, and Pielou indices if cluster processes are involved. The NN estimators (methods 4 and 5) omit many points close to the plot edge, which results in a large variability of the estimators. Their bias characteristics, however, especially those for NN1

(method 4), are very favourable except for a very small numbers of trees.

The bias analyses also clearly illustrate the difference between point patterns with dependent and independent marking. Figure A9, for example, shows that in the case of dependent marking there is a much larger differentiation between the different edge-correction methods, while the corresponding curves for the independent cases are very close together. Similar behaviour occurs with the mingling, DBH dominance, and DBH differentiation indices. The stronger the dependencies between tree locations and tree attributes the larger the bias differences of the estimators.

Especially when comparing DBH dominance and DBH differentiation indices it is evident that there are similar error patterns with indices quantifying equivalent aspects of spatial diversity. That the DBH of trees, especially, is correlated with the pattern of tree positions is well known from studies investigating competition indices (e.g., Pretzsch 2002). The error patterns of the indices accounting for species diversity and the diversity of tree positions are also similar but not as much as in the case of dimension diversity.

Conclusions

The optimal choice of edge-correction method is dependent on the algorithm of the index and on the spatial pattern of tree positions. This is particularly important when there are dependencies between the pattern of tree positions and the other tree attributes (marks), as this is often the result of competition processes and (or) woodland management. This highlights the importance of conducting sensitivity studies whenever new indices are developed or existing ones are modified.

The results clearly demonstrate the poor performance of the reflection method (2). This approach to edge correction should therefore be avoided in any investigation.

All methods that reject some sample trees (minus sampling, 3–5) usually lead to large bias values when the number of trees is small and therefore they should only be applied to samples containing a sufficiently large number of trees (≥ 100). These methods emphasize that the number of trees available for the index estimation is an important performance criterion.

Using the mathematical–statistically motivated edge-corrected NN1 estimator (method 4) with most indices leads to small bias values and can therefore be regarded as a secure option with sufficiently large numbers of trees. However, the NN1 and NN2 estimators can in many cases lead to larger RMSEs than those produced by ignoring the effect of edge bias or translation. Future research in edge-correction methods should develop the NN1 and NN2 approach by minimizing the loss of data and reducing the variability of the estimators.

Indices accounting for tree positions and especially those based on angles require edge-correction methods, no matter how large the sample size is. Translation, buffer zone, and NN1 methods are suitable methods to reduce error.

Plots to monitor indices of spatial biodiversity by means of spatial indices should be sufficiently large to minimize edge effects. This is especially crucial with long-term monitoring in forestry where the number of trees steadily decreases during the rotation period. Where small plot sizes are unavoidable, for example, in forest inventories with small-sized circular sample plots designed for other purposes (Gignoux et al. 1999), it is advisable to include the neighbours immediately outside the plot in the survey. Stoyan and Stoyan (1994) and Stoyan et al. (1995) refer to this method as plus-sampling. Plus-sampling is unbiased but requires additional measurements.

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Appendix A

Appendix appears on the following page.

Fig. A1. Uniform angle index: results of the RMSE performance of the edge-bias compensation methods 0–5 (see Table 3) in relation to the point process models.

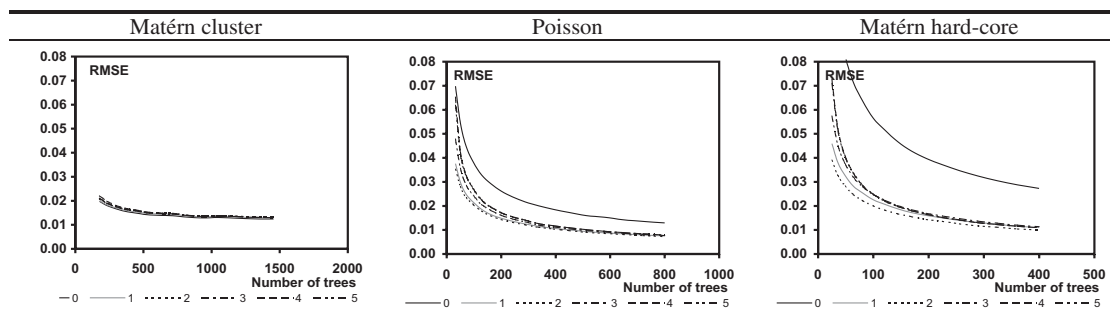


Fig. A2. Uniform angle index: results of the bias performance of the edge-bias compensation methods 0–5 (see Table 3) in relation to the point process models.

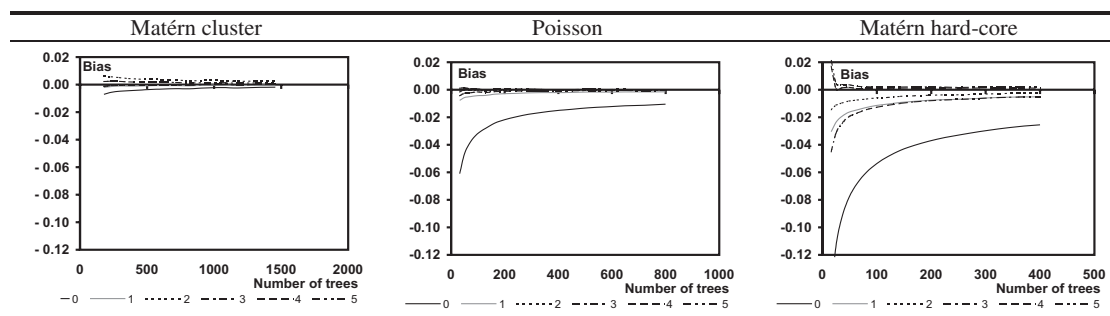


Fig. A3. Mingling index: results of the RMSE performance of the edge-bias compensation methods 0–5 (see Table 3) in relation to the point process models. “Dependent” and “independent” refers to the dependent and independent way of marking tree locations with binary species and DBH.

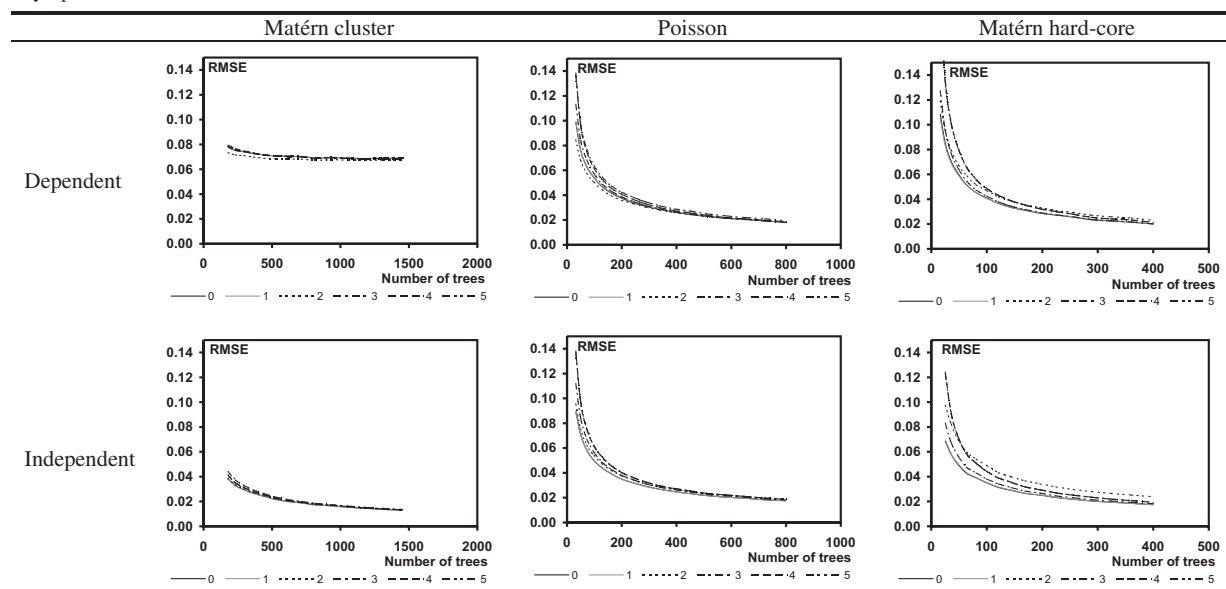


Fig. A4. DBH dominance index: results of the RMSE performance of the edge-bias compensation methods 0–5 (see Table 3) in relation to the point process models. “Dependent” and “independent” refers to the dependent and independent way of marking tree locations with binary species and DBH.

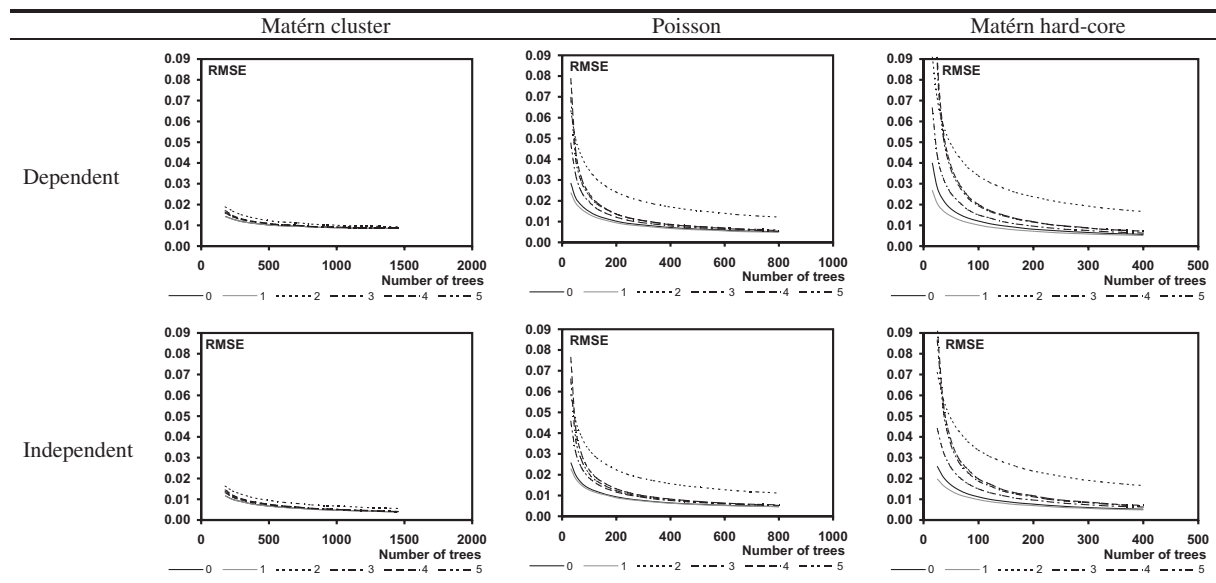


Fig. A5. DBH differentiation index: results of the RMSE performance of the edge-bias compensation methods 0–5 (see Table 3) in relation to the point process models. “Dependent” and “independent” refers to the dependent and independent way of marking tree locations with binary species and DBH.

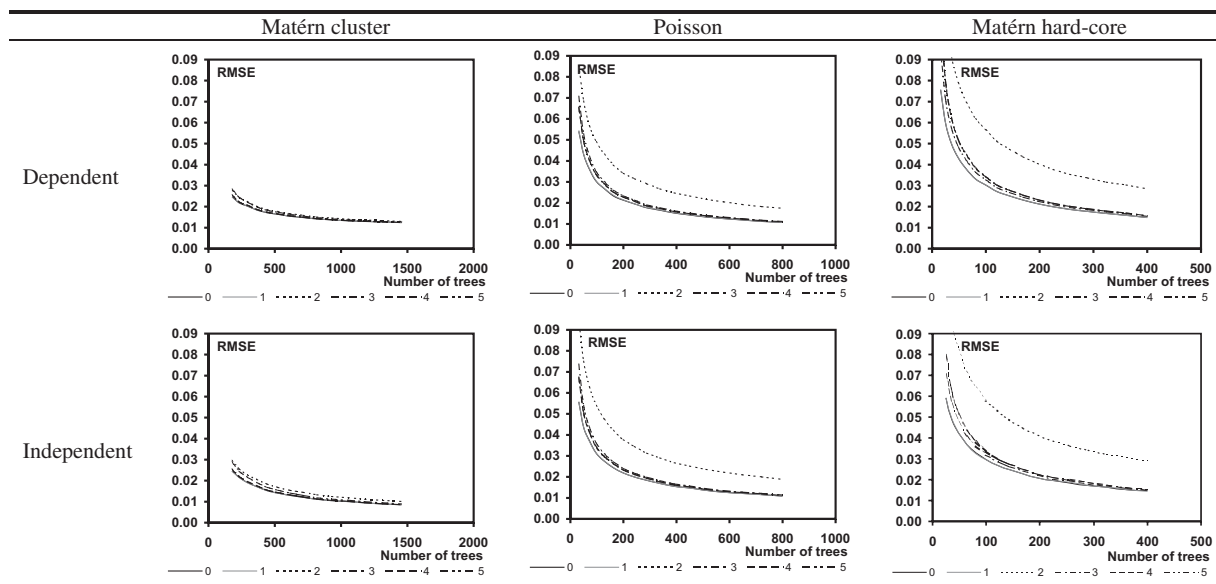


Fig. A6. Clark and Evans index: results of the RMSE performance of the edge-bias compensation methods 0–5 (see Table 3) in relation to the point process models.

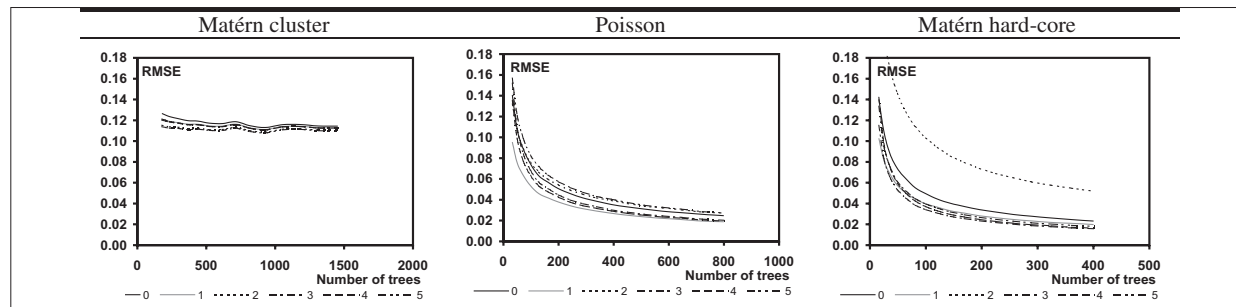


Fig. A7. Clark and Evans index: results of the bias performance of the edge-bias compensation methods 0–5 (see Table 3) in relation to the point process models.

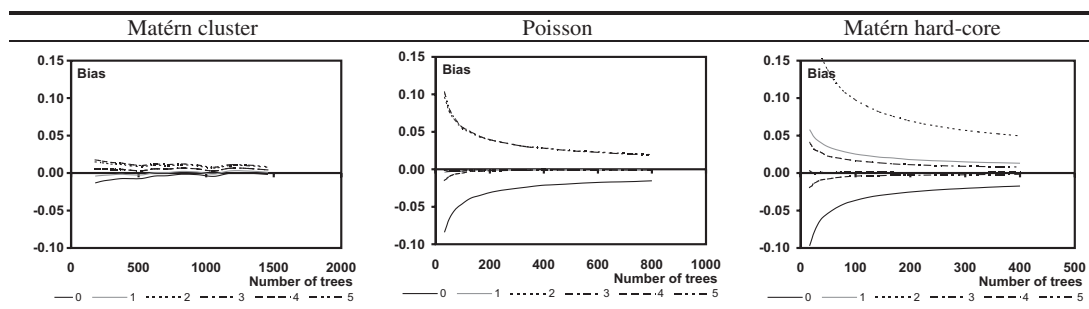


Fig. A8. Pielou index: results of the RMSE performance of the edge-bias compensation methods 0–5 (see Table 3) in relation to the point process models. “Dependent” and “independent” refers to the dependent and independent way of marking tree locations with binary species and DBH.

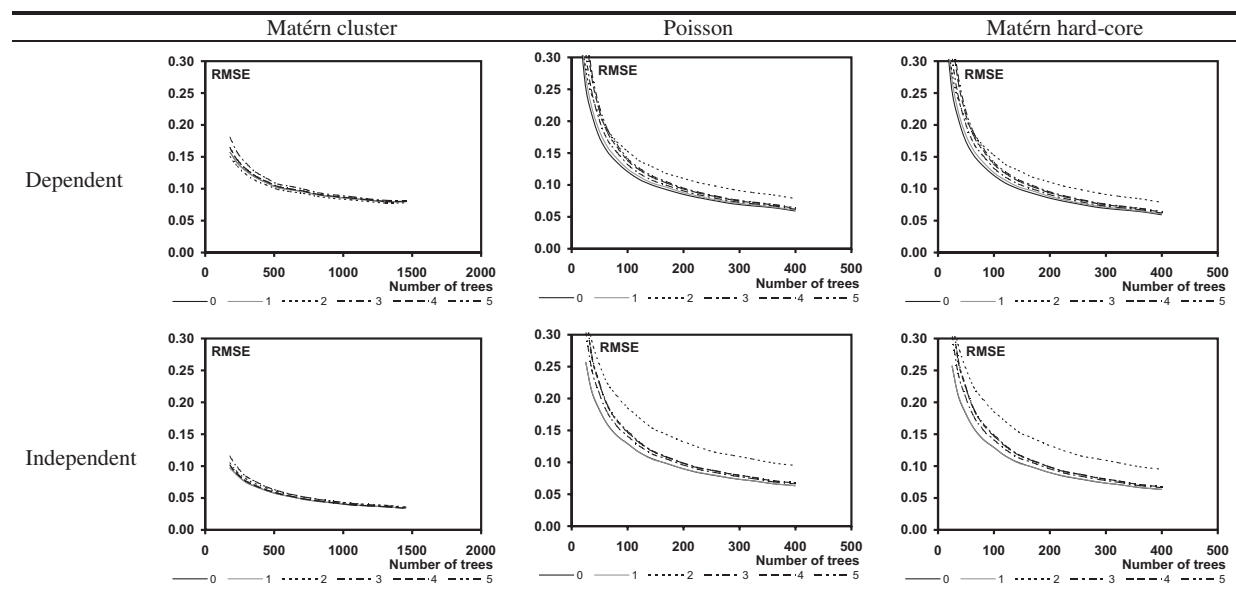


Fig. A9. Pielou index: results of the bias performance of the edge-bias compensation methods 0–5 (see Table 3) in relation to the point process models. “Dependent” and “independent” refers to the dependent and independent way of marking tree locations with binary species and DBH.

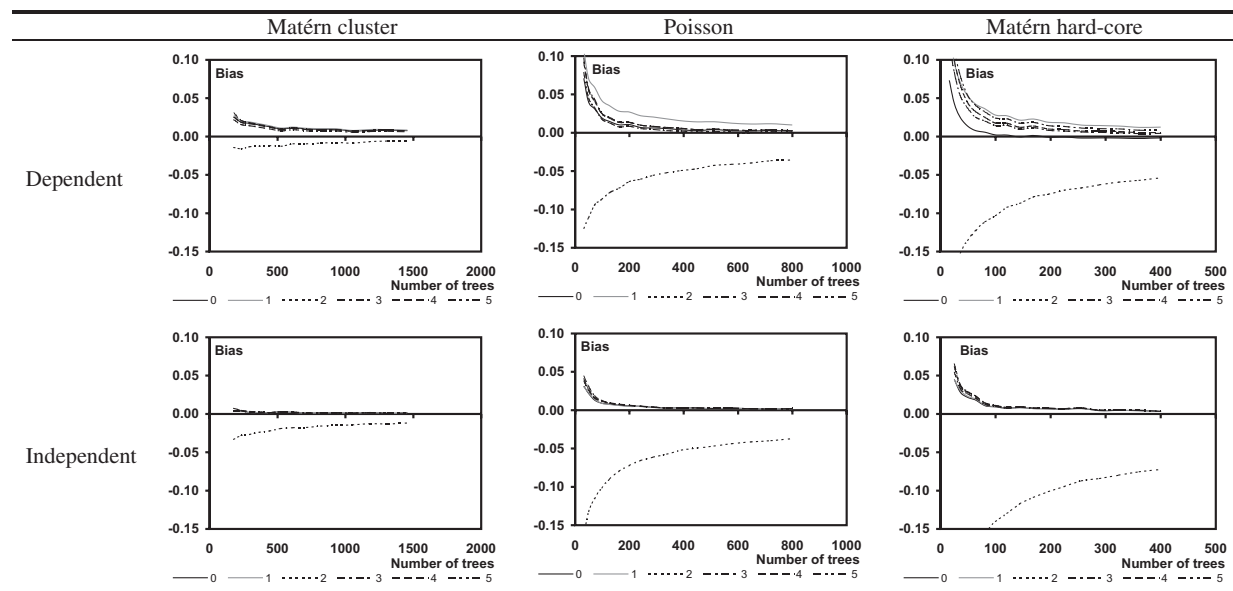


Fig. A10. Results of the RMSE performance of the edge-bias compensation methods 0–5 (see Table 3) and the uniform angle, mingling, and DBH dominance indices in relation to the two forestry data sets.

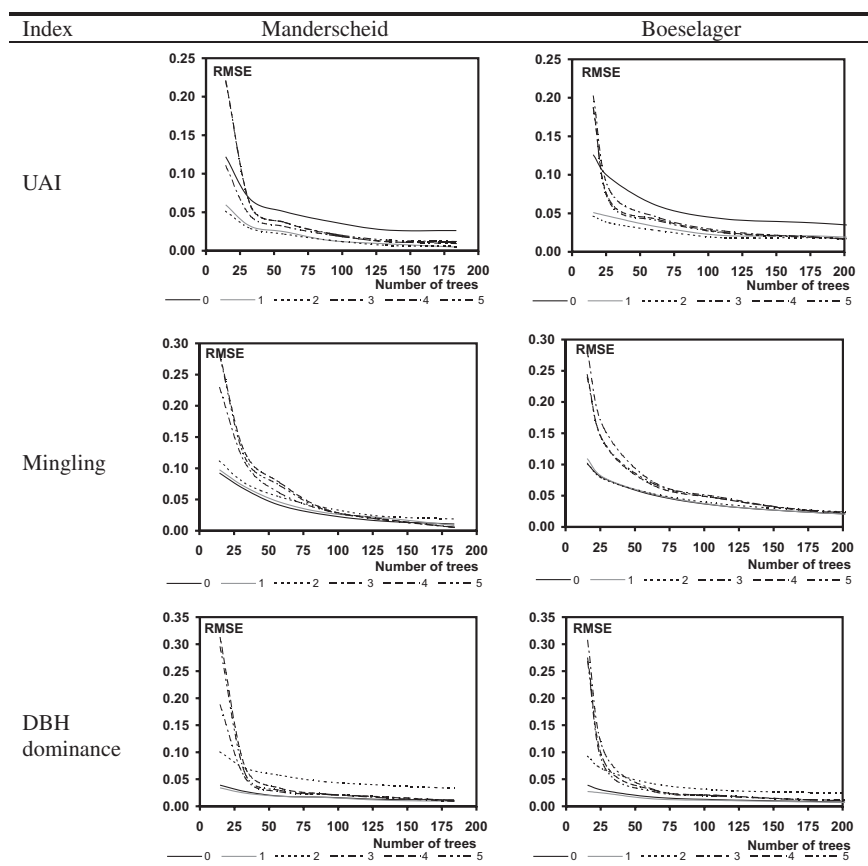


Fig. A11. Results of the RMSE performance of the edge-bias compensation methods 0–5 (see Table 3) and the DBH differentiation, Clark and Evans, and Pielou indices in relation to the two forestry data sets.

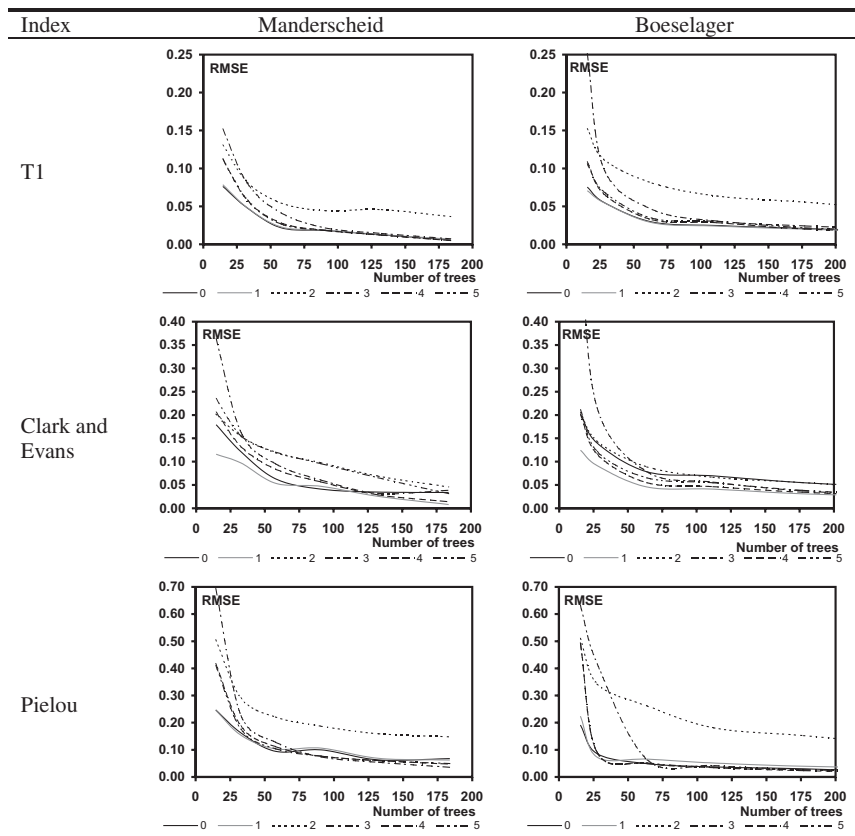
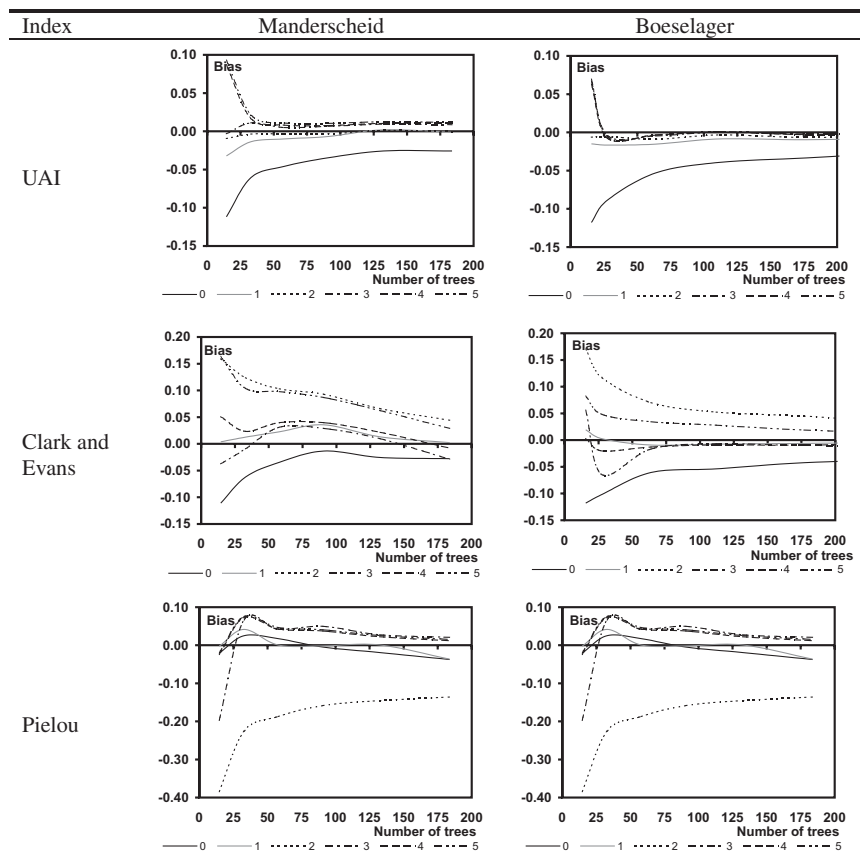


Fig. A12. Results of the bias performance of the edge-bias compensation methods 0–5 (see Table 3) and the uniform angle, Clark and Evans, and Pielou indices in relation to the two forestry data sets.



IV

Spatial Structure of
Semi-Natural and
Plantation Stands of Scots
Pine (*Pinus sylvestris* L.) in
Northern Scotland

Spatial structure of semi-natural and plantation stands of Scots pine (*Pinus sylvestris* L.) in northern Scotland

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Summary

The success of current initiatives to maintain and enhance the area of and the special habitats provided by the remnant semi-natural pinewoods of northern Scotland will depend upon foresters' ability to foster more natural structures in even-aged plantations through stand manipulation. However, there is little information on the structures and spatial patterns that can be found in Scottish pinewoods; such knowledge could be used to design appropriate silvicultural regimes. A study was carried out to compare spatial structure in three 0.8–1.0 ha plots in the Cairngorms National Park; one plot was a 78-year-old plantation stand, the other two were semi-natural stands with trees up to 300 years old. Basic mensurational data showed that the semi-natural stands were characterized by a wider range of tree sizes and more large (>50 cm d.b.h.) trees. Spatial structure was evaluated with a range of different indices: the aggregation index of Clark–Evans (CE), the uniform angle and diameter differentiation indices, Ripley's *L* function of tree spatial distribution, pair and mark correlation functions and experimental variograms of tree diameter. The CE revealed a regular distribution in the plantation with the semi-natural stands having a random pattern. Further analysis of the latter stands indicated that, in each case, the older trees in the stand were regularly distributed while the younger ones were clustered. There was little difference in uniform angle values between the stands while the diameter differentiation distributions suggested greater variety in diameter within the semi-natural stands than in the plantation. The Ripley's *L* function showed that trees in the plantation were regularly distributed at close distances but clustered over wider distances. There were differences in pattern between the semi-natural stands; in one, trees were clustered because the positions of the younger trees were influenced by past regeneration trials, whereas in the other stand a random pattern was observed. Similarly, the variogram indicated widespread homogeneity in diameter within the plantation, while the semi-natural stands showed high variation at close spacing because of competition followed by spatial autocorrelation up to ~20 m distance. Thereafter, one of these stands had a very different pattern because of a more intensive regeneration history. All the indices, apart from uniform angle, were able to discriminate between the plantation and the two semi-natural stands, but only the more detailed spatial indices were capable of identifying differences within the latter. The implications of these results for management strategies in plantations are discussed.

Introduction

Scots pine (*Pinus sylvestris* L.) is the most widely distributed conifer species in the northern hemisphere with a European range stretching from the Urals to the Atlantic and from the Barents Sea to the Mediterranean (Nikolov and Helmisaari, 1992). It is one of three conifer species native to Great Britain with a natural distribution confined to the northern Highlands of Scotland, where ~18,000 ha of ancient semi-natural (The term 'ancient semi-natural' is used (Anonymous 2003)) in Britain to identify woods composed of locally native trees and shrubs which are derived from natural regeneration rather than planting (i.e. semi-natural). A wood is classed as 'ancient' when it is known to have occurred continuously on the site since at least 1750 (in Scotland) and often considerably earlier (Anonymous, 2003)) pinewoods survive (Mason *et al.*, 2004). The area of these remnants (hereafter called 'the native pinewoods' or 'Caledonian pinewoods') represents ~1 per cent of the original area covered by the natural pine forests 4000 years ago (McVean and Ratcliffe, 1962). These forests were progressively reduced in extent over the centuries through a variety of unsustainable practices such as overgrazing, burning and exploitation (Steven and Carlisle, 1959; Smout, 1997). Despite their much reduced area, the native pinewoods are an integral part of some of the most cherished landscapes in the Scottish Highlands which attract visitors from many parts of Britain and the wider world. The need to conserve and restore these pine forests has been increasingly recognized (Stevenson and Peterken, 2004) since the publication of the first authoritative survey some 50 years ago (Steven and Carlisle, 1959). These forests contain a number of rare species of fauna and flora that, in Britain, are dependent upon the favourable habitat provided by the native pinewoods. Examples include the capercaillie (*Tetrao urogallus* L.), crested tit (*Lophophanes cristatus* L.) and twinflower (*Linnaea borealis* L.). As a result, the native pinewoods are recognized as a priority habitat under the EU Habitats Directive (Mason *et al.*, 2004) and a Pinewood Habitat Action Plan (Anonymous, 1995) has been published. The latter aimed to increase the area of the remnant pinewoods by 35 per cent by 2005 and also sought to create 25 000 ha of new native

pinewoods through a mixture of natural colonization and planting.

Because of their high conservation and landscape value, there has been very little timber production from any of the pinewood remnants during the last 50 years. Most Scots pine timber harvested in Britain is derived from 220 000 ha of plantations located on nutrient poor and free draining mineral soils throughout the country, where the species provides acceptable timber yields with moderate establishment costs (Mason, 2000). The prevailing silvicultural system used in the plantations is patch clearfelling with artificial regeneration and a rotation age of ~60–80 years. Perhaps because of the prevalence of this even-aged management system, to date little consideration has been given to the potential role of Scots pine plantations in the proposals for the restoration and expansion of the native pinewoods. This is despite there being ~100 000 ha of such plantations within the native pinewood zone in northern Scotland (Mason *et al.*, 2004). However, recent studies have shown that pine stands of plantation origin where the overstorey trees have been retained for periods of perhaps 120–150 years can provide suitable habitat for species of high biodiversity value (Humphrey *et al.*, 2003). Therefore, alternative approaches to the management of Scots pine plantations may have an important role to play in the development of an expanded pinewood ecosystem in northern Britain. Unfortunately, there is little tradition of using alternative systems to clearfelling and artificial regeneration in the management of Scots pine forests in Britain, although there are reports of shelterwood and seed tree systems on private estates with extensive pine plantations (e.g. Guillebaud, 1933; Hart, 1995).

In addition, one concern over the future of the native pinewoods involves the lack of regeneration that is apparent in many of the remnant woodlands. This is believed to be due to a variety of reasons including heavy deer pressure, severe competition from ericaceous shrubs, a lack of adequate regeneration niches and reduced seed yields because of an ageing tree population (Nixon and Cameron, 1994; Scott *et al.*, 2000; Mason *et al.*, 2004). Because many of the mature trees in these remnant stands are over 200 years of age, the habitat is increasingly vulnerable to disturbance and tree mortality and the consequent

loss of the structural features (e.g. coarse woody debris, hollows in old trees) required for many of the specialist pinewood species disturbance. The new native pinewoods created in recent decades will take many years to develop these desirable features. This again suggests that modified management of pine plantations may have a role to play in sustaining key features of the native pinewood habitat.

Appropriate application of principles derived from knowledge of stand dynamics is an important element in successful restoration of forest ecosystems with a complete range of stand structures. Pine plantations in Scotland under conventional silviculture are largely comprised of stands in the 'stand initiation' and 'stem exclusion' phases (*sensu* Oliver and Larson, 1996). By contrast, the larger native pinewoods contain structures characteristic of all four phases of stand development (Summers *et al.*, 1997; Mason *et al.*, 2004). In particular, it is argued that many of the biodiversity and spiritual values associated with the pinewoods are linked to the presence of open stands with a mosaic of large trees of 50–100 cm d.b.h. (Gill, 1995). Thus, the presence of capercaillie is positively influenced by the combination of such structures with abundant ground cover of bilberry (*Vaccinium myrtillus* L.) (Kortland, 2003). Stands of this type are considered to be characteristic of the 'old growth' phase in Scottish pine forests (Mason, 2000) and one aim of management of plantations is to foster the development of old growth features through thinning (Taylor, 1995). This is primarily achieved by regimes that increase the small-scale spatial variation by creating a mosaic of heavily thinned areas that alternate with untouched groups. These approaches have similarities to the variable density thinning (VDT) promoted in north-west America as a means of improving ecosystem diversity in second growth conifer forests (Carey, 2003). In Sweden, VDT has been proposed as means of promoting within stand structural diversity in Scots pine forests (Fries *et al.*, 1997). This approach is combined with the indefinite retention of 5–20 seed trees ha⁻¹ as a means of increasing the occurrence of old trees within regenerated stands. A similar approach termed 'group thinning' with a mixture of denser groups and a more heavily thinned matrix has been proposed for beech (*Fagus sylvatica* L.) forests in central Europe (Kato, 1979).

While the ideas behind VDT appear reasonable, there is uncertainty as to the scale and pattern in which such regimes should be applied. Most attempts to introduce VDT in pine stands in Britain have used subjective adjustment of thinning schedules used to maximize timber production in even-aged plantations (e.g. Taylor, 1995). Such approaches fail to define the desired spatial arrangement in the stands being manipulated and there is a consequent risk that the structures produced may not be the most appropriate for biodiversity and that there may be some intervening loss in timber production and quality.

One way of improving guidance on the appropriate patterns and intensity of VDT is to base this on comparative analysis of spatial patterns of tree distribution and other structural parameters in both existing 'old growth' pine stands and younger ones of plantation origin. Pommerening (2002) reviewed a number of spatial indices that could be used to characterize stand structures and having applied them to a range of German stands suggested how these might be used to measure differences between forests over space and time including the impact of differential harvesting. Such studies have indicated that more intensively managed stands tend to show a more regular arrangement of stem distribution than that occurs in more natural ones and that this regularity is particularly noticeable at comparatively short distances. However, while these findings conform to general understanding of the impact of silviculture upon Scots pine forests in northern Scotland, there are no published data to show whether the detailed results obtained in other countries can be readily extrapolated to British site types and conditions. One advantage of these indices is that they can be derived from data collected in small plots used in forest inventories and other types of monitoring (Pommerening, 2002, 2006).

In this paper, we present the results from a study of the diameter and age-class distributions and spatial structure of three Scots pine stands in north-east Scotland of different ages and with varying management histories. We compare the information provided by an aggregate index of spatial structure and two neighbourhood-based indices against the more detailed analysis provided by functions based upon known tree positions and sizes. We discuss how the results might

inform ways of promoting more natural structures in Scots pine plantations.

Materials and methods

Sample sites

The three stands are located in Glen More and Abernethy forests within the Cairngorms National Park (57° 30' N and 4° 15' W). Both these forests are considered to form part of the native pinewoods of Scotland (Steven and Carlisle, 1959) and have been dominated by Scots pine for several millennia (Mason *et al.*, 2004). The stands called 'young' and 'old' Glen More occur within 1 km of each other at an altitude of ~330 m a.s.l., while the Abernethy stand occurs ~10 km to the north-east at a similar elevation. Mean annual rainfall is ~900 mm, while mean annual temperature is ~6.0°C. The soils are typically free draining podzols with occasional weak iron pans, while the vegetation is dominated by *Calluna vulgaris* (L.) Hull, *Vaccinium myrtillus*, *V. vitis-idaea* L., *Deschampsia flexuosa* (L.) Trin. and the mosses *Polytrichum commune* Hedw. and *Hylocomium splendens* (Hedw.) Br. Eur.. This forest type is typical of the native pinewoods of north-east Scotland and is considered to be similar to the acidic pinewoods of Scandinavia (Rodwell and Cooper, 1995).

Both Glen More and Abernethy forests were exploited for their timber in previous centuries. Management practices before 1750 are little known but are likely to have involved selective exploitation of a range of tree sizes for local use (see Smout, 1997 for discussion of the history of forest management on an adjacent forest). From ~1780 until 1850, there were periods of intense harvesting in both forests which severely depleted the growing stock of mature trees (Dunlop, 1997).

The extent of the exploitation in Glen More was so extreme that by 1850 local foresters reported that few large trees of harvestable quality could be found (Dunlop, 1997). For the following half century Glen More was mainly managed as a sporting estate for deer stalking. The forest was acquired by the Forestry Commission in the early 1920s following further felling during the 1914–1918 war. The 'old' stand was an area of

~5 ha reserved in 1930 for research experimentation when it was described as having a 50–80 per cent coverage of old pine trees (MacDonald, 1952). A number of small-scale regeneration experiments were carried out between 1930 and 1938 and thereafter it has been maintained as a non-intervention reserve to examine the long-term natural dynamics of pine stands (MacDonald, 1952; Henman, 1961; Thompson *et al.*, 2003). The 0.8-ha plot described in this paper is that part of the old stand most intensively studied in recent years (Edwards and Rhodes, 2006). The 'young' Glen More plot was part of a 30-ha stand planted in 1926 on the site of an old stand that was felled during the 1914–1918 war. The stand had been thinned at least three times on an 8- to 10-year cycle between 1956 and 1996. In 2003, six 1 ha plots were installed as part of an experiment to compare the effect of different patterns and intensities of thinning upon subsequent stand structure. The data presented in this paper are for the control plot (i.e. not thinned in 2003/2004) in the experiment, selected as being typical of older pine plantation stands in the region.

Abernethy forest formed part of the Seafeld estates which were well-known in the first part of the last century for being the only example in Britain where natural regeneration of Scots pine had been successfully promoted over a long time period and on a large scale (Guillebaud, 1933). The main silvicultural system used was a uniform shelterwood with a seeding felling overstorey density of ~250 trees ha⁻¹ followed by progressive removal of the overstorey as natural regeneration developed (Guillebaud, 1933). This system was gradually abandoned after 1945 and greater emphasis was given to patch clearfelling systems. Despite this, Abernethy is the largest remnant of Caledonian pinewood forest in Scotland with considerable diversity of stand structure (Summers *et al.*, 1997). A change of ownership in the 1980s resulted in much greater emphasis on conservation and amenity objectives and there has been no systematic management for timber in the last two decades. The Abernethy plot forms part of a 20–30 ha area of similar structure which shows no evidence of timber harvesting for at least 30 years and is considered to be developing towards a structure typical of semi-natural pinewoods (C. Edwards, personal communication).

Assessments

A standard set of tree parameters was assessed in the three plots between 2000 and 2003. The plots were all either square or rectangular in layout. Diameter at breast height (1.3 m above ground: d.b.h.) and height of all trees of >7 cm d.b.h. were recorded, together with their x, y positional coordinates mapped to the nearest 10 cm using a Criterion laser supplemented by compass and tape measure as necessary. An increment core was taken from all trees at 1.0 m above ground in the old Glen More and Abernethy plots and age (at 1.0 m) calculated from counting of rings via a travelling microscope using standard procedures (Stokes and Smiley, 1968). We did not take cores in the young Glen More stand because the planting history and age could be confirmed from local records. At Abernethy and in the old Glen More stand, any dead trees or those where increment cores were unobtainable (<1 per cent of the total) were excluded from the analysis.

Spatial indices

The indices we used to analyse the data from the three stands were the following. Details of the formulae used to calculate each index are given in Table 1 together with other supporting information. In the paragraphs below we provide a brief overview of each index:

- 1 The 'aggregation index' of Clark and Evans (1954) where the average distance between trees and their nearest neighbours (r_A) is compared with the expected mean distance if the trees were randomly positioned (r_E). Values of CE lie between 0 and 2.1491; if the index is <1, there is deemed to be clustering, while values of >1 denote regularity. A completely random pattern will have a value of 1. The significance of any deviation from an index value of 1 can be tested using the SD of r_E in a randomly distributed forest of given density (Kint *et al.*, 2000).
- 2 The 'uniform angle (contagion) index' (W_i) of von Gadow *et al.* (1998) which can be used to evaluate small-scale pattern based on the regularity of the positions of the four nearest neighbours to a given tree (Pommerening, 2002; Aguirre *et al.*, 2003). Pommerening (2002) followed Albert (1999) in suggesting

that mean W_i values of <0.5 indicates a regular distribution, values of 0.5–0.6 characterize a random distribution, while those >0.6 are found with clumped distributions. Comparing the distribution of the five possible values of W_i (0, 0.25, 0.5, 0.75 and 1.0) provides more detailed insight into structural variability within a stand (Aguirre *et al.*, 2003).

- 3 The 'diameter differentiation index' (T_{ij}) which provides information on the spatial distribution of tree sizes (Pommerening, 2002). Values of the index vary between 0 and 1, where low values indicate that the neighbouring trees are similar in size to the reference tree while high values indicate variation in diameter between neighbouring trees.
- 4 Ripley's 'K function' (K_r) was developed to analyse mapped spatial point data (Ripley, 1977) and has been used in forestry to describe spatial structure on a continuous basis (Stoyan and Penttinen, 2000). It provides an indication of how many trees are within a given distance from the average tree.
- 5 We calculated the K statistic for each stand using a SAS macro SPATIAL (Moser, 1987). The statistic was estimated in half metre increments up to a distance of ~80 m and a kernel width of 1.9 m. Where required, prior to calculation the tree coordinates were rotated to form a classically orientated rectangular lattice and translated to have a (0,0) origin.
- 6 We renormalized the K function through the L function (Wiegand and Moloney, 2004) to make variance constant and easier to interpret. This approach avoids some of the problems associated with large variations in values of K at short distances where it is based on few data. Note that the standard graphical outputs from Moser (1987) used in this paper show values of $r - L(r)$ so that positive values indicate regularity and negative values signify clumping.
- 7 The 'Pair' (g_r) and 'Mark' (diameter) (k_r) 'Correlation' functions are based on point process statistics for pairs of trees and depend upon the distance between trees (r) (Penttinen *et al.*, 1992). These functions can be plotted as a graph showing either the pattern of variability of tree locations (g_r) or the mutual interaction of trees in terms of an attribute (k_r), in this case diameter (Pommerening, 2002). When trees are distributed at random, $g_r = 1$, while

Table 1: A list of the spatial indices and functions used to analyse data from three Scots pine stands in north-eastern Scotland.

Index	Formula	Where	Interpretation	Key References
Aggregation	$CE = \frac{r_A}{r_E}$	$r_E = \frac{1}{2\sqrt{\frac{N}{\lambda}}}$ the forest stand and N = number of trees	Values of CE 0-2.1491; <1 indicates clustering; >1 denotes regularity; 1 = random	Clark and Evans, 1954; Kint <i>et al.</i> , 2000.
Uniform angle (contagion) index	$W_i = \frac{1}{n} \sum_{j=1}^n v_j$	$v_i = \begin{cases} 1, & \alpha_i < \alpha_0, \\ 0 & \text{otherwise} \end{cases}$ α_i = angle with tree i , α_0 = standard angle and $0 \leq W_i \leq 1$	Values of $W_i < 0.5$ = a regular distribution, 0.5-0.6 = random, >0.6 = clumped. Calculated with $n = 4$ nearest neighbours	von Gadow <i>et al.</i> , 1998; Pommerening, 2002; Aguirre <i>et al.</i> , 2003
Diameter differentiation index	$T_{ij} = 1 - \frac{\min_i (DBH_i, DBH_j)}{\max_i (DBH_i, DBH_j)}$	DBH_i = diameter tree i DBH_j = diameter tree j $T_{ij} \in [0, 1]$	Values of T_{ij} 0-0.3 = small differentiation, 0.3-0.5 = moderate, 0.5-0.7 = large, 0.7-1.0 = very large differentiation. Calculated with the single $n = 1$ nearest neighbour	Pommerening, 2002
Ripley's L function	$L(r) = \sqrt{\frac{K(r)}{\pi}} - r$	$K(r) = \frac{1}{N\lambda} \sum_{i=1}^n \sum_{j=1}^n I_{ij}(r)$	$I_{ij}(r) = 1$ if the distance betw een the points i and $j \leq r$; 0 otherwise	Ripley, 1977; Stoyan and Penttinen, 2000; Wiegand and Moloney, 2004
Pair Correlation function	$g(r) = \frac{d}{dr} K(r) / (2\pi r)$	$r \geq 0$	<1 typically clustering >1 inhibition 1 = random	Stoyan and Penttinen, 2000
Mark (Diameter) Correlation function	$k_m(r) = \text{mean} (m_1 m_2) / m^2$	m_1 = diameter tree 1 m_2 = diameter tree 2 m^2 = squared mean diameter mean () = mean function	For diameter, often $k_m(r) < 1$ for small r	Pommerening, 2002
Experimental variogram (Diameter)	$\gamma(r) = \text{mean} \left(\frac{1}{2} (m_1 - m_2)^2 \right) / m^2$	m_1 = diameter tree 1 m_2 = diameter tree 2 m^2 = squared mean diameter mean () = mean function	$\gamma(r)$ describes tree interaction at large r and environmental factors at small r	Kuuluvainen <i>et al.</i> , 1996; Stoyan and Penttinen, 2000.

values greater than 1 indicate, for the particular distance, a higher frequency of occurrence than in a random process. Should this be found for short inter-tree distances, then this can be an indication of clustering as is found in dense groups of regeneration. Where k_r is concerned, a tendency for values to be less than unity at close distances reveals that both trees in a pair have smaller diameters than is found over the stand as a whole. This can also be a consequence of inter-tree competition in dense groups of regeneration (Pommerening, 2002).

- 8 We also explored the spatial pattern and spatial autocorrelation of tree diameter in the three stands using an 'experimental variogram' $\gamma(r)$ analysis for each site. The variograms were isotropic and had a lag spacing of 1 m. The output from this analysis is a plot of $\gamma(r)$ as a function of distance. As the distance between points increases, the diameters become increasingly dissimilar and the variance increases. This continues until the points are so far apart that they are not related to each other and the plot approaches a flat horizontal line. The distance within which samples are still autocorrelated forms the 'range' (Kuuluvainen *et al.*, 1996).

We used the 'Crancod' software package (Pommerening, 2005, 2006) to help with the analysis of the spatial data, the calculation of the various indices and the spatial functions. We explored various methods available in 'Crancod' of adjusting for edge effects that would be incurred by trees within a plot having their nearest neighbour outside the plot and therefore unrecorded. The one used for the data presented here is the NN1 technique which uses a variable buffer zone around the edge of the plot based on the spatial arrangement of the trees. One advantage of

this method is that it was designed to work with nearest neighbour techniques and it was shown to reduce bias effectively in a comparative study of several different edge correction techniques (Pommerening and Stoyan, 2006). The large size of the plots each with >100 trees will have limited, any of errors associated with this method of edge correction (Pommerening and Stoyan, 2006).

Results

Tree assessments

Tree density ranged from slightly more than 1000 stems ha^{-1} in the young Glen More stand to ~370 at Abernethy and ~270 at old Glen More (Table 2). Tree heights were similar for all sites at ~15–17 m. The highest basal area was found at Abernethy and the lowest at old Glen More. Large trees of over 50 cm d.b.h. were only found in the old stands with both sites having a few trees larger than 100 cm d.b.h.

The diameter distribution at young Glen More showed a clear peak in the 10–20 cm class with few trees in the larger d.b.h. classes (Figure 1). There was a similar, but lesser, peak in the old Glen More stand but here there were substantial numbers of trees in the larger sizes. At Abernethy, most trees were found in the 30–50 cm range with a lower percentage in the immediately adjoining size classes and a few very large trees.

The age-class distribution at old Glen More was bimodal, with ~60 per cent of trees of ~20–60 years of age and another group of between 160 and 220 years (Figure 2). At Abernethy, some 85 per cent of the trees were between 80 and 120 years, with slightly fewer than 10 per cent of the stand between 180 and 320 years.

Table 2: Basic stand and mensurational details for three experimental Scots pine stands in northeast Scotland

Name	Plot area (ha)	Date plot established	Trees (ha^{-1})	Tree age (years)	Top height (m)	Basal area ($\text{m}^2 \text{ha}^{-1}$)	Maximum d.b.h. (cm)	Number of trees >50 cm d.b.h.
Young Glen More	1.0	2003	1012	78	15.9	37.4	40	0
Old Glen More	0.8	1930	274	20–225	16.5	25.9	100	48
Abernethy	1.0	1997	368	50–315	17.9	47.3	115	39

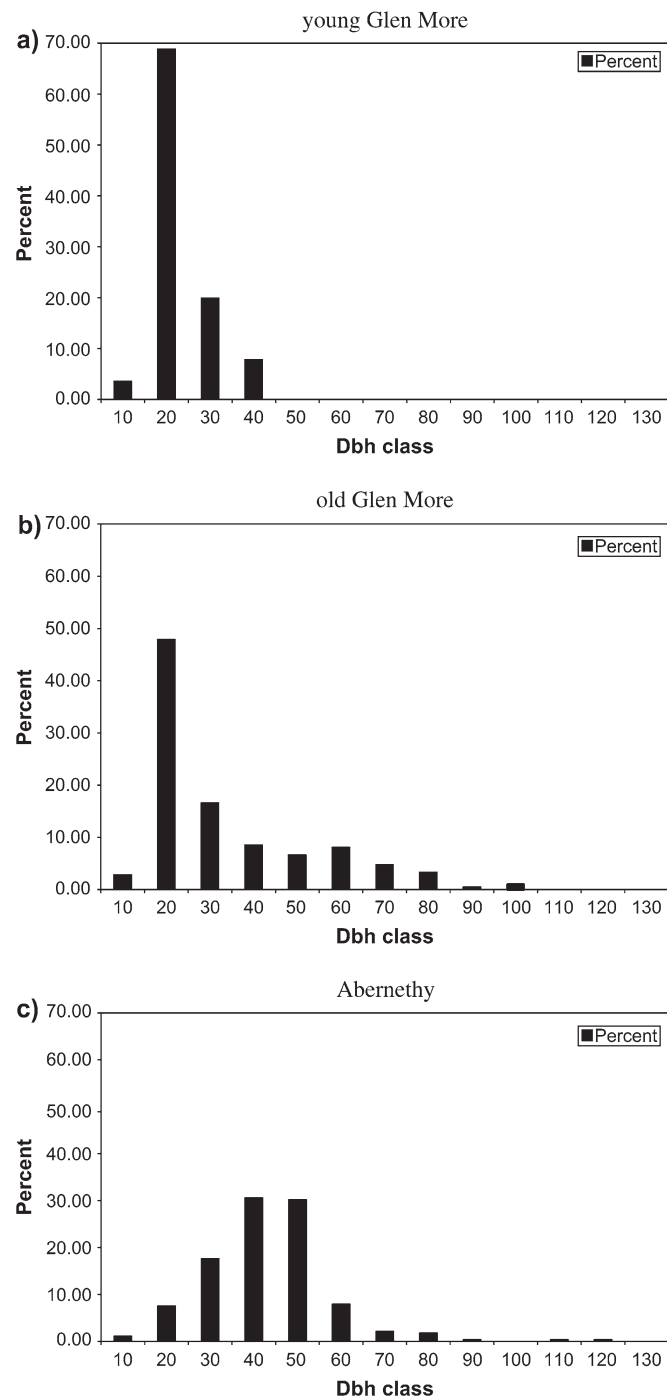


Figure 1. Percentage diameter distribution by 10 cm d.b.h. classes for the three Scots pine stands. (a) Young Glen More, (b) old Glen More and (c) Abernethy.

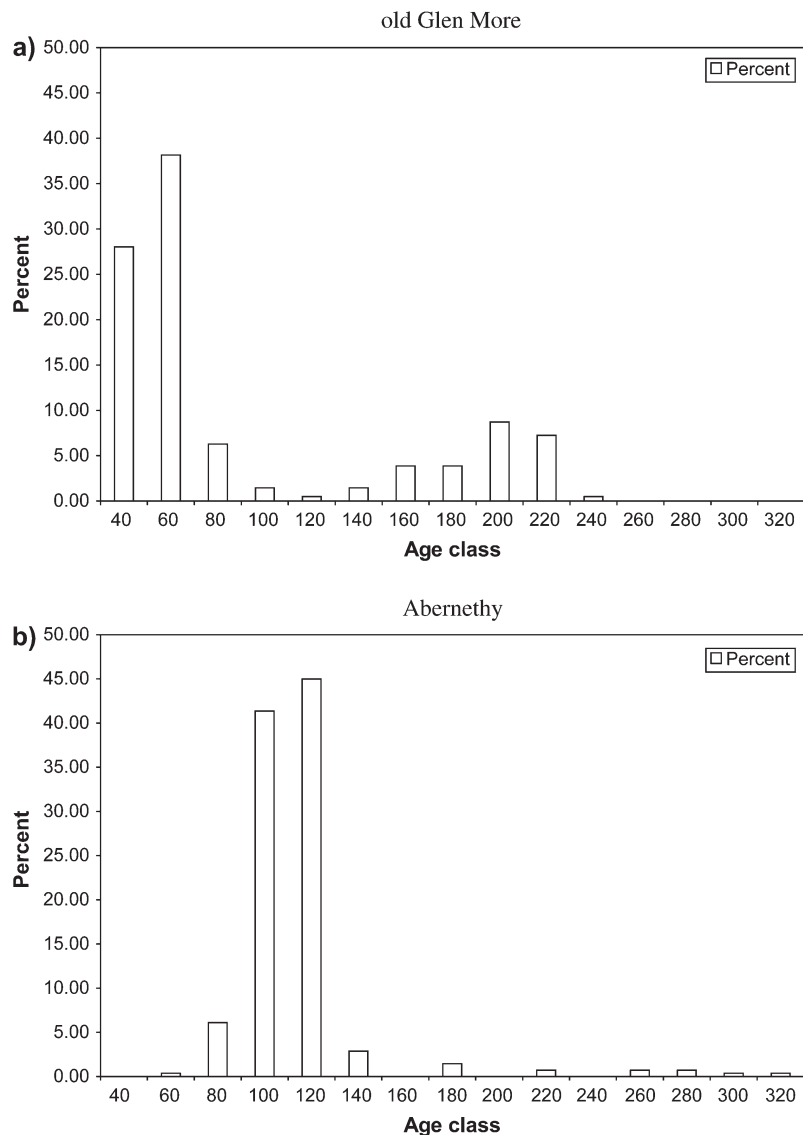


Figure 2. Percentage age-class distribution for (a) old Glen More and (b) Abernethy.

There was a positive relationship between increasing diameter and tree age at old Glen More and Abernethy with respective correlation coefficients (r) of 0.66 and 0.56. However, there was considerable scatter in the data and the relationship differed between sites. Thus, a tree of ~50 cm d.b.h. at old Glen More might range from less than 80 to over 210 years in age, whereas at Ab-

ernethy the equivalent range might be from 70 to ~130 years old.

Spatial indices

Values for aggregation, uniform angle and diameter differentiation are given in Table 3. In young

Table 3: Comparison of three different spatial indices in the experimental Scots pine stands.

Stand	Indices		
	Clark-Evans (CE)	Uniform angle (W_i)	Diameter differentiation (T_i)
Young Glen More	1.07	0.48	0.26
Old Glen More	0.90	0.53	0.30
Abernethy	0.94	0.51	0.29

Glen More the estimate for aggregation showed a regular tree distribution very highly significantly ($P < 0.001$) different from a random distribution. By contrast, the values for the stands at old Glen More and Abernethy were indicative of a clumped ($P < 0.01$) and a random distribution, respectively. However, at the latter sites, we can also separate the trees into an 'older' and 'younger' cohort based on the ages of the trees (Figure 2). At old Glen More the older cohort was defined as the trees that existed before the regeneration experiments in the 1930s and the younger one those that developed as a result of these interventions. The aggregation estimate for the former was 1.00, indicating a random pattern, while an estimate of 0.64 for the young cohort showed highly significant clumping ($P < 0.01$). At Abernethy we used 150 years to separate the older from the younger cohort. The younger trees were again clumped (0.87; $P < 0.01$), and there were only 12 trees in the older category, too few for a reliable estimate.

The mean value of the uniform angle index was similar in all three stands, although the lowest value (0.48 in the plantation; Table 3) was within the range considered typical of a regular distribution. Examination of the proportion of the different values also showed no major differences between the three stands other than a slightly higher instance of trees with value 1.0 (i.e. a clumped distribution) at Abernethy and old Glen More.

The mean diameter differentiation values also showed little variation between the three stands (Table 3). This lack of differentiation was also evident in the analysis of the proportions. The

only minor difference was that more than 60 per cent of the trees at young Glen More belonged to the first diameter differentiation class (i.e. their immediate neighbours were within 70 per cent of their own diameter) compared with 50 and 60 per cent at Abernethy and old Glen More.

Spatial functions

The actual positions of the trees in each plot are shown in Figure 3 together with the estimated values of L in relation to the value for a completely random distribution and the associated 95 per cent confidence limits.

At young Glen More, there appeared to be weakly significant regularity at very close spacings of 1–2 m. However, this trend was reversed with increasing distance so that from ~5 until 50 m there was significant clustering in tree distribution. In old Glen More, there was randomness at very short distances, but thereafter significant clustering between ~3 and 20 m and again beyond 40 m. Between 20 and 40 m, the pattern of tree distribution did not seem to deviate from randomness. At Abernethy, there was no evidence of significant deviation from randomness at any distance up to 50 m.

We explored patterns of distribution with age of tree by separating the trees at old Glen More and Abernethy into older and younger cohorts using the same age discriminators as in the aggregation analysis (see above). At Glen More the younger cohort was significantly clustered at all distances from 2 to 50 m, with this being particularly pronounced at ~10 m (Figure 4). There was much higher variation associated with the distribution of the older trees, presumably because of fewer individuals, but there was again clustering between ~5 and 35 m. At Abernethy, the results revealed few significant differences from randomness.

Pair and mark correlation functions

The pair correlation function indicated appreciable differences in spatial pattern between the three stands. In young Glen More, there was only a suggestion of clustering at very close spacing (0.5 m), but thereafter the pattern approximated to a random distribution (Figure 5a). Old Glen

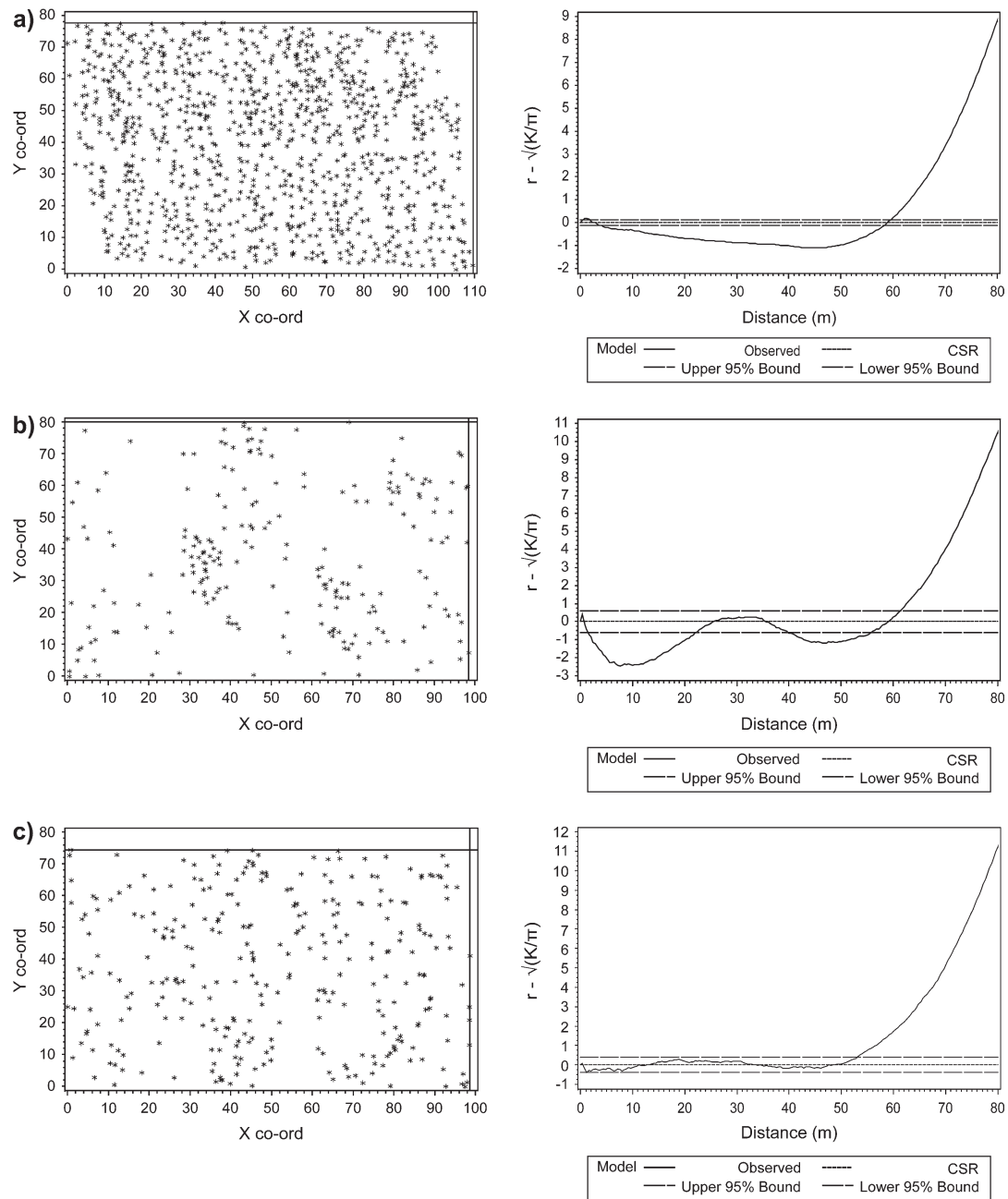


Figure 3. Spatial stem distribution and Ripley's L function for (a) young Glen More (rotated 32°), (b) old Glen More and (c) Abernethy (rotated 15°).

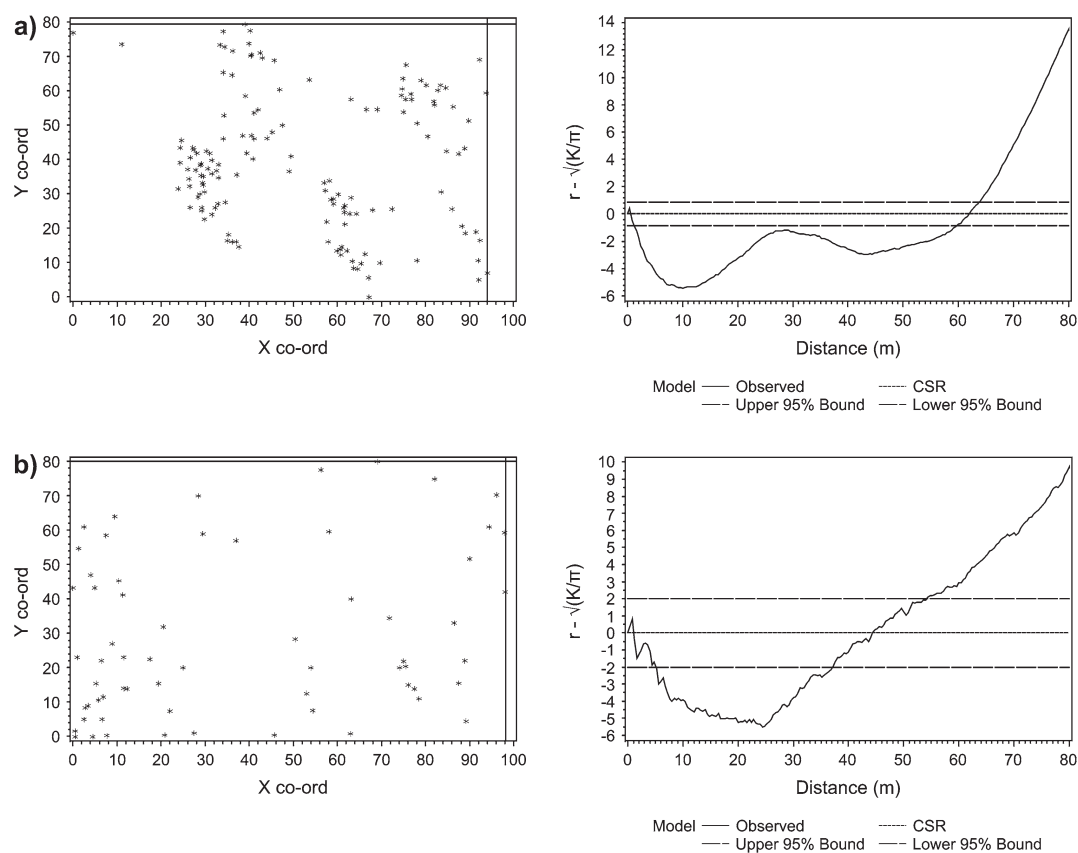


Figure 4. Spatial stem distribution and Ripley's L function for (a) the young trees at old Glen More and (b) the old trees at old Glen More.

More (Figure 5b) showed much greater evidence of clustering at all distances up to 12–13 m. There was also an indication of fewer inter-tree distances at ~20–25 m, possibly revealing some inhibition process between trees. At Abernethy (Figure 5c), clustering was found over a range of inter-tree distances up to ~3–4 m, but thereafter a random pattern prevailed.

In the plantation origin stand (young Glen More), there was little evidence of spatial correlation of diameters at any distance. By contrast, at Abernethy and particularly at old Glen More, the trees at closer distances tended to have smaller diameters than would be expected for a random process. The distances where such differences were evident were very similar to those for which a clustering effect was evident.

Experimental variograms

The patterns of variance in diameter with increasing spacing showed appreciable difference between the three stands (Figure 6). Young Glen More had the most homogeneous pattern with comparatively little difference in diameter with a high variance value at ~2 m distance, slight auto-correlation between 3 and perhaps 20 m where a sill seemed to occur. The old Glen More stand had a very different pattern with high variance at the closest 1 m spacing, strong positive auto-correlation between ~5 and 25 m, a decline in variance until ~35 m, followed by moderate auto-correlation until ~50 m where a sill may occur. At Abernethy, the pattern had many similarities to old Glen More at the closer distances, but there

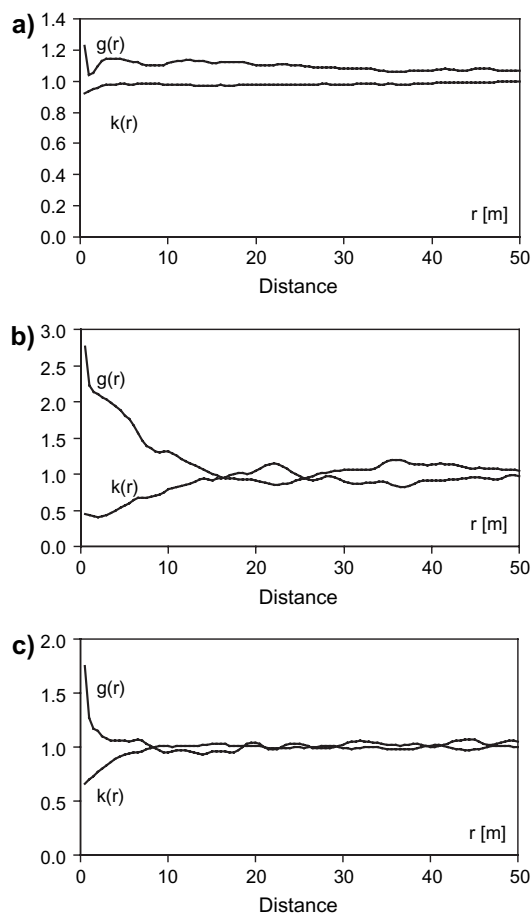


Figure 5. Correlation functions for tree position (pair – $g(r)$) and mark i.e. diameter ($k(r)$) in relation to distance (r) for (a) young Glen More, (b) old Glen More and (c) Abernethy.

appeared to be a sill reached at ~15–20 m. The exponential model provided a reasonable fit to the data at the young Glen More and Abernethy stands, but did not describe the pattern evident in the old Glen More stand.

Discussion

There is increasing recognition that the spatial pattern of tree positions and sizes within stands can be a major influence on the flow of benefits provided by forests. However, most traditional

assessments used in forestry to describe stands (e.g. basal area, diameter distributions) do not give any information on such patterns and the assessments are interpreted through guidance in stand management tables which are predicated on maximizing timber production. Other patterns may be more appropriate in situations where amenity or biodiversity are major objectives of management and so better understanding of spatial structure could be helpful in providing improved guidance for multi-purpose management (Pommerening, 2002).

The three stands selected for this study occurred within close proximity on similar sites, were of similar height and yet had contrasting structures. The young Glen More stand showed the restricted range of diameter and comparatively high stocking density characteristic of an even-aged stand being managed for timber production with thinning at regular intervals to remove a percentage of the trees. It exemplifies a type of structure widely found in financially mature Scots pine plantations in northern Scotland. The two semi-natural stands had a number of common features such as a much wider range of ages than the young Glen More stand with older trees exceeding 200 years (Glen More) or 300 years (Abernethy). The diameter distributions were also much wider with each having ~15 per cent of trees with diameters more than 50 cm and a few of 100 cm or more. Thus, both stands possessed structural features such as large old trees and a wide range of tree sizes considered characteristic of old Caledonian pine-wood stands (Steven and Carlisle, 1959; Mason, 2000). Similar characteristics have been reported from natural pine stands in Finland where studies of diameter distributions have shown a wider range of diameters and more large (>40 cm diameter) trees in natural compared with managed pinewoods (Rouvinen and Kuuluvainen, 2005).

There were also substantial differences between the two semi-natural stands, in common with other recent studies in Scottish pinewoods (Edwards and Mason, 2006). The Glen More stand, with a lower basal area, had a diameter distribution which, with the exception of the smallest size class, showed similarities to the negative exponential ('inverse J') pattern considered to be characteristic of irregular stands that are thought to be self-sustaining (Hett and Loucks, 1976; Nyland, 1996). Puumalainen *et al.* (2002) reported that

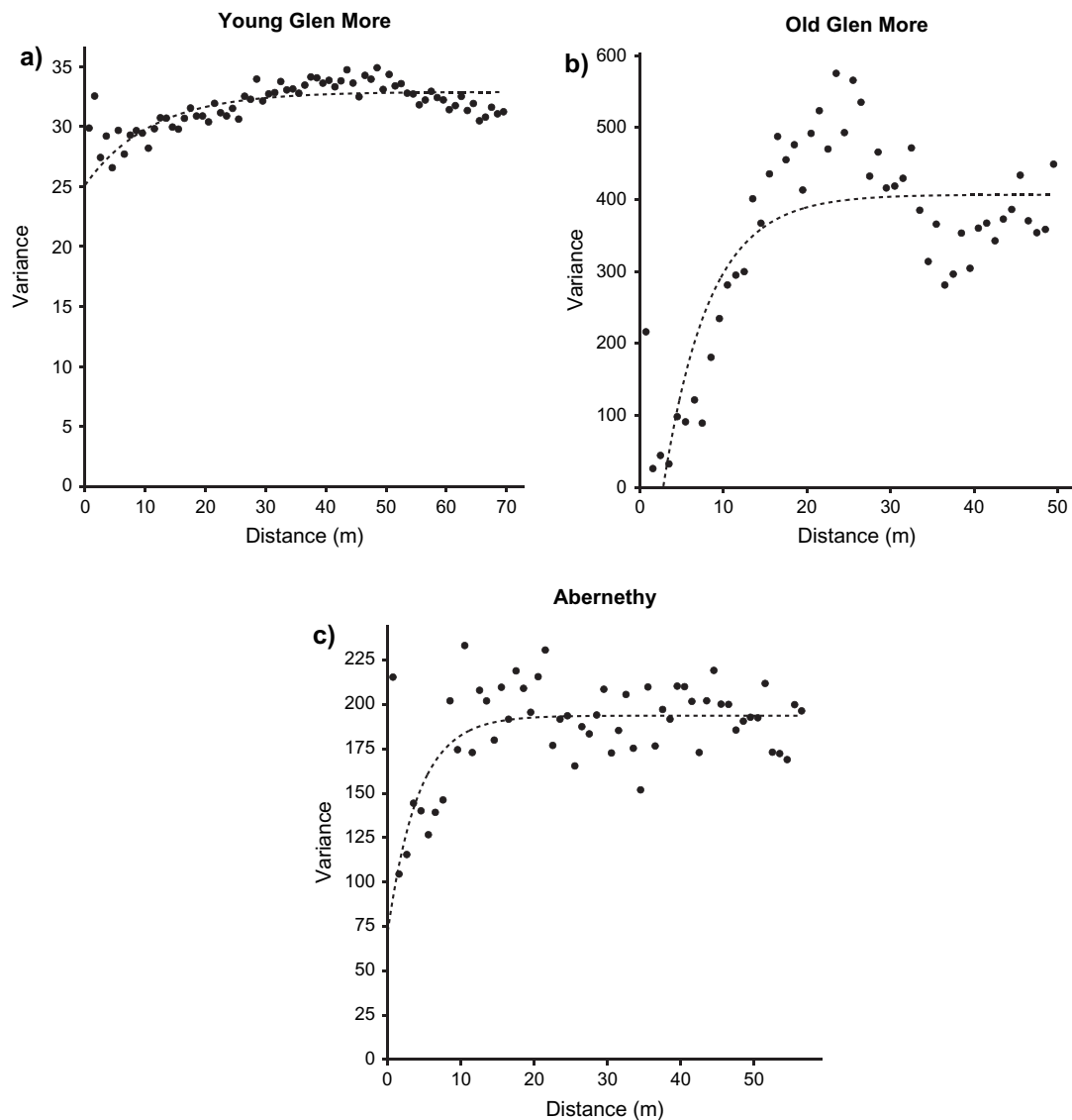


Figure 6. Experimental variogram showing variance of tree diameter as a function of distance for (a) young Glen More, (b) old Glen More and (c) Abernethy.

a similar distribution has been found in a range of boreal forest types. They suggest that this is a successional stage that occurs several decades after an event promoting a cohort of regeneration (e.g. fire, scarification). By contrast, at Abernethy there was a much higher basal area and the diameter distribution showed a peak in the

middle sizes but with a wider spread than found in the young Glen More stand. The diameter distributions did not reflect the age-class distribution at either site, in line with other reports than have shown wide variation around the expected positive correlation between diameter and age in Scottish pinewoods (Nixon and Cameron, 1994;

Edwards and Nixon, 1997; Edwards and Mason, 2006).

Initial examination of the spatial distribution of the trees showed a few differences between the plantation and the semi-natural stands. The aggregation index suggested a regular pattern in the young Glen More stand and a random or clumped distribution in the two older stands. Kint (2003) used the same index to show that Scots pine stands of 70–160 years in forest reserves in The Netherlands tended towards either a regular or a random pattern, with the latter characteristic of lower stand densities. Pommerening (2002) also reported regularity in two younger stands of conifers and broadleaves with randomness in an older mixed broadleaved stand. There was little difference between the three stands in the mean value of the uniform angle index. Other studies (Pommerening, 2002; Aguirre *et al.*, 2003) report similar mean values in the range of 0.51–0.57. Neumann and Starlinger (2001) also found this index to be less sensitive to differences in neighbourhood pattern.

Kent and Dress (1980) used a modelling approach to show that regular patterns of tree distribution would become random over time, assuming random mortality and no regeneration. However, where regeneration occurs and becomes established, it may be necessary to separate the various cohorts within a stand since in both semi-natural stands the aggregation index showed a regular distribution in the older cohort and evidence of clumping in the younger one. Similar patterns of clumping in younger cohorts have been reported for natural Scots pine stands in Latvia (Brumelis *et al.*, 2005).

More detailed analysis of tree spatial distribution using Ripley's L function highlighted additional differences between stands. At young Glen More, a more complex pattern was revealed than was apparent using the spatial indices. There was evidence of regularity at very close spacing presumably reflecting the original planting pattern of ~1 m spacing that was the norm in the 1920s. The clustered distribution revealed from ~5 to 50 m is thought to be a consequence of the local thinning regime which involved selective removal of better quality co-dominant stems for telegraph poles (D. Jardine, personal communication). This will have had the effect of allowing groups of subdominant trees to survive longer than might be expected under normal low thinning regimes.

This pattern is different from that at old Glen More where there was a 'wave-like' distribution with greatest evidence of clustering at ~10 and 45 m distances. This pattern appeared to reflect the distributions of the older and younger cohorts. Clustering in the older trees was most evident at a distance of 15–25 m, and this pattern suggests that these trees were not retained for silvicultural reasons as 'seed trees' since a more regular pattern would have been evident. The 'wave-like' pattern appeared to be caused by the distribution of the younger cohort that was affected by the layout of the 1930s regeneration trials. The latter used a plot size of 20 by 20 m but only those treatments with simulated strip cultivation at 1 m spacing were successful (Thompson *et al.*, 2003; Edwards and Rhodes, 2006). Thus, the peak at 10 m appears to be caused by plot size with groups of regeneration in the cultivated treatments, while that at 50 m reflects the distribution of cultivated plots. Over time, increased competition among the regeneration may result in a move towards randomness or even regularity as a consequence of self-thinning (Kent and Dress, 1980; Stoyan and Penttinen, 2000).

By contrast, in Abernethy the distribution of all stems appeared to be random. Examination of the younger cohort gave a suggestion (very weakly significant at best) of clumping at 5–10 m. This probably reflects groups of successful regeneration occurring away from the light and moisture competition exerted by the mature trees (Valkonen, 2000). Kuuluvainen *et al.* (1996) found a random pattern in a natural Norway spruce stand in southern Finland, while a managed stand showed regularity at distances of up to 4 m because of the effect of low thinning.

Further evidence of clustering in these stands is provided by the pair correlation function. In all three cases, the tendency for more trees at close spacing suggests an effect due to the occurrence of groups of regeneration. This is surprising for the plantation stand and appears to indicate that there was also some natural seeding on this site. The general trend is distinct from that reported from a number of plantations in continental Europe where there is often inhibition at close spacings because of the effect of planting distance and subsequent thinning (Penttinen *et al.*, 1992; Pommerening, 2002). At young Glen More and Abernethy, the range of distances over which

interaction between trees was observed is less than 10 m in line with other reports (Stoyan and Penttinen, 2000). However, at old Glen More, the range at 15 m is appreciably greater as a result of the impact of the regeneration experiment.

The mean values of diameter differentiation indicated comparatively little difference between stands. Pommerening (2002) recorded a much higher mean value for a two-storied mixed broadleaved stand (0.42) compared with two regular conifer and broadleaved stands (0.25 and 0.21, respectively). It is surprising that there was not a greater difference between the young stand and the two older ones, given that the latter had greater variations in diameter and age. This may reflect the light-demanding nature of Scots pine, in that regeneration develops in clumps in gaps in the canopy, but is rarely found under an intact canopy (Malcolm, 1995). The dominant trees in the clumps grow comparatively rapidly towards the canopy so that it is unusual to find more than two layers within a stand. Kuuluvainen *et al.* (1998) also noted a bimodal distribution of heights (i.e. only small and large trees) in a natural Scots pine stand in eastern Finland. A consequence of this gap-based regeneration process is that nearest neighbours will tend to be of more similar size than might be found in more intimate stand structures containing more shade-tolerant species. The mark correlation function also indicated that all three stands tended to have more similar diameters at close spacing, probably as a result of the pattern of regeneration.

Examination of spatial distributions of diameters using variograms indicated similarities but also differences in patterns between the sample stands. In all cases there was high variance at close distances of 1–2 m (Figure 6). This trend has been reported in other studies (Kuuluvainen *et al.*, 1996, 1998; Kint *et al.*, 2003) and is considered to be the consequence of asymmetric competition for light between trees growing in dense clumps of regenerated trees. The occurrence of this peak in the planted stand (young Glen More) is slightly surprising since Kuuluvainen *et al.* (1996) did not find high variance at close spacing in a managed Norway spruce (*Picea abies* Karst.) stand. This is probably an artefact of the local thinning regime as discussed above. Beyond 3–5 m, at all sites there was positive autocorrelation up to a distance of ~12–20 m, suggesting a pattern of

neighbouring trees of roughly equal size competing with each other. The correlation was greater in the two natural stands than in the planted and managed stand, suggesting that a consequence of thinning was to reduce neighbourhood competition while making the overall stand more homogeneous. Beyond 20–25 m, there was little evidence of systematic pattern in either the young Glen More or Abernethy stands, but in old Glen More there was high variance at ~25 m followed by positive autocorrelation between 35 and 50 m. This has similarities to the 'wave pattern' of tree distribution noted above and reflects the effect of the regeneration plots established in the 1930s. The high variance at 25 m suggests asymmetric competition between the mature overstorey trees and the regenerating cohort. The overall pattern from Abernethy was similar to that reported from natural pine and spruce stands in Finland (Kuuluvainen *et al.*, 1996, 1998).

One outcome of this study is to indicate the variety of structural pattern that can be found within semi-natural stands in the Caledonian pinewoods. The reasons for such differences may often be a consequence of site history. At Glen More, the oldest trees (160 years or more of age) are likely to have regenerated following the heavy fellings of the early nineteenth century (Smout, 1997). The sizeable cohort of younger trees (40–80 years of age) reflects the favourable regeneration conditions provided by strip cultivation in the experiments carried out in the 1930s (Edwards and Rhodes, 2006). By contrast, at Abernethy there are 12 trees of over 150 years of age, while the majority of the trees in the stand are between 90 and 130 years old. This distribution would be compatible with Guillebaud's description (1933) of the uniform shelterwood system used on Seafeld estates. Thus, a series of seeding fellings would have been started in the 1880s with progressive removal of the overstorey as the regeneration developed. The reason for the retention of the few remaining old trees is unclear, but it is possible that these were destined for a local market requiring very large dimension timber. These differences suggest that it would be unwise to assume that there is a single model for the development of old pinewood stands. There may be a range of stand structures appropriate to this phase of stand development. Pommerening (2002) also pointed out that there might not be

uniformity in 'natural' stand structures between reserves, let alone between forest types.

Conclusions

In these semi-natural pinewood stands, measurements based on tree dimensions alone do not give sufficient insight for informed analysis of past history and evaluation of future management. Particularly in older stands with few records of past interventions, it is essential to complement tree measurements with information on tree age to give adequate understanding of stand history. Without the tree age data, it would be possible to misinterpret the diameter distribution at old Glen More as an example of an all aged irregular stand when it is mainly composed of two cohorts with an age difference of ~100 years (see also Kerr and O'Hara, 2000).

When combined with age data, spatial indices have the potential to provide greater understanding of stand development as shown by the different values for the aggregation index of the two cohorts at old Glen More and at Abernethy. However, in this study, neither the uniform angle nor the diameter differentiation index proved sensitive to variation in stand spatial structure, although alternative formulations such as expressing diameter indices in relation to diameter class may be more effective (A. Pommerening, personal communication). A problem with spatial indices is that they provide a snapshot of average dispersion within a stand but are less effective in revealing the detailed pattern and causal processes (Stohlgren, 2002; Kint *et al.*, 2003). They do have the advantage of not requiring specific tree positions to be determined and the data can be collected in routine inventories.

The various spatial functions and the variogram of tree diameter provide greater insight since they revealed substantial differences between the underlying patterns in the two older stands. They also indicated subtle differences in spatial pattern in the plantation stand that were not evident from the indices. The disadvantage with such functions is the cost of obtaining tree positional data. The advent of remote sensing technology may start to make such intensive measurements more realistic, at least in research programmes dealing with applied stand dynamics.

An ultimate aim of this study was to see whether the information could be used to formulate preliminary guidance on how to develop more natural structures within pine plantations. Current pine plantation management is producing stands with more uniformity of tree sizes and a different spatial pattern than can be found in semi-natural stands. Yet these results, and other studies (Edwards and Mason, 2006), reveal the extent to which the 'natural' structure of old Caledonian pinewood stands in Scotland are often a legacy of forest management decades or even centuries ago. Therefore, the issue is not whether plantations can be converted to more natural structures but how this is best achieved. One feature of this study is that the major differences in spatial structure between stands occur at distances of less than ~50 m. This suggests that any attempt to foster greater variation in plantation stands (e.g. through VDT) should take place at a comparatively intimate scale of perhaps 10–20 m. A possible approach compatible with this scale would be to use a modified crown thinning regime which progressively removed the competitors to 50–100 potential dominant trees per hectare. One consequence could be to increase the range of diameters compared with current plantations while avoiding the development of the regular structure characteristic of thinning from below (Pretzsch, 1998). Such an approach would also promote trees with the potential to develop into the large diameter specimens characteristic of the native pinewoods. This would be compatible with greater use of shelterwood or seed tree silvicultural systems in pine plantation management. Indeed, evidence from this study is that the larger trees characteristic of semi-natural pinewood stands may often be the remnants of past seeding fellings (e.g. Abernethy). Studies in Scots pine forests in central Spain also suggest that shelterwood systems can increase structural and spatial diversity (Montes *et al.*, 2005). Thus, greater use of silvicultural systems associated with Continuous Cover Forestry (Mason *et al.*, 2004) can be seen as a means of developing an integrated management of the whole pinewood resource in northern Scotland.

However, this hypothetical approach still lacks information on the appropriate intensity and timing of thinning to produce a more natural

structure from a younger regular plantation stand. To achieve this goal, two aspects need to be pursued. Firstly, further spatial analysis of both young and especially old pinewood stands needs to be undertaken. The latter is of particular importance because of the substantial difference that this study has revealed between two semi-natural stands growing in the same region. Such investigations should provide a better understanding of the range of structures that can be found in old stands in the native pinewoods. However, we also need to confirm that the patterns found in the plantation stand are an adequate reflection of those found in other pine plantations in northern Scotland. Secondly, simulations of the effect of different patterns of thinning should be undertaken to see the outcome of a range of potential VDT strategies. Such simulations are needed because of the lack of long-term datasets showing how stands respond to different interventions. They should be based on the use of single-tree, distance-dependent growth models to 'grow' alternative stand structures over time and check their suitability for purpose (Stoyan and Penttinen, 2000). Without the knowledge gained from such research, there is a danger of silvicultural regimes being implemented to 'naturalize' pine plantations that do not produce desired structural outcomes and fail to optimize habitat provision, amenity value or timber quality.

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Conflict of Interest Statement

None declared.

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Reconstructing spatial tree
point patterns from nearest
neighbour summary statistics
measured in small
subwindows

Reconstructing spatial tree point patterns from nearest neighbour summary statistics measured in small subwindows

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Abstract: Spatial tree data are required for the development of spatially explicit models and for the estimation of summary statistics such as Ripley's K function. Such data are rare and expensive to gather. This paper presents an efficient method of synthesizing spatial tree point patterns from nearest neighbour summary statistics (NNSS) sampled in small circular subwindows, which uses a stochastic optimization technique based on simulated annealing and conditional simulation. This nonparametric method was tested by comparing tree point patterns, reconstructed from sample data, with the original woodland patterns of three structurally different tree populations. Analysis and validation show that complex spatial woodland structures, including long-range tree interactions, can be successfully reconstructed from NNSS despite the limited range of the subwindows and statistics. The influence of the NNSS varies depending on the woodland under study. In some cases, the sampling results can be improved by reconstruction. Furthermore, it is clearly shown that it is possible to estimate second-order characteristics such as Ripley's K function from small circular subwindows through the reconstruction technique. The results offer new opportunities for adding value to woodland surveys by making raw data available for further work such as growth projections, visualization, and modelling.

Résumé : Les données spatiales au sujet des arbres sont nécessaires pour développer des modèles spatialement explicites et pour estimer les statistiques sommaires telles que la fonction K de Ripley. De telles données sont rares et coûteuses à acquérir. Cet article présente une méthode efficace pour synthétiser les patrons de points qui représentent des arbres dans l'espace à partir des statistiques sommaires du plus proche voisin, échantillonnées dans de petites sous-fenêtres circulaires. Cette méthode utilise une technique d'optimisation stochastique basée sur le recuit simulé et la simulation conditionnelle. Cette méthode non paramétrique a été testée en comparant des patrons de points représentant des arbres, reconstitués à partir de données d'échantillonnage, avec la structure forestière originale de trois populations d'arbres différentes du point de vue structural. L'analyse et la validation montrent que la structure spatiale complexe de la forêt, incluant les interactions à long terme entre les arbres, peut être reconstituée avec succès à partir des statistiques sommaires du plus proche voisin malgré l'étendue limitée des sous-fenêtres et des statistiques. L'influence des statistiques sommaires du plus proche voisin varie selon la forêt à l'étude. Dans certains cas, les résultats de l'échantillonnage peuvent être améliorés par la reconstitution. De plus, il est clairement démontré qu'il est possible d'estimer les caractéristiques de second ordre telles que la fonction K de Ripley à partir de petites fenêtres circulaires par la technique de reconstitution. Les résultats offrent de nouvelles opportunités pour valoriser les inventaires forestiers en rendant disponible des données brutes pour d'autres travaux tels que des projections de croissance, de la visualisation et de la modélisation.

[Traduit par la Rédaction]

Introduction

To understand many problems in woodland ecology and management, detailed spatial tree data are a necessary prerequisite. For example, computer-based habitat and landscape models need such information (Letcher et al. 1998; Merrill et al. 1999) and are becoming important tools in environmental and conservation planning (Seppelt and Voinov 2002; Niklitschek and Secor 2005; Carroll and Miquelle 2006). Spatially explicit tree growth models, sampling simulators, and landscape visualization tools also require detailed data including tree characteristics and tree positions (Pommerening 2000; Moravie and Robert 2003; Bauer et al. 2004;

Hasenauer 2006; Pommerening 2006). In many biological and ecological studies, Ripley's K function is used (e.g., Penttinen et al. 1992; Moer 1993). This second-order characteristic is normally estimated from fully mapped tree point patterns in large observation windows. (The term "observation window" in the context of this paper refers to a two-dimensional area or tract of trees.) Methods of simulating spatial patterns are also used to study the performance of summary characteristics (Torquato 2002; Pommerening 2006).

A full enumeration and mapping of trees in large woodlands is usually not feasible because of the costs involved, and spatially explicit tree data are collected on a sample ba-

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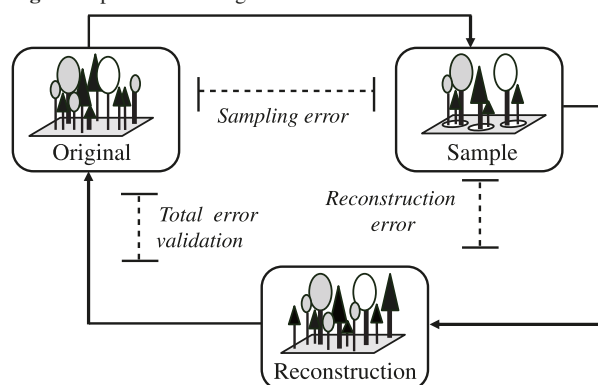
sis, for example, from circular sample plots. Therefore, when simulating woodland structure, methods are required that allow a full and detailed spatial analysis of woodlands from the limited information available. If the model parameters can be estimated from the sample data, point process models are capable of providing a partial solution to this problem and classic examples are Cox process models (Stoyan and Penttinen 2000) and Gibbs process models (Tomppo 1986; Degenhardt and Pofahl 2000). However, these classical approaches are perhaps too simplistic, do not provide great detail, and therefore are not necessarily the best solution to the problem discussed in this paper. Also, heterogeneity in the natural environment makes the application of stationary process models questionable. Because the original sample points are not used in modelling, other than providing the parameters, much valuable information is lost and the simulation requires additional software. Ideally, models would make direct use of the sample data using the original points from the subwindows and only generating new points outside the windows. Such simulations are said to be “conditional” and maximize the use of original data. Although not difficult, such conditional simulations are rarely described in the literature.

Reconstruction methods offer much potential for detailed simulation and several methods have already been proposed. Pretzsch (1997) developed a method that is based on an empirical function estimating the distance to the nearest neighbour and a set of probability functions combining an inhomogeneous Poisson process and a hardcore process. Other authors have tried to decrease the role of model assumptions and empirical parameters: Lewandowski and Gadow (1997) constructed a three-phase model approach using nearest neighbour summary statistics (NNSS), Pommerening (2000) developed a method, based on Pretzsch (1997), that generalized the empirical functions by deducting spatial information from sample data, Biber (1999) suggested a method of simulating spatial forest structure to compensate edge effects in growth simulations, and Pommerening (2006) presented an approach using NNSS and cellular automata. With the method introduced in this paper, the authors follow the latter philosophy of reducing the effect of model assumptions on the reconstruction process to a minimum to allow wider application. Based on the encouraging results achieved by Lewandowski and Gadow (1997) and Pommerening (2006), the reconstruction process of this study is also driven by NNSS.

The authors adopt the idea of reconstruction methods as developed in statistical physics (Torquato 2002). These are general stochastic optimization techniques based on “simulated annealing” (Kirkpatrick et al. 1983; Černý 1985). This approach is adapted to point processes as proposed in Tscheschel and Stoyan (2006). The general idea in the context of forestry data is to extrapolate spatial woodland structures, which have been observed in small subwindows, over the whole study area.

The aim of this paper is to show that the partial information obtained from woodland surveys can be effectively exploited by using them to reconstruct the original spatial woodland structure of the whole study area. To achieve this, data from three large, fully enumerated and mapped woodlands were resampled on a computer using circular

Fig. 1. Experimental design.



sample plots as subwindows and the original forest structures were then recreated using data from the sample trees only. The experiments include a thorough error analysis, which considers “total errors”, “sampling errors”, and “reconstruction errors” (Fig. 1).

The patterns obtained by the reconstruction method were validated using various correlation functions (Stoyan and Penttinen 2000), which not only consider short-range correlations (which are characterized by the NNSS) but also include long-range correlations.

Materials and methods

The reconstruction method presented in this study is based on a specific sampling design, which is described in the next section. Then the generic reconstruction algorithm is explained in detail. The method is tested and validated for three example woodlands.

Sampling design

The data for the tree point pattern reconstruction originate in a partial sampling of two-dimensional tree locations, diameters at breast height (DBH), and species information in large windows of observation. Instead of sampling them totally, only subsets are analysed. Because of its widespread application and efficiency, systematic sampling of circular subwindows is considered in this paper. However, modifications using other geometric shapes, such as rectangles, squares, and line transects (e.g., see Krebs 1999), to define subwindows are also possible. The radius of the circular subwindows depends on window size and forest structure. The centres of subwindows have to be determined such that the subwindows are completely within the window. The examples considered show that the total area of all circular subwindows should be around 25% of the area of the large windows.

The trees in the circular subwindows are used for estimating various statistical summary characteristics. Since the subwindows offer only short interpoint distances, the spatial summary characteristics should be related to nearest neighbour distances. The NNSS used are defined in Table 1. They are all based on reference trees indexed by i and their n nearest neighbours, ordered with respect to distance; thus, for example, index $j = 2$ corresponds to the second-nearest neighbour.

Table 1. Overview of the nearest neighbour summary statistics (NNSS) used as histograms for the contrast measure in this study, corresponding publications, their diversity aspects, and formulae, characteristics, ranges of neighbours ($j, \dots, n$), and weights used.

Eq.	NNSS (reference(s))	Diversity of:	Formula	Variable definition	$j, \dots, n$	Weight in contrast measure (eq. 2)
1.1	Distance to 1st, ..., n th neighbour (Pommerening, 2006)	Tree positions	Histograms with class width 0.5m	The distance D_j between each tree i and its j th neighbour tree is allocated to a class of the distance distribution	1.7	0.3
1.2	Mean directional index (Ullian et al. 2008)	Tree positions	$R_i = \sqrt{\left(\sum_{j=1}^n \cos \alpha_{ij}\right)^2 + \left(\sum_{j=1}^n \sin \alpha_{ij}\right)^2}$	α_{ij} is the angle between a line pointing away from the reference tree i to neighbours j and a reference bearing (e.g., due north)	3.7	0.3
1.3	Species mingling (Füldner 1995; Aguirre et al. 2003)	Tree species	$M_i = \frac{1}{n} \sum_{j=1}^n m_j$	$m_j = \begin{cases} 1, \text{species}_j \neq \text{species}_i \\ 0, \text{otherwise} \end{cases}$	3.7	0.1
1.4	DBH differentiation (1) (Füldner 1995; Pommerening 2002)	Tree dimensions	$TD_i = 1 - \frac{\min(\text{DBH}_i, \text{DBH}_j)}{\max(\text{DBH}_i, \text{DBH}_j)}$	n th neighbour tree of i	1.7	0.1
1.5	DBH differentiation (2) (Füldner, 1995; Gadow, 1999)	Tree dimensions	$TDm_i = 1 - \frac{1}{n} \sum_{j=1}^n \frac{\min(\text{DBH}_i, \text{DBH}_j)}{\max(\text{DBH}_i, \text{DBH}_j)}$	$j = 1, \dots, n$ neighbour trees	2.7	0.1
1.6	DBH dominance (Gadow and Hui 2002; Aguirre et al. 2003)	Tree dimensions	$U_i = \frac{1}{n} \sum_{j=1}^n u_j$	$u_j = \begin{cases} 1, \text{DBH}_i > \text{DBH}_j \\ 0, \text{otherwise} \end{cases}$	3.7	0.1

Equation 1.1 in Table 1 refers to the histogram of distance to the j th nearest neighbour in discretized form; in this study, the class width is 0.5m, which is appropriate for forestry data.

Equation 1.2 concerns the mean directional index R_i . This is the length of the sum of the n unit vectors pointing from the reference tree to its nearest neighbours. This index takes large values in the case of clustering and small ones in the case of repulsion between the points. It is a good variability characteristic for nonmarked point patterns.

Equation 1.3 is the species mingling index M_i , which is defined as the proportion of the n nearest neighbours that differ from the reference tree i in terms of tree species. If M_i takes a large value, the species of the reference tree and its neighbours tend to be different, i.e., the trees are intermingled at short distances. Otherwise, there is segregation of points with different marks.

Equation 1.4 gives the diameter differentiation index TD_i , which describes the mark differences of neighbouring trees on a continuous scale. Here, the marks are diameters, but other choices are also possible. If all marks are equal, $TD_i = 0$, and if the mark differences are very large, then TD_i can approach 1.

Equation 1.5 is the diameter differentiation TDm_i that is the mean of the size difference of n neighbouring trees. It follows the principle of eq. 1.4.

Equation 1.6 gives the diameter dominance U_i , which is defined as the proportion of the n nearest neighbours that are smaller than the reference tree. Large values of U_i describe dominant trees, and small values indicate dominated or suppressed trees.

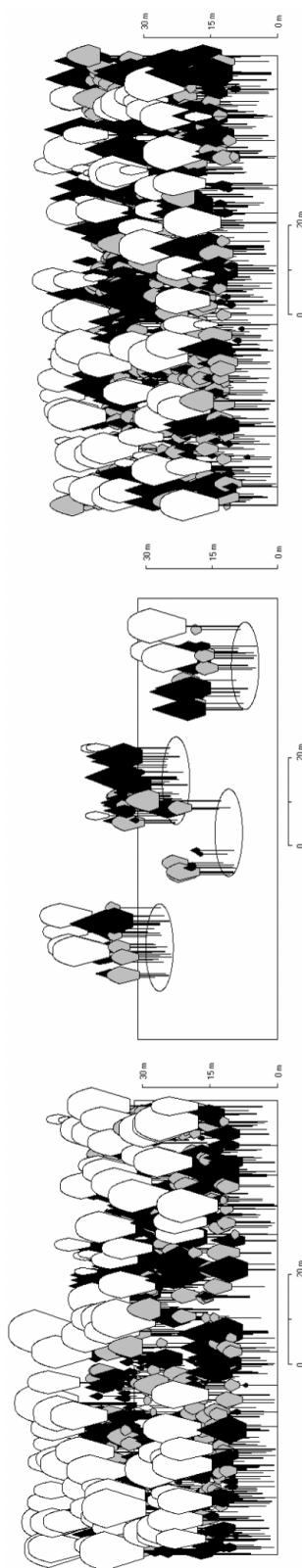
For the reconstruction, histograms of the nearest neighbour distances of eq. 1.1 and of the characteristics defined in eqs. 1.2–1.6 are used as summary characteristics because they provide more information than simple measures of location such as arithmetic means or measures of dispersion (Stoyan and Stoyan 1994; Pommerening 2006). The histograms are based on all points in the subwindows and detailed information on how to estimate them is provided in Aguirre et al. (2003), Gadow and Hui (2002), and Pommerening (2006).

Reconstruction method

Using the information from the subwindows and the summary characteristics described above, point patterns in the whole window were generated numerically. The aim is that these are as similar as possible to the true point pattern in the whole window. The reconstruction is performed as “conditional simulation”, i.e., the positions of the points in the subwindows are fixed during the whole reconstruction process (Fig. 2). Outside the subwindows, new points are simulated in such a way that the whole point pattern has NNSS as close as possible to those estimated for the circular subwindows. This is carried out by an iterative optimization procedure.

A convenient choice of the initial configuration is a completely random point distribution with marks (species and diameters) copied randomly (following the bootstrap method and sampling with replacement described by Schreuder et al. (1993) and Krebs (1999)). The total number of points in the whole window is calculated as the point density estimated from the subwindows multiplied by the window area. Thus, the number of points outside the subwindows is this product less the combined number of all trees in the subwindows.

Fig. 2. Illustration of the conditional simulation technique applied to the Białowieża woodland (Poland): original forest (left), sample with the circular subwindows serving as “structural islands” (centre), and reconstructed forest (right). Understorey: open, pendunculate oak; shaded, Scots pine. Overstorey: solid, Norway spruce; open, birch; shaded, hornbeam.



An improved initial configuration in the case of repulsion between the trees is a modified Matérn hardcore process (Stoyan et al. 1995), which is a simple point process model with some degree of inhibition between the points expressed by the hardcore distance h (= minimum inter-point distance), which can be estimated from the subwindows. This initial configuration is then subsequently improved by a stochastic optimization technique (Torquato 2002; Tscheschel and Stoyan 2006).

The reconstruction technique is based on the simulated annealing method, which has found various applications also in ecology and forestry (Jørgensen et al. 1992; Liu et al. 2000; Baskent and Glen 2002; Chen and Gadow 2002; Öhman and Ljusk 2002). It can be considered as an application of the so-called Joshi–Quiblier–Adler (JQA) device, which has originally been applied in physics for heterogeneous microstructures (Rice 1945). A detailed description is given in Torquato (2002), and Tscheschel and Stoyan (2006) discuss its modification for the reconstruction of point patterns, which is used in this paper. The algorithm works as follows.

(1) A given marked point configuration is characterized by a contrast measure C_{old} , which quantifies the difference between the NNSS estimated from the subwindows and the reconstructed point pattern in the whole window.

(2) To improve this configuration, one randomly selected marked point is moved to a random “candidate” position and the corresponding new contrast measure value C_{new} is calculated.

(3) If $C_{\text{new}} < C_{\text{old}}$ and $h_{\text{new}} \geq h_{\text{sample}}$ (the new minimum interpoint distance h_{new} in the simulated pattern is not smaller than the hardcore distance h_{sample} observed in the subwindows), the candidate tree position is accepted and replaces the old position and C_{new} is set as C_{old} . Otherwise, the candidate tree position is rejected.

(4) Steps 2 and 3 are repeated until the current contrast measure C_{old} falls below a preset value or a maximum number of iterations is exceeded.

The reconstruction method mainly depends on the choice of the contrast measure C (also referred to as energy function), which is at the heart of and fuels and controls the simulation process. As usual in physics (see Torquato 2002), it is recommended that C be defined as a least-squares error function, which can be written as

$$[1] \quad C = \sum_{i=1}^k \left(\hat{f}(x_i) - f(x_i) \right)^2$$

where k is the discrete number of classes of a histogram, $\hat{f}(x)$ is the histogram estimated from the reconstructed data, $f(x)$ is the histogram estimated from the data of the circular subwindows, and x values are class means.

In this paper, the total contrast measure C consists of a number, $m = 6$, of individual contrast measures, C_i , corresponding to histograms of the different summary characteristics of Table 1 multiplied by specific weights, w_i , which evaluate the importance of the various NNSS:

$$[2] \quad C = \sum_{i=1}^m C_i w_i$$

The weights w_i used in this study were determined through

preparatory trial simulations and reflect the relative importance of the NNSS in the reconstruction process. They are given in Table 1.

Once the reconstruction is finished, a point pattern in the whole window has been established. Inside the subwindows, it is identical to the pattern originally observed. Outside the subwindows, it is irregular but has NNSS similar to those of the true pattern. Also, it is expected that other summary characteristics of the reconstructed pattern are similar to those of the original pattern. For example, this may be true for second-order characteristics such as Ripley's K function or the pair correlation function. These functions can be simply obtained from a large number of replications, e.g., 100, and the final K function estimated by averaging the functions of all replications. If a number of replications of the same forest pattern are required, independent runs of the algorithm with the same parameters can be performed.

Testing the reconstruction algorithm

The reconstruction method described above was tested by applying it to three different woodland patterns for which all two-dimensional tree coordinates, DBH, and species information were given in large observation windows. For these patterns, the sampling with circular subwindows was simulated, the reconstruction algorithm was applied, and various second-order characteristics were determined. These estimates were then compared with their exact counterparts from the large original windows.

Woodland data

Because the methodology was not based on model assumptions, it is applicable to data from a wide range of plant communities. In this paper, it was used for data of very different mixed-species forests, from three European countries, with various degrees of spatial heterogeneity. The summary characteristics of the three woodlands are given in Table 2.

The site Białowieża is located in the famous Białowieża Forest in eastern Poland in the border zone between Poland and Belorussia. As well as containing one of the largest virgin old-growth forests in lowland Europe, and forming a refuge for the last free-living herd of European bison, it is increasing in importance because of long-term research into its ecology (Peterken 1996). The woodland consists of a pedunculate oak (*Quercus robur* L.) overstorey interspersed with Scots pine (*Pinus sylvestris* L.) and an understorey of Norway spruce (*Picea abies* (L.) Karst.), hornbeam (*Carpinus betulus* L.), and silver birch (*Betula pendula* Roth). It is situated at an elevation of 165 m above sea level and is part of a silvicultural experiment, outside the protected virgin forest reserve, exploring the possibility of retaining the virgin forest structure and species composition by mimicking natural disturbances during commercial timber production (Brzeziecki and Drozdowski 2005).

The Hidegvizvölgy woodland is located in the buffer zone of the Hidegvizvölgy forest reserve in the hills around the northwest Hungarian city of Sopron (near the border with Austria). Before being designated as a forest reserve, the area was part of the restricted border zone of the Iron Curtain for almost 50 years and remained undisturbed for a comparatively long time. Hidegvizvölgy has a mixed overstorey of European beech (*Fagus sylvatica* L.), sessile oak

(*Quercus petraea* (Matt.), European larch (*Larix decidua* Mill.), and Norway spruce. The understorey comprises hornbeam, silver birch, and large-leaved lime (*Tilia platyphyllos* Scop.). The forest is at an elevation of 410 m above sea level (Puttkamer 2005).

Walsdorf is a management demonstration site situated in a community woodland in the German federal state Rhineland-Palatinate. It lies within the Kalkeifel Mountains at an elevation of 560–580 m above sea level. The overstorey is formed by 103-year-old beech, a few sessile oak trees of the same age, and 85-year-old Norway spruce. The understorey consists of 47-year-old beech (Pommerening 1997).

These three research sites formed the tree populations of this study.

Experimental design

The behaviour of the reconstruction algorithm was investigated by simulation. The radius of the circular subwindows was chosen mainly for technical reasons as a compromise between the ratios of the sides of the original forests and standard recommendations in forestry text books (Loetsch et al. 1973); radii of 10 and 13 m were selected (see Table 2). The trees in the subwindows were sampled according to the plus-sampling method (Stoyan and Stoyan 1994; Stoyan et al. 1995; Pommerening and Stoyan 2006) in which off-plot neighbour trees are used in the calculation of NNSS for the trees within but close to the edge of the subwindows to eliminate edge effects. The locations of the subwindow centres were the grid points of systematic grids, which were randomly located within the window of observation.

One thousand sampling replications were carried out for each of the three original forests. The simulation experiment was conducted for each original forest in the following five steps: (1) calculation of the NNSS of the original woodland, (2) simulation of the circular subwindows of the i th sampling replication, (3) estimation of the NNSS from the data of step 2, (4) reconstruction of the original woodland in the whole window using the NNSS estimated in step 3, and (5) statistical analysis of sampling, reconstruction and total errors. Here, i is an index variable denoting the replication under study with $i = 1, \dots, 1000$.

Error and validation statistics

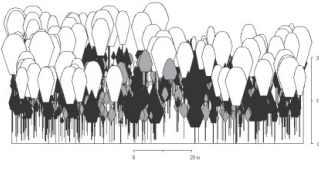
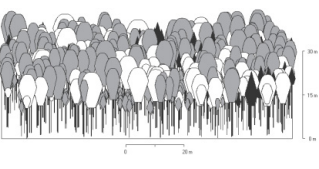
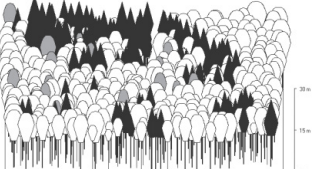
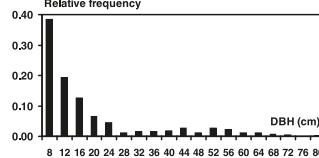
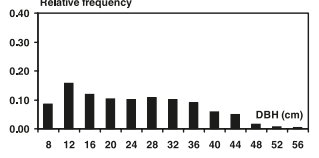
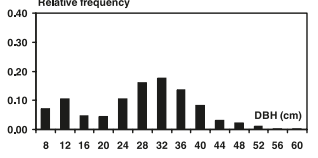
As indicators of errors, the arithmetic means of the NNSS were used, e.g., mean distances to the 1st, ..., n th neighbour (eq. 1.1 in Table 1) or the mean mingling index $\bar{M}$ (eq. 1.3 in Table 1). For the separate evaluation of sampling, reconstruction, and total errors, the relative bias (r.Bias) and relative root mean squared error (r.RMSE) were estimated for the arithmetic mean NNSS:

$$[3] \quad \text{r. Bias} = \frac{\hat{\theta} - \theta}{\theta}$$

$$[4] \quad \text{r. RMSE} = \frac{\sqrt{\frac{1}{s-1} \sum_{i=1}^s (\hat{\theta}_i - \theta)^2}}{\theta}$$

where θ is the arithmetic mean NNSS for the original forest under study, $\hat{\theta}_i$ is the arithmetic mean NNSS estimated from

Table 2. Characteristics of the Białowieża, Hidegvizvölgy, and Walsdorf woodlands.

	Białowieża (Poland)				Hidegvizvölgy (Hungary)				Walsdorf (Germany)			
Schematic												
Location	50°43'0"N, 23°50'0"E				47°32'60"N, 17°37'0"E				50°17'0"N, 6°42'0"E			
Survey date	1996				2005				1996			
Forestry characteristics	Species	<i>G</i>	<i>N</i>	<i>dg</i>	Species	<i>G</i>	<i>N</i>	<i>dg</i>	Species	<i>G</i>	<i>N</i>	<i>dg</i>
	NS	16.1	430.4	21.8	SOK	22.8	259.7	33.4	BE	18.9	346.9	26.3
	HBM	3.2	214.7	13.7	HBM	3.2	219.4	13.5	NS	15.1	143.4	36.6
	POK	14.6	83.3	47.3	BE	10.8	192.5	26.7	SOK	1.0	12.7	31.4
	BI	0.1	16.7	8.5	EL	5.5	77.6	30.1	—	—	—	—
	SP	1.0	3.9	55.4	Other	2.1	32.8	28.5	—	—	—	—
	Total	35.0	749.0	24.4	Total	44.4	782.0	26.9	Total	35.0	503.0	29.8
DBH distribution												
Age (years)	20–200				81				47–103			
Area (ha)	1.02				0.67				1.26			
Clark and Evans	0.87				0.94				1.23			
Pielou	0.25				0.38				0.58			
<i>D</i> ₁ (m)	1.64				1.68				2.73			
MDI	1.95				1.76				1.45			
Mingling	0.47				0.59				0.23			
TD	0.43				0.31				0.36			
TDm	0.43				0.36				0.34			
Dominance	0.51				0.51				0.50			
Boundaries (m)	101 × 101				100 × 67				101 × 125			
No. of trees	764				524				635			
% sample trees	33				18				20			
Grid width (m)	21.0				26.2				34.3			
Plot radius (m)	10				10				13			
No. of plots	11.5				3.8				4.9			

Note: *G*, basal area per hectare (m²); *N*, number of trees per hectare; *dg*, quadratic mean diameter (cm); DBH distribution, frequency distribution of diameters at breast height (1.3 m above ground level); age (years); approximate stand age; Clark and Evans (1954), aggregation index; Pielou (1977), coefficient of species segregation; *D*₁, mean distance to the first neighbour (eq. 1.1 in Table 1); MDI, mean directional index (eq. 1.2 in Table 1); mingling, species mingling with *n* = 4 neighbours (eq. 1.3 in Table 1); TD, DBH differentiation (1) with *n* = 1 neighbour (eq. 1.4 in Table 1); TDm, DBH differentiation (2) with *n* = 4 neighbours (eq. 1.5 in Table 1); dominance, DBH dominance with *n* = 4 neighbours (eq. 1.6 in Table 1); boundaries, boundaries of the research sites (populations); no. of trees, total number of trees of the populations; % sample trees, percentage of trees sampled out of the total number; grid width, distance between sample grid points = distance between subwindow centres; plot radius, radius of circular subwindows; no. of plots, average number of circular subwindows; BE, beech (open); BI, birch (open); EL, European larch (solid); HBM, hornbeam (shaded); NS, Norway spruce (solid); POK, pendunculate oak (open); SOK, sessile oak (shaded); SP, Scots pine (shaded). NN1 (Pommerening and Stoyan 2006) was the edge correction method used for the estimation of spatial measures.

replication *i*, and $\hat{\theta}$ is the arithmetic mean NNSS of the *s* = 1000 replications

As there are two distinctive processes, sampling and reconstruction, involved in this study, the errors were split into two different components (see also Fig. 1) to investigate how they relate to each other. The statistics referring to the

sampling error were calculated by comparing the estimated NNSS of the whole original forests with those of the simulated circular subwindows. The statistics relating to the reconstruction error were calculated by comparing the NNSS estimated from the circular subwindows with those of the whole reconstructed woodlands. A comparison of the NNSS

of the whole original forests with those of the reconstruction replications provided the total error. The results will show whether the common hypothesis that total error = sampling error + reconstruction error holds.

Figure 3 shows the results of the sampling analysis through the arithmetic means of the nearest neighbour summary characteristics of Table 1. The solid regression curves were fitted to the data of all six NNSS together.

As expected, there is generally a decreasing tendency of the r.RMSE (eq. 4) with increasing sample size; the larger the sample size, the more accurately the NNSS can be reestimated. Between the three woodlands, there are marked differences that highlight that the relationship between r.RMSE and sample size depends on the spatial structure of the woodland considered. The Białowieża woodland, for example, shows a higher average r.RMSE level (curve in Fig. 3) than the other forests. In the Walsdorf Forest, the mingling index estimator (M) is affected by a large r.RMSE caused by bias, a result that has already been reported in previous studies (Pommerening 1997). Note that if tree locations and other tree attributes (e.g., diameters, species) were stochastically independent, the arithmetic mean r.RMSE across all NNSS could be expressed as $\text{r.RMSE} = x^{-0.5}$, where x is the sample size. The deviations in the regression curves of Fig. 3 from this expected behaviour are the result of correlations between tree variables. Such correlations have been identified in many studies (e.g., Penttinen et al. 1992; Pommerening 2002).

For the reconstruction experiment, the sample size was determined as the sample size corresponding to a mean r.RMSE (eq. 4) of 5% across the six NNSS. Sample size, grid width, and plot radii in Table 2 correspond to this 5% target.

Since the NNSS used in the reconstruction only quantify short-range interactions between points, an important question to be answered was whether long-range interactions were established adequately. This question was investigated by applying a range of correlation functions and second-order characteristics (Penttinen et al. 1992; Moeur 1993; Stoyan and Penttinen 2000) to the original and the reconstructed forest data. These functions were not used during the reconstruction process and quantify the relationship between *all* pairs of points in the observation window, i.e., they explicitly go beyond the scale of nearest neighbours. For this study, the authors have selected the pair $g(r)$ and mark $k_{mm}(r)$ correlation functions, the L function $L(r)$, its marked counterpart $L_{mm}(r)$, and the mark variogram $\gamma(r)$. For the mark correlation functions, tree diameters were used as quantitative marks. To answer the question whether the species-specific interaction between trees was reconstructed correctly, the mark connection function $p_{ij}(r)$ and the partial pair correlation function $g_{ij}(r)$ were computed. For these last two functions, the trees' species attributes were used as discrete, qualitative marks. In a similar way, the nearest neighbour distance functions $D_{ij}(r)$ and $D_{ji}(r)$ were estimated for the main species. Details about the characteristics used are given in Stoyan and Penttinen (2000) and Stoyan and Stoyan (1994).

As with Degenhardt and Pofahl (2000) and Stoyan and Stoyan (1994), confidence envelopes were estimated from the minimum and maximum values of the 1000 reconstruction replications.

Results

Both the error and the validation statistics in Figs. 4–9 suggest that all three original forests were successfully reconstructed. Białowieża and Hidegvizvölgy forests were particularly well reconstructed. This is demonstrated by comparatively low r.Bias and r.RMSE values (Figs. 4 and 5) and by the graphs of the correlation functions of the original woodlands being located well inside the confidence envelopes throughout their range (Figs. 7 and 8). The correlation functions (Fig. 9) also indicate good results for the Walsdorf woodland (with the slight exception of the mark correlation function $k_{mm}(r)$ for small r), although the mingling index is affected by a high bias (Fig. 6).

The graphs of the marked L function $L_{mm}(r)$, the mark variogram $\gamma(r)$, the mark connection function $p_{ij}(r)$, the partial pair correlation function $g_{ij}(r)$, and the nearest neighbour distance functions $D_{ij}(r)$ and $D_{ji}(r)$ also indicated very good reconstruction results but are not displayed in detail. However, examples of their performance are given in Fig. 10.

These results are of great interest, as they show that the reconstruction method is able to establish realistic long-range point interactions despite the exclusive use of only NNSS.

The use of different numbers, n , of nearest neighbours had only a minor effect on the reconstruction results. Some of the statistics used are better with larger n and others worse, and based on the fact that none of the results were extremely sensitive to different values of n , five neighbours appears to be a reasonable recommendation. However, the number of neighbours can be chosen with due regard to the sampling techniques and computational power available to the individual analyst.

The r.RMSE of individual NNSS did not exceed 10% in most cases (Figs. 4–6). NNSS with maximum errors that sometimes exceed the 10% limit are “mingling” (M) (eq. 1.3 in Table 1) and the “mean directional index” (MDI) (eq. 1.2 in Table 1). The NNSS with the smallest errors is “diameter dominance” (DOM) (eq. 1.6 in Table 1). The changes in r.Bias and r.RMSE with increasing n vary with the choice of statistic and site. It is also worth noting that the error statistics of the mingling and the mean directional NNSS were negatively correlated. While r.RMSE of M in the Białowieża (Fig. 4), Hidegvizvölgy (Fig. 5), and Walsdorf (Fig. 6) woodlands decreased with increasing n , the r.RMSE of the MDI increased.

It is difficult to make a general statement about the size and relationship of sampling error, reconstruction error, and total error. Interestingly, the relationship between the three errors did not always reflect the simple expectation total error = sampling error + reconstruction error. For example, in Białowieża Forest, sampling error was often highest (Fig. 4). The reconstruction error was frequently the smallest of the three (e.g., Białowieża, Fig. 4; Hidegvizvölgy, Fig. 5). Occasionally, the total error was even smaller than one of the other two errors, especially when $n = 7$ (e.g., TD and TDM in Białowieża, DOM in Hidegvizvölgy). This finding suggests that reconstruction can potentially be used to improve or even to correct sampling results.

Figure 10 shows the results for the partial pair correlation function $g_{ij}(r)$, mark variogram $\gamma(r)$, and the nearest neighbour distance functions $D_{ij}(r)$ for the Białowieża, Hidegviz-

Fig. 3. r .RMSE (curve) as an average of the nearest neighbour summary statistics of Table 2 (regardless of tree species) and corresponding sample size for the Białowieża (left), Hidegvizvölgy (centre), and Walsdorf (right) forests. D , distance distribution (eq. 1.1 in Table 1); DOM, diameter dominance (eq. 1.6 in Table 1); M , mingling (eq. 1.3 in Table 1); MDI, mean directional index (eq. 1.2 in Table 1); TD, DBH differentiation (1) (eq. 1.4 in Table 1); TDm, DBH differentiation (2) (eq. 1.5 in Table 1). x is the percentage of total trees sampled.

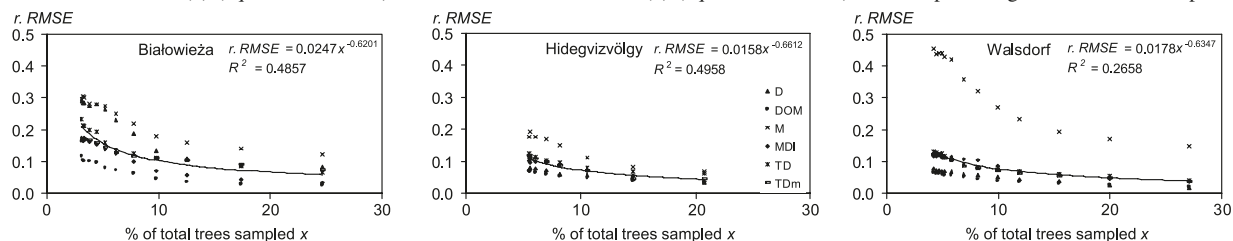


Fig. 4. r .Bias and r .RMSE for the Białowieża woodland (Poland). D , distance distribution (eq. 1.1 in Table 1); DOM, diameter dominance (eq. 1.6 in Table 1); M , mingling (eq. 1.3 in Table 1); MDI, mean directional index (eq. 1.2 in Table 1); TD, DBH differentiation (1) (eq. 1.4 in Table 1); TDm, DBH differentiation (2) (eq. 1.5 in Table 1). Open bars, sampling error; shaded bars, reconstruction error; solid bars, total error. n is the number of nearest neighbours. r .Bias and r .RMSE were calculated for $n = 3, 5$, and 7 and all species together.

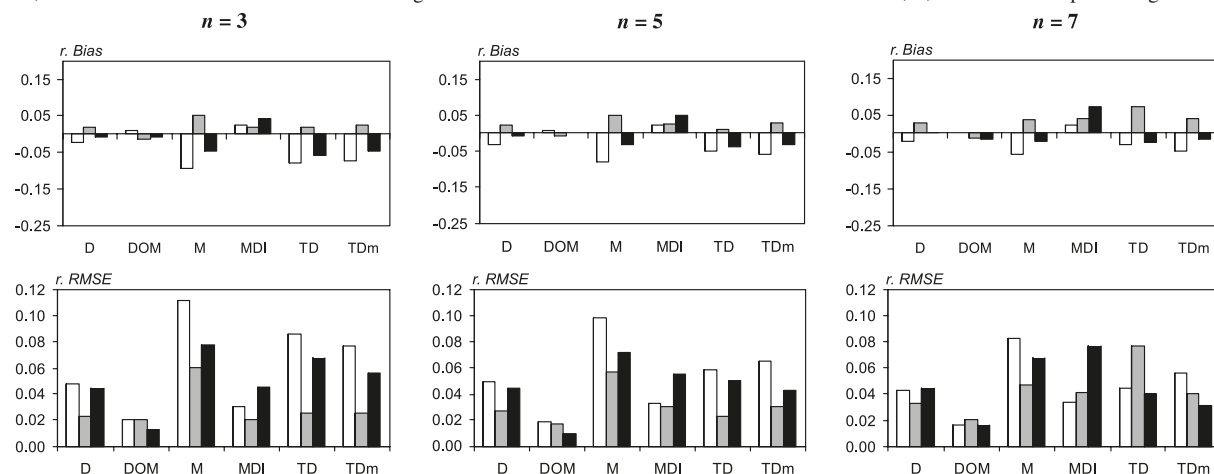


Fig. 5. r .Bias and r .RMSE for the Hidegvizvölgy woodland (Hungary). D , distance distribution (eq. 1.1 in Table 1); DOM, diameter dominance (eq. 1.6 in Table 1); M , mingling (eq. 1.3 in Table 1); MDI, mean directional index (eq. 1.2 in Table 1); TD, DBH differentiation (1) (eq. 1.4 in Table 1); TDm, DBH differentiation (2) (eq. 1.5 in Table 1). Open bars, sampling error; shaded bars, reconstruction error; solid bars, total error. n is the number of nearest neighbours. r .Bias and r .RMSE were calculated for $n = 3, 5$, and 7 and all species together.

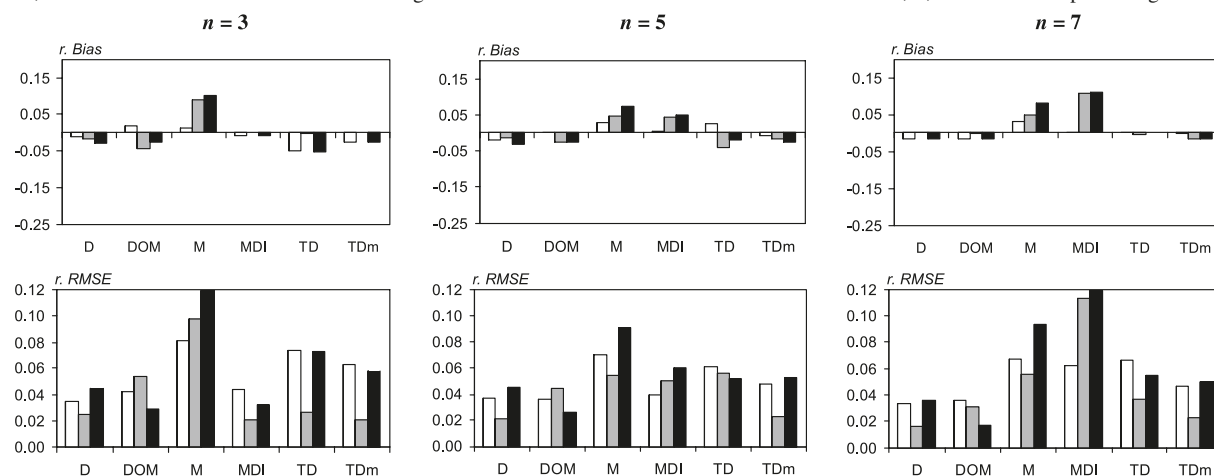


Fig. 6. r .Bias and r .RMSE for the Walsdorf woodland (Germany). D , distance distribution (eq. 1.1 in Table 1); DOM, diameter dominance (eq. 1.6 in Table 1); M , mingling (eq. 1.3 in Table 1); MDI, mean directional index (eq. 1.2 in Table 1); TD, DBH differentiation (1) (eq. 1.4 in Table 1); TDm, DBH differentiation (2) (eq. 1.5 in Table 1). Open bars, sampling error; shaded bars, reconstruction error; solid bars, total error. n is the number of nearest neighbours. r .Bias and r .RMSE were calculated for $n = 3, 5$, and 7 and all species together.

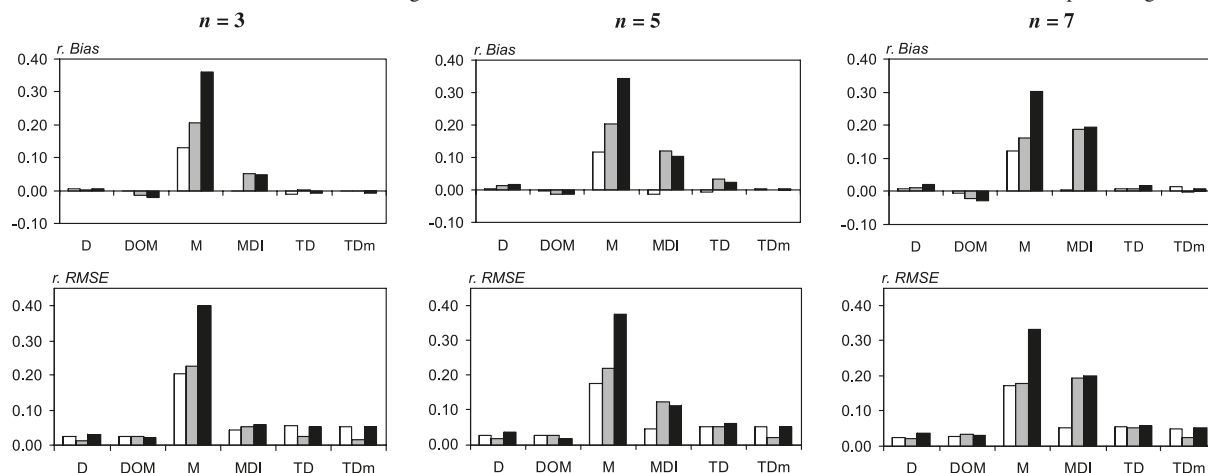
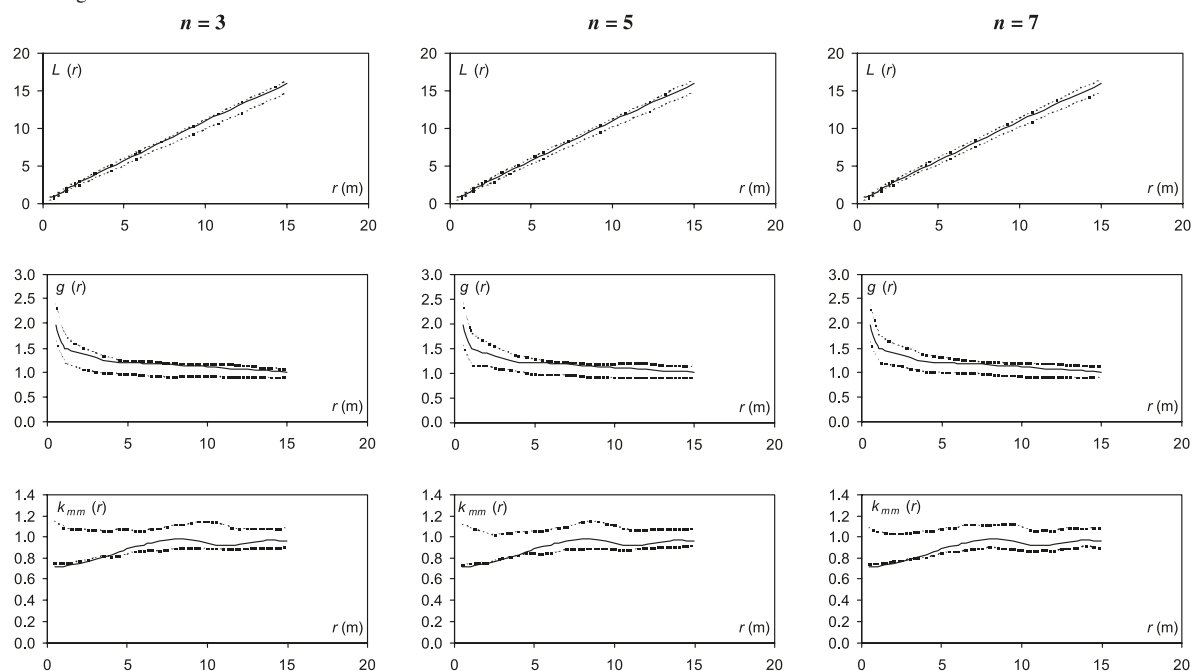


Fig. 7. L function $L(r)$, pair correlation function $g(r)$, and mark correlation function $k_{mm}(r)$ (r is intertree distance) for the Białowieża woodland (Poland). The mark used for the estimation of $k_{mm}(r)$ is tree DBH. The solid curves were estimated from the original forest. The confidence envelopes (broken curves) represent the maximum and minimum values of the 1000 reconstruction replications. n is the number of nearest neighbours.



völgy, and Walsdorf woodlands and the case $n = 5$. The graphs of the three functions of the original forests are inside the confidence envelopes with the slight exception of $g_{ij}(r)$ of the Walsdorf woodland at short distances. This is a bit surprising, as the NNSS should yield good short-range information; perhaps this is related to the large r .Bias and r .RMSE values associated with the mingling index for this forest (Fig. 6).

Discussion and conclusions

The results of testing and validating the reconstruction approach clearly show that the method worked successfully for three quite different woodlands. All NNSS are precisely duplicated in the reconstruction, which is to be expected, since these are used in the contrast measures. However, the second-order characteristics of the reconstructed patterns are

Fig. 8. L function $L(r)$, pair correlation function $g(r)$, and mark correlation function $k_{mm}(r)$ (r is intertree distance) for the Hidegvizvölgy woodland (Hungary). The mark used for the estimation of $k_{mm}(r)$ is tree DBH. The solid curves were estimated from the original forest. The confidence envelopes (broken curves) represent the maximum and minimum values of the 1000 reconstruction replications. n is the number of nearest neighbours.

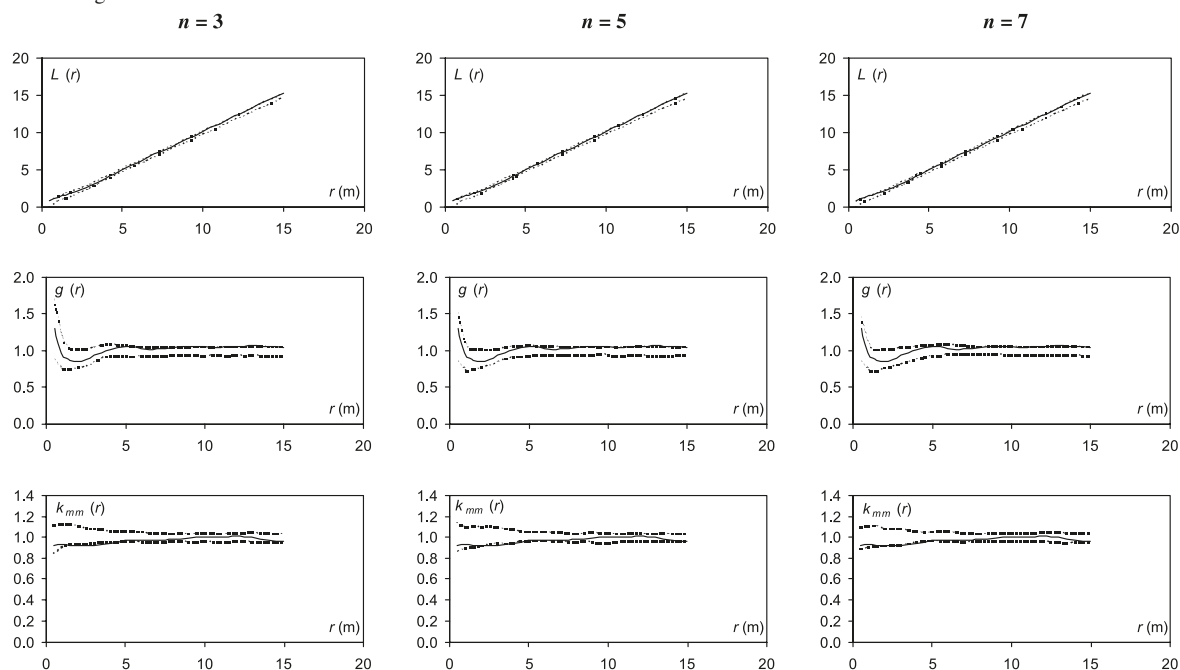


Fig. 9. L function $L(r)$, pair correlation function $g(r)$, and mark correlation function $k_{mm}(r)$ (r is intertree distance) for the Walsdorf woodland (Germany). The mark used for the estimation of $k_{mm}(r)$ is tree DBH. The solid curves were estimated from the original forest. The confidence envelopes (broken curves) represent the maximum and minimum values of the 1000 reconstruction replications. n is the number of nearest neighbours.

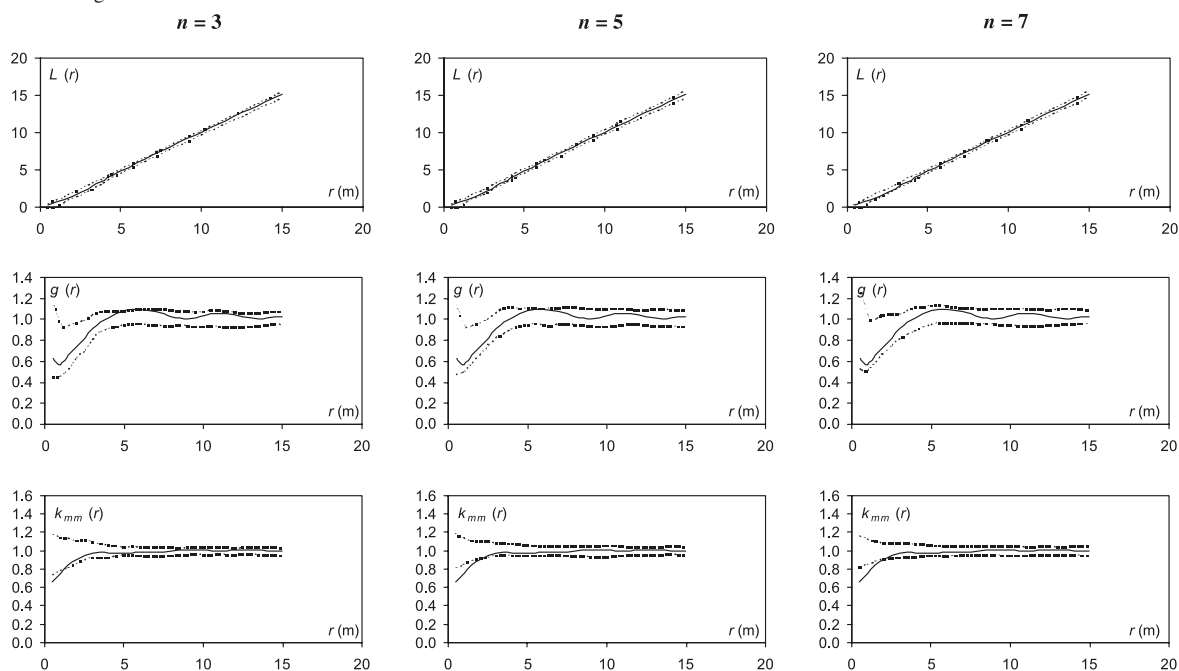
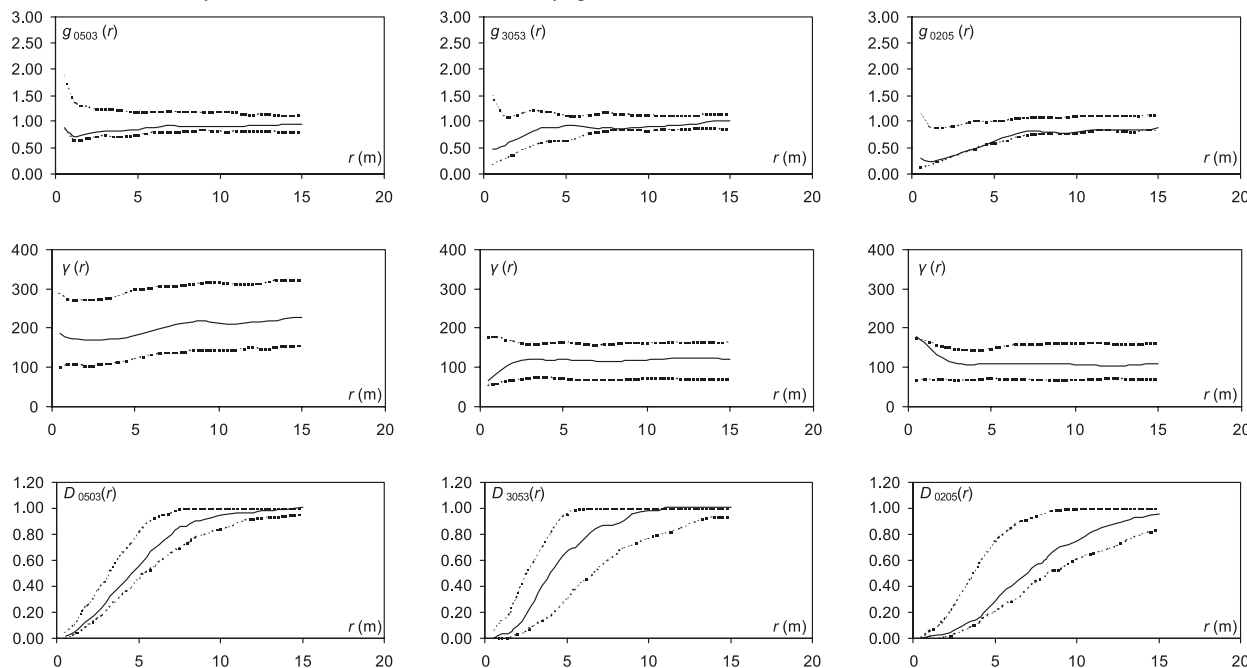


Fig. 10. Partial pair correlation function $g_{ij}(r)$, the mark variogram $\gamma(r)$, and the nearest neighbour distance function $D_{ij}(r)$ (r is intertree distance) for the Białowieża (left panels), Hidegvizvölgy (middle panels), and Walsdorf (right panels) woodlands. n , the number of nearest neighbours, is 5. The mark used for the estimation of $\gamma(r)$ is tree DBH. The solid curves were estimated from the original forest. The confidence envelopes (broken curves) represent the maximum and minimum values of the 1000 reconstruction replications. The number codes in the diagrams of the partial pair correlation function $g_{ij}(r)$ and nearest neighbour distance function $D_{ij}(r)$ refer to the main species of the woodlands under study: 02, beech; 03, hornbeam; 05, Norway spruce; 30, sessile oak; 53, hornbeam.



also close to their observed counterparts. This is explained by the close relationship between Ripley's K function and the nearest neighbour distribution functions:

$$[5] \quad \lambda K(r) = \sum_{j=1}^{\infty} D^{(j)}(r)$$

where $D^{(j)}(r)$ is the distribution function of the distance to the j th neighbour (Stoyan and Stoyan 1994, p. 267, eq. 14.52). The success of the reconstruction also demonstrates the value of the NNSS used, highlighting the fact that they contain important information regarding tree distributions. It is possible to reconstruct woodland patterns based only on short-range information provided by NNSS. The good performance of the reconstruction method presented can be explained by the fact that the NNSS take care of small-scale variation while the systematically placed sample plots control global variation and possibly heterogeneity of the whole data. The successful estimation of confidence envelopes from the 1000 reconstruction replications (Figs. 7–10) and their good fit with the second-order characteristics, such as Ripley's K function, from the large window data demonstrate that reconstruction can be used to estimate these long-range statistics from small sub-windows. The quality of estimation of second-order characteristics also depends on how well the systematic sample covers and reflects the original woodland structures.

This suggests that short-range and in particular nearest

neighbour tree interactions contribute most of the information necessary for the understanding of overall spatial forest structure. This is of great interest from an ecological perspective and leads to the recommendation that NNSS techniques in conjunction with systematic sampling should be used also in other reconstruction applications.

Standard recommendations for the radii of circular sub-windows can be taken from forestry text books (e.g., Loetsch et al. 1973).

The percentage of points sampled is likely to fall when larger populations are used. Preliminary reconstruction simulations in larger populations (not reported in detail here) have revealed that, to retain a constant sampling error, the number of trees that need to be sampled relative to the total number in the population decreases as the population size increases.

Finally, the reconstruction method is flexible so that it can be used with practically no modification to construct tree patterns in areas of larger size than the original window of observation.

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VI

The contribution of structural indices to the modelling of Sitka spruce (*Picea sitchensis*) and birch (*Betula* spp.) crowns



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The contribution of structural indices to the modelling of Sitka spruce (*Picea sitchensis*) and birch (*Betula* spp.) crowns

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ABSTRACT

Crown dimensions are important for the quantification of tree interactions in some growth models. This study investigates the potential for structural indices and other spatial measures to improve the prediction of crown radius and crown length for birch (*Betula* spp.) and Sitka spruce (*Picea sitchensis* (Bong.) Carr.) in forests in Wales. Crown dimensions were measured for 125 birch and 154 spruce in six fully stem-mapped research plots. These data were used to test the performance of a crown radius model and a crown length model which estimated crown dimensions on the basis of allometric relationships with stem dimensions. Spatial data from the six plots were used to calculate the structural indices mean directional index, diameter correlation index, species mingling, and dbh and height dominance, as well as the Hegyi competition index, and basal area of neighbours and larger neighbours, for each crown measurement sample tree, using various numbers of nearest neighbours. Two non-spatial indices, BAL and BALMOD, were also calculated for all sample trees for comparison. These spatial and non-spatial variables were then incorporated into modified crown dimension models. Model performances, in terms of efficiency and relative bias, were compared to determine whether the inclusion of spatial or non-spatial variables resulted in any improvements over models using tree dimensions alone. Crown length and radius were found to be correlated with most of the spatial measures studied. Models incorporating spatial variables gave improvements in performance over allometric models for every data set, and performed more consistently than models containing non-spatial variables. The greatest improvements were achieved for suppressed birch in unthinned forests which had irregularly shaped and strongly displaced crowns. The spatial variable contributing to the most efficient model for each data set varied widely. This points to the complexity of tree spatial interactions and indicates that there is a great deal of scope for investigating other structural indices and crown dimension model forms.

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1. Introduction

The crown of a tree may be defined as “the system of its photosynthetic organs together with the non-photosynthetic organs by which they are physically and physiologically supported, translocating carbohydrates to the stem” (Ottorini et al., 1996, p. 394). Crown dimensions – length, width, surface area or volume – can be used to approximate the scale of the photosynthetic apparatus (e.g. Smith, 1994), which determines the capacity of the tree to produce dry matter (Kozłowski et al., 1991). The sizes, shapes and relative locations of crowns both determine and respond to the shading and constriction effects that characterise above-ground interactions between trees (von Gadow and Hui, 1999). Consequently, crown dimensions have often been used to

model competition and growth (e.g. Biging and Dobbertin, 1992; Moravie and Robert, 2003). Such modelling approaches may require information on crown length, radius and possibly profile (shape) as inputs (e.g. Pretzsch et al., 2002) and, as they are rarely measured in forest surveys, these variables must themselves be modelled.

Crown models range in sophistication from simple, deterministic models based on allometric relationships with stem dimensions (e.g. Hasenauer, 1997) to complex models incorporating stochastic elements (e.g. Biging and Gill, 1997), explicitly modelling crown asymmetry (e.g. Cescatti, 1997), or even modelling branch architecture (e.g. Berezovskaya et al., 1997). For the purposes of modelling inter-tree interaction, simple models usually suffice (von Gadow, 1996). Allometric models tend to perform well because of the direct dependence of stem growth on crown size, but the intensely spatial nature of tree crown interactions means that variables describing the immediate neighbours of a tree may be able to improve model predictions. Simple non-spatial indices can be used to quantify a tree's

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competitive status within a stand. More sophisticated indices of spatial structure can be used to quantify the regularity of the arrangement of a tree's neighbours, their relative sizes or the degree of species mingling (von Gadow and Hui, 2002; Pommerening, 2002; Aguirre et al., 2003), and stand density in the area immediately surrounding a tree, such as within a radius proportional to tree size, can be measured. All of these factors are likely to influence crown development.

This paper considers approaches to crown modelling suitable for inclusion in single-tree, spatially explicit growth models for Sitka spruce (*Picea sitchensis* (Bong.) Carr.) and birch (silver birch *Betula pendula* Roth and downy birch *B. pubescens* Ehrh.) in Wales. Sitka spruce is the main commercial timber species in Great Britain and birch is a significant contributor to biodiversity in plantation forests (Humphrey et al., 1998). The study focuses on crown length and radius as the most important biometric characteristics of tree crowns. Relationships between these crown dimensions and non-spatial and spatial variables describing local stand density and structure are examined, and crown models incorporating these variables are tested with the aim of improving on the predictions of allometric models. It is hypothesised that models containing non-spatial or spatial variables will be able to account for a greater proportion of variation in crown dimensions than those based on stem dimensions alone, and that spatial models will perform better than non-spatial models because of the structural complexity of the stands studied.

2. Materials and methods

2.1. Data

Data for crown modelling were collected in six permanent research plots in North Wales, covering a range of stand structures (Table 1). These plots were established in three state-owned forests: plots CLG1, CLG2 and CLG7 are located in Clocaenog Forest, plots CYB1 and CYB2 are located in Coed y Brenin, and plot GWY1 is located in Gwydyr Forest. Management input in the sampled

stands has been minimal until the recent past, and as a result inter-tree interactions have occurred relatively unmodified by human intervention. In the Coed y Brenin plots, this has led to marked suppression of birch by the more competitive Sitka spruce. In each plot, for all trees equal to or greater than 5 cm diameter outside bark at breast height, 1.3 m (dbh), the species, dbh and height to tree tip (height) were recorded and tree main stem positions were mapped in three dimensions using an electronic theodolite.

Crown measurements were made on random sub-samples of trees in each plot. Altogether 125 birch (in plots CLG7, CYB1, CYB2 and GWY1) and 154 Sitka spruce (in plots CLG1, CLG2, CYB1, CYB2 and GWY1) trees were sampled. For each crown-measured sample tree, two measurements of height and two of height to crown base were made, separated by a horizontal angle of 180°. Crown base was defined as the point at which the lowest live branch left the main stem, excluding epicormic and adventitious shoots and sparsely foliated branches not contiguous with the main part of the crown. Arithmetic mean height and height to crown base were calculated for each tree, and crown length was calculated by subtracting mean height to crown base from mean height.

Eight measurements of crown radius were made for each tree in the cardinal and intercardinal directions, using a vertical sighting tube (Hale, 2004) to sight to the crown margin and a 30 m tape to measure the distance between the centre of the tree bole and the edge of the crown, following the guidelines for the objective identification of the crown edge offered by Ayhan (1977). Quadratic mean (or root mean square) crown radius (Eq. (1)) was calculated for each tree (Hasenauer, 1997), as this gives the most accurate estimate of cross-sectional area (Siostrzonek, 1958) for tree interaction modelling.

$$r_{\text{rms}} = \sqrt{\frac{1}{n} \sum_{i=1}^n r_i^2} \quad (1)$$

where r_{rms} is the root mean square crown radius (m), r_i the i th crown radius measurement (m) and n is the number of

Table 1
Summary data for permanent research plots

Plot	CLG1	CLG2	CLG7	CYB1	CYB2	GWY1
Location (lat./long.)	53°4'25"N 3°25'53"W	53°4'32"N 3°25'42"W	53°4'7"N 3°25'9"W	52°48'26"N 3°54'1"W	52°48'39"N 3°54'2"W	53°6'16"N 3°49'4"W
Elevation (m.a.s.l.)	390	400	350	210	250	230
Elevation range within plot (m)	9.5	10.1	9.3	6.8	12.3	8.8
Area (ha)	1.00	1.00	1.00	0.10	0.26	1.00
Planting year	1951	1951	1985	1970	1972	1931
Planted crop (species)	SS	SS, LP	SS	SS	SS, RC	SP, SS
Natural regeneration (main species by basal area)	SS	SS, ROW	BI, NS, ROW, WIL	BI, DF, OK, WIL	BI, OK, DF, LP	BI, ROW, OK, WH
Stocking (stems per ha)						
BI	–	–	1,094	1,040	738	177
SS	289	251	172	1,120	1,088	46
Other spp.	–	68	600	500	416	271
Total	289	319	1,866	2,660	2,242	494
Basal area (m ² per ha)						
BI	–	–	10.2	5.8	5.6	1.0
SS	29.9	27.0	3.1	35.1	31.7	6.0
Other spp.	–	2.7	3.8	3.5	8.1	23.7
Total	29.9	29.7	17.1	44.4	45.4	30.7
Top height (m)						
BI	–	–	11.4	13.8	12.5	7.4
SS	28.6	27.0	11.2	17.3	17.7	27.3

Plot abbreviations are explained in the text. BI is birch (*Betula pendula* Roth, *B. pubescens* Ehrh.), DF is Douglas fir (*Pseudotsuga menziesii* (Mirb.) Franco), LP is lodgepole pine (*Pinus contorta* Douglas ex Loudon), NS is Norway spruce (*Picea abies* (L.) Karst), OK is oak (*Quercus* spp.), ROW is rowan (*Sorbus aucuparia* L.), SP is Scots pine (*Pinus sylvestris* L.), SS is Sitka spruce (*Picea sitchensis* (Bong.) Carr.), WH is western hemlock (*Tsuga heterophylla* (Raf.) Sarg.), and WIL is willow (*Salix* spp.). Stocking, basal area and top height are reported separately for birch and Sitka spruce. Top height was calculated as the arithmetic mean height of the 100 trees of largest diameter at breast height per hectare, except for GWY1 spruce where there were only 46 spruce per hectare; in this case, the average height of all 23 overstorey trees is given.

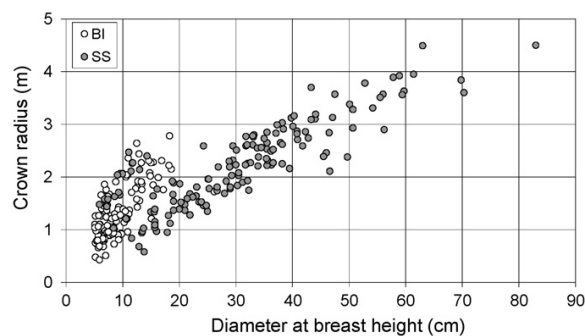


Fig. 1. Scatter plot of crown radius against stem diameter at breast height for all birch (BI) and Sitka spruce (SS) crown measurement sample trees.

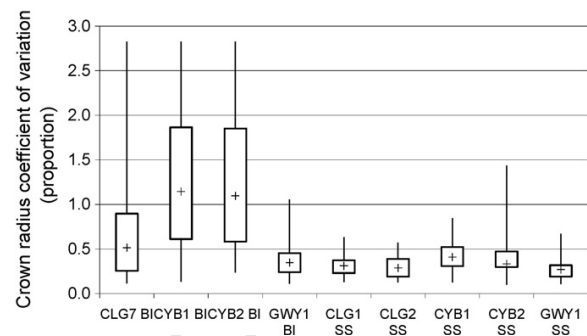


Fig. 3. Medians, minima, maxima and interquartile ranges of crown radius coefficient of variation for all birch (BI) and Sitka spruce (SS) crown measurement samples.

measurements (in this case, $n = 8$). The coefficient of variation of the radius measurements for each sample tree was calculated as a quantification of uniformity in crown shape (Francis, 1986), with a value of zero indicating a perfectly circular crown. In addition, relative canopy displacement, a measure of crown eccentricity defined as “the distance between stem position and canopy centre of mass divided by the mean of the eight canopy extent measurements” (Muth and Bazzaz, 2003, p. 1325), was calculated for each tree. Muth and Bazzaz (2003, p. 1325) state that “a value of zero represents a tree with its canopy centered directly above its stem base. For a canopy the shape of a regular polygon, a value greater than one represents a situation in which the canopy is displaced entirely from the stem base. For the majority of forest trees, relative canopy displacement values tend to range between zero and one, indicating that the canopy is displaced but that the stem base is still positioned at some location beneath the canopy.”

Fig. 1 plots crown radius against dbh for all crown measurement sample trees and Fig. 2 plots crown length against height. Data ranges are particularly wide for spruce because of the range of ages and stand densities represented in the permanent research plots. Two broad groups of trees can be recognised in terms of crown radius (Fig. 1). The majority of the spruce conform to an obvious, more or less linear relationship between dbh and crown radius. The birch and the few understorey spruce from CLG2 and GWY1 show a less obvious but steeper linear trend between 5 and 20 cm dbh, demonstrating that these trees have proportionally broader crowns. In the case of the understorey spruce, this is presumably the result of less physical constriction and of increased lateral growth at the expense of apical growth in order to improve light capture. Two distinct trends are also apparent in crown length

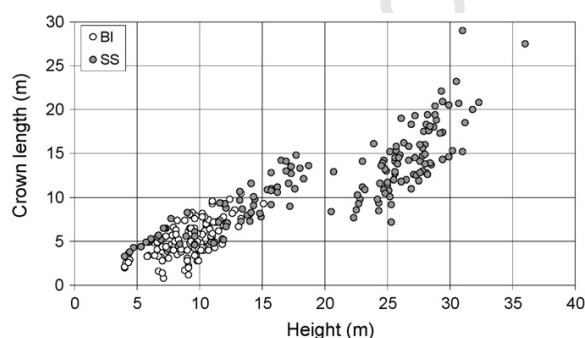


Fig. 2. Scatter plot of crown length against tree height for all birch (BI) and Sitka spruce (SS) crown measurement sample trees.

(Fig. 2); the crowns of trees greater than 20 m in height are proportionally shorter than those of shorter trees. The trees above 20 m height are all planted spruce in plots CLG1, CLG2 and GWY1, and have relatively short crowns presumably because of a lack of thinning in these stands in the past. Initial spacing may have played a role; plots CLG1, CLG2 and GWY1 were established at 1.7 m spacing, whereas CYB1 and CYB2 were planted at 2 m spacing. These trends in crown dimensions indicate that, while there may be strong allometric relationships between stem and crown dimensions, they are also heavily influenced by stand structure.

Fig. 3 demonstrates that the relatively suppressed Coed y Brenin birch have far higher coefficients of variation of crown radius than any of the other trees measured. This cannot be ascribed solely to differences in branching patterns between broadleaves and conifers, as the GWY1 birch appear very similar to the spruce data sets; rather, it seems that physical constriction results in considerable crown asymmetry in this species. CYB1 and CYB2 birch also have the highest values of relative canopy displacement (Fig. 4), with medians greater than one (1.12 and 1.04, respectively) suggesting that the crowns of most trees in these data sets are so heavily displaced that no part of the crown is directly above the stem base. These observations suggest that there may be some difficulty in modelling these crowns as simple, rotationally symmetrical shapes, but quantifications of spatial interrelations may help.

2.2. Tree crown models

Simple crown models based on dbh and tree height were chosen from seven crown radius models and four crown length

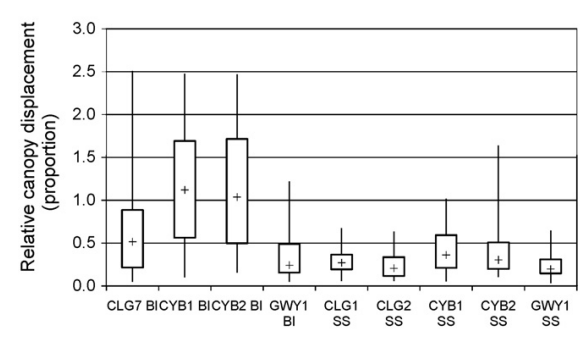


Fig. 4. Medians, minima, maxima and interquartile ranges of relative canopy displacement for all birch (BI) and Sitka spruce (SS) crown measurement samples.

models tested in a previous study (Davies, 2006). Models were chosen on the basis of their predictive power, their simplicity, and the ease with which they could be modified to incorporate spatial variables. The crown radius model (Eq. (2)) was modified from one presented by Hasenauer (1997), and the crown length model (Eq. (3)) was modified from a crown base height model presented by Nagel et al. (2002).

$$\hat{r} = e^{a+b \ln(\text{dbh})} \quad (2)$$

$$\hat{L} = h e^{-a/(h/\text{dbh})} \quad (3)$$

where $\hat{r}$ is the predicted crown radius (m), $\hat{L}$ the predicted crown length (m), dbh the diameter at breast height (cm), h the tree height (m), e the base of natural logarithms, and a and b are the regression coefficients. Eq. (2) returns positive values of $\hat{r}$ for all values of regression coefficients a and b . For Eq. (3) to return crown length values between zero and tree height, the exponential term should be equal to or less than zero; this is accomplished by constraining the coefficient a to be equal to or greater than zero. These basic model forms were modified to include spatial variables (Eqs. (4)–(7)) so that their relative predictive powers could be compared. The modifier approach was favoured because of its simplicity and parameter parsimony (Vanclay and Skovsgaard, 1997; Schmidt et al., 2006).

$$\hat{r} = e^{a+b \ln(\text{dbh})+cx} \quad (4)$$

$$\hat{r} = e^{a+b \ln(\text{dbh})+c \ln(x+1)} \quad (5)$$

$$\hat{L} = h e^{-\max(0, a/(h/\text{dbh})+bx)} \quad (6)$$

$$\hat{L} = h e^{-\max(0, a/(h/\text{dbh})+b \ln(x+1))} \quad (7)$$

where x is the independent variable to be tested, and c is a regression coefficient. Eqs. (4) and (5), like Eq. (2), give positive crown radius values regardless of the values of the regression coefficients. Eqs. (6) and (7) are adapted from Eq. (3) so that the regression coefficients need not be constrained and the directions of crown length relationships with height:diameter ratio (h/dbh) and the independent variables can vary freely.

2.3. Spatial variables

Nine spatial variables were calculated for each crown measurement sample tree. Six spatial diversity indices were calculated (Table 2): mean directional index, MDI (Eq. (8)); species mingling,

M (Eq. (9)); diameter dominance, U_{dbh} (Eq. (10)); height dominance, U_h (Eq. (11)); diameter differentiation, T (Eq. (12)); and diameter correlation index, DCI (Eq. (13)). The inclusion of individual tree elevation in the height dominance index, U_h , reflects the importance of topography in determining vertical canopy stratification and dominance (see Table 1 for the elevation ranges observed in the various plots). These spatially explicit structural indices share the same conceptual principles as spatially explicit competition indices (Pommerening, in press). Therefore, the Hegyi competition index, H_g (Table 2, Eq. (14)), was also calculated for each sample tree. The total basal area of all neighbours (BA, m^2) and the total basal area of neighbours of larger dbh than the subject tree (BAL, m^2) were also calculated.

Spatial variables were calculated using several different numbers of nearest neighbours for each subject tree, based on Euclidean distances. All variables were calculated using the three, four, five and seven nearest neighbours (Fig. 5), and diameter differentiation, T , was also calculated for one nearest neighbour. Variables calculated using different numbers of neighbour trees were identified using numerical subscripts. For example, the species mingling index calculated using three neighbours would be written as M_3 , and the same index calculated using four neighbours would be written as M_4 .

Neighbour selection could result in trees outside plot boundaries being identified as neighbours. Edge correction was therefore required for unbiased estimation of spatial variables. Neither translation nor reflection edge correction (Pommerening and Stoyan, 2006) was deemed suitable, as the height dominance index (U_h) depends heavily on the elevation of individual trees, which could not be objectively modelled beyond plot boundaries. Instead, the random selection of crown measurement sample trees excluded buffer strips (Pommerening and Stoyan, 2006) in each plot. These strips were 10 m wide in the 1 ha plots and 6 m wide in the smaller plots CYB1 and CYB2. In three of the spruce data sets (CLG1, CLG2 and GWY1), however, distances to the seventh nearest neighbours of some reference trees exceeded buffer strip width. This was the case for five reference trees in CLG1, six in CLG2 and four in GWY1; in addition, the distance to the fifth nearest neighbour of one reference tree in CLG2 exceeded buffer strip width. To determine whether this resulted in systematic variation in spatial variables, these were examined for significant non-parametric (Spearman's) correlations with the shortest distance to a plot boundary for each sample tree. The relationships observed were not considered sufficiently numerous or consistent to give cause for concern.

Non-parametric (Spearman's) correlation analyses were carried out in SPSS 11.5 (SPSS Inc.) using all data for each species to

Table 2
Individual tree indices used in this study

Index	Diversity of ...	Formula
Mean directional index (Pommerening, in press)	Tree positions	(8) $MDI_i = \sqrt{\left(\sum_{j=1}^n ((X_j - X_i)/(HDist_{ij}))\right)^2 + \left(\sum_{j=1}^n ((Y_j - Y_i)/(HDist_{ij}))\right)^2}$
Species mingling (von Gadow and Hui, 2002)	Tree species	(9) $M_i = \frac{1}{n} \sum_{j=1}^n w_j \quad w_j = \begin{cases} 1, & \text{species}_i \neq \text{species}_j \\ 0, & \text{otherwise} \end{cases}$
dbh dominance (Aguirre et al., 2003)	Tree dimensions	(10) $U_{\text{dbh}_i} = \frac{1}{n} \sum_{j=1}^n u_{\text{dbh}_j} \quad u_{\text{dbh}_j} = \begin{cases} 1, & \text{dbh}_i > \text{dbh}_j \\ 0, & \text{otherwise} \end{cases}$
Height dominance (Aguirre et al., 2003)	Tree dimensions	(11) $U_{h_i} = \frac{1}{n} \sum_{j=1}^n u_{h_j} \quad u_{h_j} = \begin{cases} 1, & h_i > Z_j + h_j \\ 0, & \text{otherwise} \end{cases}$
dbh differentiation (Pommerening, 2002)	Tree dimensions	(12) $T_i = \frac{1}{n} \sum_{j=1}^n \left(1 - \frac{\min(\text{dbh}_i, \text{dbh}_j)}{\max(\text{dbh}_i, \text{dbh}_j)}\right)$
Diameter correlation index (Pommerening, in press)	Tree dimensions	(13) $DCI_i = \frac{\text{dbh}_i \sum_{j=1}^n \text{dbh}_j}{n \text{dbh}_i^2}$
Hegyi index (Hegyi, 1974)	Tree dimensions	(14) $H_{g_i} = \sum_{j=1}^n \left(\frac{\text{dbh}_j}{\text{dbh}_i HDist_{ij}}\right)$

i is the reference tree, j is a neighbour tree, n is the number of neighbours, dbh is diameter at breast height (cm), $\overline{\text{dbh}}$ is the arithmetic mean diameter at breast height of all trees in the plot (cm), h is the tree height (m), $HDist_{ij}$ is the horizontal distance between trees i and j (m), and X , Y and Z are Cartesian co-ordinates (m). Values of the species mingling, dbh dominance and height dominance indices (Eqs. (9)–(11)) are distributed between 0 and 1.

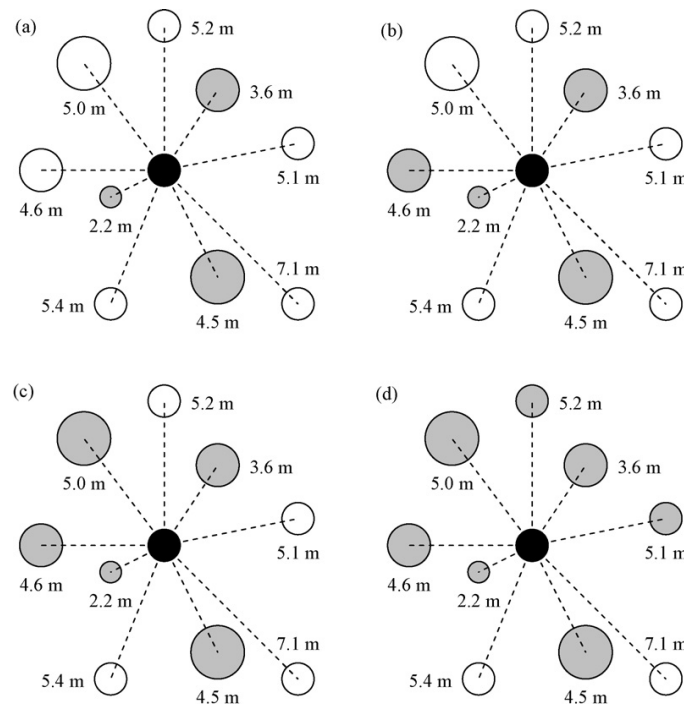


Fig. 5. Selection of the (a) three, (b) four, (c) five and (d) seven nearest neighbours of the reference tree (black circle) based on Euclidean distances, showing the trees identified as neighbours (grey circles) and surrounding trees not identified as neighbours (white circles).

investigate relationships between crown dimensions and spatial variables.

2.4. Non-spatial variables

Although it was assumed that spatially explicit variables would be required to quantify tree interactions in stands which were, despite their plantation origins, relatively irregular in structure, model tests were also carried out with non-spatial variables for comparison. Two common variables, BAL (Eq. (15)) and BALMOD (Eq. (16)), were calculated for each tree as described by Schröder and von Gadow (1999). BALMOD was originally applied to pure, uniform stands, and includes an index of relative spacing (Eq. (17)) based on stand dominant height. Although most of the stands in this study are mixed and uneven-aged, dominant height was still calculated using data from all trees in each plot, rather than separately by species or cohort, as this was considered to be more representative of the overall stocking of the stands.

$$BAL_{all,i} = BA_{all}(1 - p_i) \quad (15)$$

$$BALMOD_i = \frac{(1 - p_i)}{RS} \quad (16)$$

$$RS = \frac{\sqrt{10,000/N}}{H} \quad (17)$$

where $BAL_{all,i}$ is the basal area per unit area of all trees in a given plot larger than reference tree i ($m^2 ha^{-1}$), BA_{all} the basal area per unit area of all trees in a given plot ($m^2 ha^{-1}$), p_i the basal area percentile of reference tree i , $BALMOD_i$ the modified BAL for reference tree i , RS the relative spacing index, N the number of trees per unit area for a given plot ($stems ha^{-1}$), and H is the stand dominant height (m).

As with spatial variables, non-parametric correlations analyses were carried out in SPSS to investigate relationships with crown dimensions.

2.5. Model parameterisation and validation

Data from each crown measurement sample (i.e. all of the birch or spruce for a given research plot) were randomly split into parameterisation and validation data sets of roughly equal size for model testing. In addition, all of the crown measurement data for each species were used to produce a combined data set, randomly split in the same way. For each data set, basic crown models (Eqs. (2) and (3)) and modified models (Eqs. (4)–(7)) were parameterised using Solver (Frontline Systems, Inc.) in Microsoft Excel 2002 (Microsoft Corporation) to minimise the sum of squared errors between measured and predicted values. The modified models were parameterised for all 39 independent variables, the 37 combinations of spatial variables and numbers of nearest neighbours, and the non-spatial variables BAL_{all} and $BALMOD$.

Model validations, comparing observed and predicted values of crown dimensions, were also carried out in Excel, calculating values of model relative bias and efficiency (von Gadow et al., 2003):

$$B (\%) = \frac{\sum_{i=1}^n (\hat{y}_i - y_i)}{n\bar{y}} \times 100 \quad (18)$$

$$E = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (19)$$

where B (%) is relative bias, E the efficiency, $\hat{y}_i$ the i th prediction (modelled crown radius or crown length), y_i the i th observation (measured crown radius or crown length), $\bar{y}$ the mean observation, and n is the number of observations. A t -test was used to determine

whether bias was equal to zero. Efficiency values approach one with improving model performance; a value of zero indicates that the model explains no more variation than the mean value alone (von Gadow et al., 2003). Modified models containing spatial variables were ranked for each data set according to efficiency (and relative bias in the case of ties), as were models containing non-spatial variables. The validation results of the best models were compared with those of the basic models to determine whether the inclusion of spatial or non-spatial variables improved model performance.

3. Results

3.1. Relationships between crown dimensions, and spatial and non-spatial variables

The results of correlation analyses are summarised in Table 3. Many factors influence crown dimensions, particularly stand age, stocking density, thinning history, and species, and the interpretation of correlations should be considered in this context. In the plots studied, crown size generally increases with subject tree dominance (Udbh and Uh) and decreases with increasing competitive pressure (as quantified by the Hegyi index, Hg, the basal area of larger trees, BAL, or the non-spatial variables BAL_{all} and BALMOD). There are no significant correlations with MDI, suggesting that the regularity of the arrangement of neighbouring trees does not have a direct or simple effect on crown size. The negative relationships with species mingling (M) in spruce reflect the fact that most of the largest crowns are in plots CLG1 and CLG2, where mingling is minimal, and most of the smallest crowns are in plots CYB1 and CYB2, where mingling is high, but it should be noted that the CYB stands are both younger and more densely stocked than the CLG stands (Table 1). Diameter differentiation (T) is positively correlated with birch crown radius; birch crowns are therefore wider where tree sizes are more heterogeneous, possibly representing the occupancy of different canopy strata. Conversely, spruce crowns are larger in areas where tree sizes are more homogeneous; these negative correlations are strongly influenced by the relatively small naturally regenerated trees in CLG2 and GWY1, which show exceptionally high differentiation with their neighbours. The behaviour of the diameter correlation index DCI is complex. All else being equal, values increase as reference tree size increases. For a given reference tree size, however, they increase as average neighbour size increases, whereas for a given group of reference tree and neighbours they increase as stand mean dbh decreases. The positive relationships observed here between DCI and crown dimensions are likely to be due to the strong dependence of DCI on reference tree dbh (for both birch and spruce, DCI is significantly positively correlated with dbh). The positive correlations between crown dimensions and BA are perhaps counter-intuitive. For a given stand, a larger basal area of neighbouring trees might be expected to indicate a greater level of competition, with a corresponding negative effect on crown size. Over the range of plots sampled, however, the generally positive relationship between BA and crown size simply reflects the fact that the basal area of neighbours tends to be greater in plots where sample trees are larger and more widely spaced.

3.2. Alternative crown radius models

Table 4 shows the maximum efficiency values achieved for the basic crown radius model (Eq. (2)) and modified versions incorporating non-spatial and spatial variables (Eqs. (4) and (5)) for all data sets. Regardless of the modelling approach, efficiency values are generally poorest for CYB1 and CYB2 data sets, particularly for birch. Basic model efficiencies for individual plot

Table 3

Non-parametric (Spearman's) correlations significant at the 0.05 level between crown dimensions (crown radius, r , and length, L) and spatial and non-spatial variables for birch (BI, 125 trees) and Sitka spruce (SS, 154 trees)

	r_{BI}	L_{BI}	r_{SS}	L_{SS}
MDI ₃				
MDI ₄				
MDI ₅				
MDI ₇				
M ₃			–	–
M ₄			–	–
M ₅		–	–	–
M ₇		–	–	–
Udbh ₃	+	+	+	+
Udbh ₄	+	+	+	+
Udbh ₅	+	+	+	+
Udbh ₇	+	+	+	+
Uh ₃	+	+	+	+
Uh ₄	+	+	+	+
Uh ₅	+	+	+	+
Uh ₇	+	+	+	+
T ₁	+		–	–
T ₃	+		–	–
T ₄	+		–	–
T ₅	+		–	–
T ₇	+		–	–
DCI ₃	+	+	+	+
DCI ₄	+	+	+	+
DCI ₅	+	+	+	+
DCI ₇	+	+	+	+
Hg ₃	–	–	–	–
Hg ₄	–	–	–	–
Hg ₅	–	–	–	–
Hg ₇	–	–	–	–
BA ₃	+		+	+
BA ₄	+		+	+
BA ₅	+		+	+
BA ₇	+		+	+
BAL ₃		–		
BAL ₄		–		
BAL ₅		–		
BAL ₇		–		
BAL _{all}	–	–	–	–
BALMOD	–	–	–	–

Symbol '+' indicates a positive correlation, '–' a negative correlation.

data sets are significantly negatively correlated with median crown radius coefficient of variation (Fig. 3) and relative canopy displacement (Fig. 4) at the 0.05 significance level (non-parametric correlations, two-tailed, $n = 9$, $p = 0.030$ and 0.036 , respectively), demonstrating that crown radius model performances are poorest in stands where crowns are most irregular in shape and most heavily displaced as a result of suppression. The models with non-spatial variables give slight efficiency gains for five data sets, but in some cases actually give worse results than the basic model. Spatial models not only give efficiency gains in all cases, these gains are consistently greater than those for non-spatial models. Gains are especially great for CYB1 and CYB2 birch.

Details of the spatial models giving the highest efficiency values for each data set are shown in Table 5. None of these models is significantly biased at the 0.05 level. The Hegyi index (Hg), height dominance index (Uh) and basal area of neighbours (BA) are the most commonly represented independent variables, and seven is the most common number of neighbours selected. Most variables are not logarithmically transformed. The CLG1 spruce model features the mean directional index, MDI, despite the fact that this variable was not significantly correlated with crown dimensions.

Table 4

Results of crown radius model tests for birch (BI) and Sitka spruce (SS) data sets, showing for each data set the maximum efficiency achieved by including non-spatial and spatial variables, and the gain in efficiency over the basic model

Data set (and number of trees)	Maximum efficiency (and gain over the basic model)		
	Basic model	With non-spatial variables	With spatial variables
CLG7 BI (50)	0.67	0.67 (+0.00)	0.76 (+0.09)
CYB1 BI (25)	−0.31	−0.19 (+0.12)	0.27 (+0.58)
CYB2 BI (25)	0.02	−2.11 (−2.13)	0.67 (+0.65)
GWY1 BI (25)	0.48	0.56 (+0.08)	0.62 (+0.14)
All plots BI (125)	0.40	0.42 (+0.02)	0.65 (+0.25)
CLG1 SS (50)	0.57	0.55 (−0.02)	0.61 (+0.04)
CLG2 SS (25)	0.49	0.54 (+0.05)	0.57 (+0.08)
CYB1 SS (25)	0.11	−0.11 (−0.22)	0.28 (+0.17)
CYB2 SS (25)	0.26	0.25 (−0.01)	0.47 (+0.21)
GWY1 SS (29)	0.75	0.79 (+0.04)	0.88 (+0.13)
All plots SS (154)	0.75	0.75 (+0.00)	0.77 (+0.02)

Table 5

Crown radius models incorporating spatial variables giving the highest efficiency values for birch (BI) and Sitka spruce (SS) data sets

Data set (and number of trees)	Model form	Independent variable	Efficiency	Relative bias (%)
CLG7 BI (50)	Eq. (5)	Hg ₇	0.76	−1.28
CYB1 BI (25)	Eq. (4)	Uh ₇	0.27	3.42
CYB2 BI (25)	Eq. (4)	Hg ₇	0.67	4.30
GWY1 BI (25)	Eq. (4)	T ₃	0.62	1.96
All plots BI (125)	Eq. (5)	BA ₇	0.65	2.93
CLG1 SS (50)	Eq. (4)	MDI ₅	0.61	0.41
CLG2 SS (25)	Eq. (5)	BAL ₇	0.57	0.65
CYB1 SS (25)	Eq. (5)	Udbh ₃	0.28	7.60
CYB2 SS (25)	Eq. (4)	BA ₅	0.47	3.26
GWY1 SS (29)	Eq. (4)	Uh ₇	0.88	−4.90
All plots SS (154)	Eq. (4)	Hg ₇	0.77	−0.95

The models for the two full species data sets are given in Eqs. (20) and (21), and their behaviours are illustrated in Figs. 6 and 7. Both show increases in crown radius with increases in variables which might be expected to represent levels of competition (BA₇ and Hg₇). The spruce crown radius model (Fig. 7) returns extremely high values when both dbh and Hg₇ are high; in practice, however, such high values of Hg₇ are not associated with the larger trees.

$$\hat{r}_{BI} = e^{-1.34520 + 0.70302 \ln(\text{dbh}) + 0.90728 \ln(\text{BA}_7 + 1)} \quad (20)$$

$$\hat{r}_{SS} = e^{-2.17846 + 0.82286 \ln(\text{dbh}) + 0.10903 \text{Hg}_7} \quad (21)$$

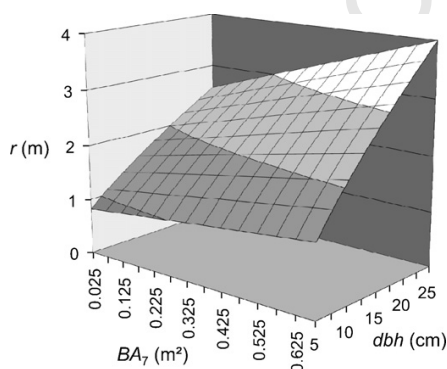


Fig. 6. Crown radius model behaviour for the combined birch data set over the approximate observed range of predictor variables, showing crown radius (r) in relation to reference tree dbh (dbh) and the basal area of the seven nearest neighbours (BA₇).

where $\hat{r}_{BI}$ is the predicted birch crown radius (m) and $\hat{r}_{SS}$ is the predicted Sitka spruce crown radius (m).

3.3. Alternative crown length models

As with crown radius models, crown length model efficiencies are lowest for CYB1 and CYB2 data sets (Table 6). In general, however, efficiencies are higher than for crown radius models. Non-spatial models give efficiency gains in most cases, and these are greatest for CYB1 and CYB2 birch. Models incorporating spatial variables give gains in all cases, though these are lower than those for non-spatial models for CYB1 and GWY1 birch; again, the greatest gains are for CYB1 and CYB2 birch.

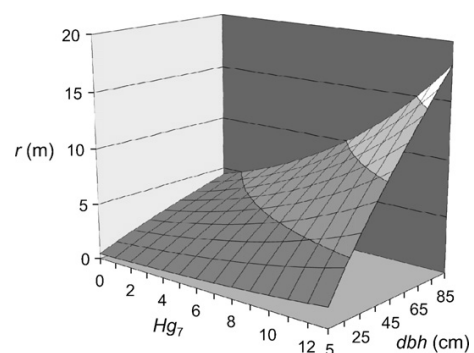


Fig. 7. Crown radius model behaviour for the combined Sitka spruce data set over the approximate observed range of predictor variables, showing crown radius (r) in relation to reference tree dbh (dbh) and the Hegyi index calculated using the seven nearest neighbours (Hg₇).

Table 6

Results of crown length model tests for birch (BI) and Sitka spruce (SS) data sets, showing for each data set the maximum efficiency achieved by including non-spatial and spatial variables, and the gain in efficiency over the basic model

Data set (and number of trees)	Maximum efficiency (and gain over the basic model)		
	Basic model	With non-spatial variables	With spatial variables
CLG7 BI (50)	0.69	0.66 (–0.03)	0.71 (+0.02)
CYB1 BI (25)	0.42	0.62 (+0.20)	0.55 (+0.13)
CYB2 BI (25)	0.31	0.47 (+0.16)	0.53 (+0.22)
GWY1 BI (25)	0.81	0.87 (+0.06)	0.85 (+0.04)
All plots BI (125)	0.59	0.60 (+0.01)	0.61 (+0.02)
CLG1 SS (50)	0.64	0.66 (+0.02)	0.70 (+0.06)
CLG2 SS (25)	0.65	0.70 (+0.05)	0.78 (+0.13)
CYB1 SS (25)	0.54	0.61 (+0.07)	0.63 (+0.09)
CYB2 SS (25)	0.55	0.50 (–0.05)	0.56 (+0.01)
GWY1 SS (29)	0.89	0.92 (+0.03)	0.92 (+0.03)
All plots SS (154)	0.67	0.65 (–0.02)	0.74 (+0.07)

Table 7

Crown length models incorporating spatial variables giving the highest efficiency values for birch (BI) and Sitka spruce (SS) data sets

Data set (and number of trees)	Model form	Independent variable	Efficiency	Relative bias (%)
CLG7 BI (50)	Eq. (6)	BA ₇	0.71	4.75
CYB1 BI (25)	Eq. (7)	M ₅	0.55	–11.22
CYB2 BI (25)	Eq. (7)	Hg ₇	0.53	–8.27
GWY1 BI (25)	Eq. (7)	Hg ₇	0.85	3.94
All plots BI (125)	Eq. (6)	BA ₇	0.61	–0.01
CLG1 SS (50)	Eq. (7)	Hg ₅	0.70	–2.10
CLG2 SS (25)	Eq. (7)	M ₃	0.78	5.30
CYB1 SS (25)	Eq. (6)	BAL ₃	0.63	9.09
CYB2 SS (25)	Eq. (6)	Uh ₃	0.56	1.49
GWY1 SS (29)	Eq. (6)	Hg ₇	0.92	3.42
All plots SS (154)	Eq. (6)	BA ₅	0.74	2.53

None of the spatial crown length models (Table 7) is biased at the 0.05 significance level. The Hegyi index (Hg) and basal area of neighbour trees (BA) are the most commonly included independent variables, with seven the most common number of neighbours. Approximately half of the models include logarithmic transformation of the spatial variable. Although most of the variables selected can be considered to represent competitive effects or reference tree dominance, the species mingling index features in one birch model and one spruce model, suggesting that crown interactions do not depend on tree size alone.

The models for the species combined data sets are given in Eqs. (22) and (23). Fig. 8 shows the behaviour of the birch model, with crown length increasing with basal area of neighbouring trees (BA₇) and decreasing with height:diameter ratio of the reference

tree. Crown length also decreases with height:diameter ratio in the spruce model (Fig. 9) and additionally decreases with basal area of neighbouring trees (BA₅).

$$\hat{L}_{BI} = he^{-\max(0, 0.54513(h/dbh) - 0.28043BA_7)} \quad (22)$$

$$\hat{L}_{SS} = he^{-\max(0, 0.58557(h/dbh) + 0.30910BA_5)} \quad (23)$$

where $\hat{L}_{BI}$ is the predicted birch crown length (m) and $\hat{L}_{SS}$ is the predicted Sitka spruce crown length (m).

4. Discussion

Our results have shown that simple crown radius and crown length models based on stem dimensions alone perform relatively

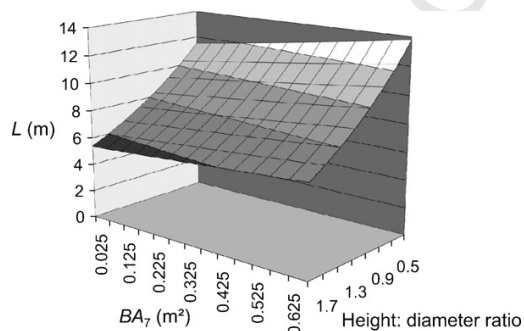


Fig. 8. Crown length model behaviour for the combined birch data set over the approximate observed range of predictor variables, showing crown length (L) of a tree 15 m in height in relation to reference tree height:diameter ratio (height (m) divided by dbh (cm)) and the basal area of the seven nearest neighbours (BA_7). Note that the height:diameter axis has been reversed.

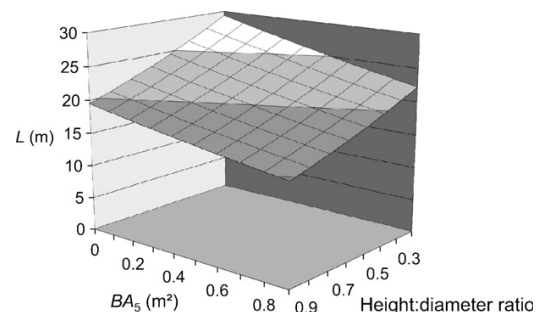


Fig. 9. Crown length model behaviour for the combined Sitka spruce data set over the approximate observed range of predictor variables, showing crown length (L) of a tree 35 m in height in relation to reference tree height:diameter ratio (height (m) divided by dbh (cm)) and the basal area of the five nearest neighbours (BA_5). Note that the height:diameter axis has been reversed.

poorly, in terms of model efficiency, in situations where trees are heavily suppressed due to excessive stocking. In these situations, the normally strong relationships between stem and crown dimensions become weaker, and crowns may be very irregular in shape, as quantified by the coefficient of variation of crown radius, and strongly displaced from stem bases, as quantified by relative canopy displacement. In this study, these effects have been particularly pronounced for birch.

Correlation analyses have revealed definite relationships between crown dimensions and a range of spatially explicit and non-spatial indices of stand structure. The inclusion of non-spatial indices can improve model performances in many cases, but greater and more consistent improvements can be achieved using spatially explicit structural indices and other spatial variables. This may indicate that the stands studied are too structurally complex for tree interactions to be modelled using non-spatial variables. Particularly large improvements in model performance have been achieved for the suppressed Coed y Brenin birch data sets. There is still much scope for improvement in the modelling of suppressed crowns, however, as absolute values of model efficiency remain low in some cases, particularly for CYB1 crown radius models. The relationships observed between crown dimensions and spatial variables highlight the intensely spatial nature of crown growth and interactions. It may be that, in the most extreme cases of competition, more sophisticated methods of explicitly modelling asymmetric crown shapes and crown displacement may be required (e.g. Cescatti, 1997; Umeki, 1995).

The variety of models produced for different data sets hints at the complexity of the underlying relationships. All of the spatial variables tested, with the exception of the diameter correlation index, featured in the most efficient models, and no one variable has been shown to be superior in accounting for the interaction of tree crowns. The variability of the models limits their wider use. At present, the most promising models for widespread application are those developed for the combined data sets for each species (Eqs. (20)–(23)). However, as with any models, these should be applied with caution outside the range of their parameterisation data, particularly since the relationships they show with measures of competition or local stand density are generally the reverse of what might be expected within a given stand (Figs. 6–8; cf. Fig. 9).

There is considerable scope for further study in this field. The potential range of model forms, structural indices and neighbour selection methods is vast (Pommerening, in press), and future research may produce robust, effective models that can be applied with confidence across a broad range of stand structures. Most of the spatial variables tested in this study quantify the diversity of tree dimensions, but variables quantifying the diversity of tree positions (mean directional index, MDI) and species (species mingling, *M*) also contributed to viable models and similar indices could be explored. Many existing crown models could be used as the basis for further tests (e.g. Gill et al., 2000; Pretzsch et al., 2002). The models presented in this study are static, and the potential role of spatial diversity indices in dynamic crown models, such as crown base recession models, should also be investigated. Research on heavily suppressed crowns may reveal models that can account for the effects of extreme competition for space and light.

5. Conclusions

This study of the use of spatial diversity indices in crown dimension models has shown that these and other spatial variables can contribute to effective models of crown radius and crown length for birch and Sitka spruce growing in a variety of stand structures, and can yield particularly great gains in model

performance in overstocked stands where interactions between crowns are exceptionally intense. The complexity of tree spatial interactions means that, at present, these models are largely stand specific due to their variability, but further work may produce more widely applicable indices and model forms, and may reveal new ways of modelling the crowns of trees facing extreme competition pressure.

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